

A Zeroth-Order Paradigm for LLM Preference Alignment

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Abstract

Direct preference alignment methods are widely used to align large language models (LLMs) with human preferences because of their computational and memory efficiency. However, likelihood displacement motivates alternative ways to extract information from preference pairs with small likelihood margins. In this paper, we propose and analyze Comparison-based Preference Optimization (ComPO), a zeroth-order alignment method based on comparison oracles. ComPO extracts directional information from these pairs without directly optimizing a differentiable preference loss on them. We establish a convergence guarantee for its basic offline scheme under smoothness, gradient sparsity, and compatibility between the oracle and a latent objective. We further introduce online ComPO, which retains the offline comparison mechanism and uses unlabeled policy generations for reverse-KL control relative to a reference policy. Following the coverage perspective of preference fine-tuning, we establish a performance guarantee for a basic constrained scheme under local coverage and in-distribution pairwise reward accuracy. Experiments on Mistral, Llama, Gemma-2, Qwen3, and Gemma-3 models demonstrate improvements over existing direct alignment methods, including length-controlled win rates, with pair-level diagnostics providing evidence consistent with mitigating likelihood displacement.

Keywords: Preference alignment, comparison oracles, zeroth-order optimization, KL regularization, local coverage

1 Introduction

Generative AI has become an increasingly important tool for building and managing intelligent systems across academia, industry, and government. Large language models (LLMs) are a core part of this progress, with strong capabilities in data organization, retrieval, reasoning, and analysis (Brown et al., 2020; Chowdhery et al., 2023; Touvron et al., 2023;

用于大型语言模型偏好对齐的零阶范式

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摘要

直接偏好对齐方法因其计算和内存效率而被广泛用于对齐大语言模型（大语言模型）与人类偏好。然而，似然位移促使人们寻找其他从具有较小似然边界的偏好对中提取信息的方法。在本文中，我们提出并分析了基于比较或acles的零阶对齐方法ComPO（基于比较的偏好优化）。ComPO从这些对中提取方向信息，而无需直接优化它们上的可微偏好损失。在平滑性、梯度稀疏性和或acles与潜在目标之间的兼容性下，我们为其基本离线方案建立了收敛保证。我们进一步引入了在线ComPO，它保留了离线比较机制，并使用无标签策略生成相对于参考策略的逆KL控制。遵循偏好微调的覆盖视角，在局部覆盖和分布内成对奖励准确性的基本约束方案下，我们建立了性能保证。在Mistral、Llama、Gemma-2、Qwen3和Gemma-3模型上的实验表明，与现有的直接对齐方法相比，包括长度控制的胜率在内，成对级别的诊断提供了与减轻似然位移一致的证据。

关键词：偏好对齐、比较预言机、零阶优化、KL正则化、局部覆盖率

1 引言

生成式人工智能已成为构建和管理学术界、工业界及政府智能系统的重要工具。大语言模型（LLMs）是这一进程的核心部分，在数据组织、检索等方面具有强大能力。

推理、分析（布朗等人，2020；乔杜里等人，2023；图尔沃等人，2023；

(Achiam et al., 2023; Bubeck et al., 2023). Since these models are trained on large and heterogeneous corpora, they need further alignment with human preferences so that their responses are helpful, harmless, and reliable (Bai et al., 2022). A prominent approach is *reinforcement learning from human feedback* (RLHF) (Christiano et al., 2017; Stiennon et al., 2020), which first learns a reward model from human preference pairs and then optimizes the policy using reinforcement learning. Despite its empirical success (Ziegler et al., 2019; Ouyang et al., 2022; Touvron et al., 2023; Achiam et al., 2023), RLHF requires a multi-stage training pipeline and can be expensive in memory and computation. This motivates direct alignment methods, e.g., direct preference optimization (DPO) (Rafailov et al., 2023) and its variants (Azar et al., 2024; Ethayarajh et al., 2024; Park et al., 2024; Xu et al., 2024a; Tang et al., 2024a; Meng et al., 2024; Chen et al., 2025a; Zhao et al., 2025), which directly optimize the policy using preference pairs and avoid separately training a reward model.

Direct alignment methods are appealing because of their simplicity and stability. Yet, they suffer from a critical issue known as likelihood displacement. *Likelihood displacement* refers to the counter-intuitive situation where training increases the likelihood of preferred responses relative to dispreferred ones, but decreases the absolute probability of the preferred responses, leading to “unintentional unalignment” (Pal et al., 2024; Tajwar et al., 2024; Rafailov et al., 2024b; Pang et al., 2024; Liu et al., 2024b; Yuan et al., 2025; Razin et al., 2025). For example, training a model to prefer NO over NEVER can sharply increase the likelihood of YES. Practically, this issue can harm LLM behavior by shifting probability mass to unsafe responses. When the prompt asks for steps for a terrorist organization to infiltrate a government agency, Gemma-2B-it initially generates refusal responses, while DPO training can make the model comply with the unsafe request because likelihood displacement shifts probability mass away from refusal responses; see Razin et al. (2025, Table 18). Another related issue is *verbosity*, which refers to the tendency of models fine-tuned with RLHF (Singhal et al., 2024; Kabir et al., 2024) or direct alignment methods (Park et al., 2024; Amini et al., 2024; Rafailov et al., 2024a) to generate longer responses without a corresponding improvement in quality, resulting in lower efficiency and higher consumption of hardware resources.

Recent works have suggested that likelihood displacement is related to preference pairs whose preferred and dispreferred responses are similar under model-dependent measures (Pal et al., 2024; Razin et al., 2025). We refer to such small margin pairs as *noisy preference pairs* in this paper (see Eq. (10)). Existing methods have tried to mitigate likelihood displacement by adding additional regularization (Pal et al., 2024; Rafailov et al., 2024b). More recently, Razin et al. (2025) proposed to measure the similarity between preferred and dispreferred responses using the centered hidden embedding similarity (CHES) score, and empirically showed that filtering out preference pairs identified by the CHES score as problematic can be more effective for mitigating likelihood displacement than adding supervised fine-tuning (SFT) regularization. This finding highlights the role of data geometry in direct alignment. However, filtering noisy pairs also removes them from training entirely, even though these pairs may still contain useful comparative information.

While DPO provides a computationally convenient framework by maximizing a certain log-likelihood margin between preferred and dispreferred responses, this objective function can be viewed as a *proxy* for the true goal of alignment. This proxy is effective when preference pairs clearly distinguish better responses from worse responses. However, when

阿奇亚姆等人, 2023; 布贝克等人, 2023)。由于这些模型是在大型异构语料库上训练的, 它们需要进一步与人类偏好对齐, 以便它们的回复是有帮助的、无害的且可靠的 (白等人, 2022)。一种突出方法是来自人类反馈的强化学习 (RLHF) (克里斯蒂安诺等人, 2017; 斯蒂尼昂等人, 2020), 它首先从人类偏好对中学习奖励模型, 然后使用强化学习优化策略。尽管它在经验上取得了成功 (齐格勒等人, 2019; 欧阳等人, 2022; 图尔沃等人, 2023; 阿奇亚姆等人, 2023), 但RLHF需要一个多阶段训练流程, 并且在内存和计算上可能很昂贵。这促使直接对齐方法, 例如直接偏好优化 (DPO) (拉菲洛夫等人, 2023) 及其变体 (阿扎尔等人, 2024; 埃塞亚拉贾等人, 2024; 帕克等人, 2024; 徐等人, 2024a; 唐等人, 2024a; 孟等人, 2024; 陈等人, 2025a; 赵等人, 2025), 这些方法直接使用偏好对优化策略, 并避免单独训练奖励模型。

直接对齐方法因其简单性和稳定性而具有吸引力。然而, 它们存在一个关键问题, 称为似然位移。似然位移指的是一种反直觉的情况: 训练会增加首选响应相对于不受欢迎响应的似然, 但会降低首选响应的绝对概率, 从而导致“非故意的不对齐” (Pal 等人, 2024; Tajwar 等人, 2024; Rafailov 等人, 2024b; Pang 等人, 2024; Liu 等人, 2024b; Yuan 等人, 2025; Razin 等人, 2025)。例如, 训练模型优先选择 No 而非 NEVER 可能会显著增加 YES 的似然。实际上, 这个问题会通过将概率质量转移到不安全的响应中来损害 LLM 的行为。当提示要求一个恐怖组织渗透政府机构的方法时, Gemma-2B-it 最初会生成拒绝响应, 而 DPO 训练可以使模型服从不安全的要求, 因为似然位移将概率质量从拒绝响应中移开; 参见 Razin 等人 (2025, 表 18)。另一个相关问题是冗长性, 它指的是使用 RLHF (Singhal 等人, 2024; Kabir 等人, 2024) 或直接对齐方法 (Park 等人, 2024; Amini 等人, 2024; Rafailov 等人, 2024a) 微调的模型倾向于生成更长但质量没有相应提高的响应, 从而导致效率降低和硬件资源消耗增加。

近期研究表明, 似然位移与偏好对有关, 这些偏好对在模型依赖的度量下, 其首选和不受欢迎的响应相似 (Pal 等人, 2024; Razin 等人, 2025)。我们在本文中将这类小差距对称为噪声偏好对 (参见公式 (10))。现有方法试图通过添加额外的正则化来缓解似然位移 (Pal 等人, 2024; Rafailov 等人, 2024b)。最近, Razin 等人 (2025) 提出使用中心隐藏嵌入相似度 (CHES) 分数来测量首选和不受欢迎响应之间的相似性, 并实证表明, 过滤掉被 CHES 分数识别为有问题的偏好对, 比添加监督微调 (SFT) 正则化更有效地缓解似然位移。这一发现突出了数据几何在直接对齐中的作用。然而, 过滤噪声对也会将它们完全从训练中移除, 尽管这些对可能仍然包含有用的比较信息。

虽然DPO通过最大化偏好和不受欢迎响应之间的一定对数似然差距, 提供了一个计算上便捷的框架, 但这个目标函数可以被视为对齐真实目标的<样式 id=1'>代理</样式>。当偏好对能清晰区分更好响应和更差响应时, 这个代理是有效的。然而, 当

faced with noisy pairs – *where the preference signal is weak or ambiguous under model-based similarity measures* – optimizing a fixed DPO-style objective can lead to adverse effects such as likelihood displacement. In such cases, the pair may still provide useful local information, even if it is not suitable for direct optimization by a margin-based loss. This motivates a comparison-oracle view of preference alignment. Explicitly defining alignment as a single optimizable mathematical objective function is exceptionally challenging. Instead of pursuing such an explicit objective, we ask whether a nearby policy perturbation improves the local behavior of the model on preference pairs. A favorable perturbation should increase the likelihood of the preferred response and decrease the likelihood of the dispreferred response. In this way, noisy preference pairs are treated as comparison signals about a latent alignment objective, rather than as direct samples for a fixed loss function.

In this paper, we propose a zeroth-order preference alignment method based on comparison oracles, called ComPO. Our approach perturbs the current policy, evaluates whether each perturbation increases the likelihood of preferred responses and decreases that of dispreferred responses, and aggregates the resulting one-bit signals to estimate a normalized update direction. This allows low-margin pairs, designated as noisy, to contribute to alignment without directly optimizing a differentiable preference loss on them, complementing standard direct alignment methods applied to clean pairs. We further extend ComPO to control policy deviation using unlabeled online generations, while retaining offline preference pairs as the source of comparison signals. The motivation follows the coverage perspective of Song et al. (2024b): reverse KL can be estimated from generations of the policy being evaluated, and constraining it permits a performance analysis based on the coverage within a prescribed neighborhood of the reference policy rather than over the full policy class. Coverage remains a separate assumption and is not implied by the KL constraint.

Our empirical also examines length-related effects which have been studied in the literature (Gao et al., 2023; Dubois et al., 2023; Park et al., 2024; Amini et al., 2024; Xu et al., 2024a; Meng et al., 2024; Pang et al., 2024). Although ComPO is not specifically designed to control verbosity, we evaluate its length-controlled (LC) win rates and examine pair-level likelihood changes. We interpret higher LC win rates as improved judged performance after adjustment for response length, rather than direct evidence of shorter responses.

Contributions. Our contributions can be summarized as follows:

1. We develop ComPO, a comparison-based method that uses low-margin preference pairs to refine an aligned policy without directly optimizing a differentiable preference loss on those pairs. Its practical offline implementation uses output-layer perturbations and entry-wise thresholding. The online extension retains the same comparison mechanism and uses unlabeled current-policy generations to adapt the step size.
2. We establish a best-iterate convergence guarantee for the basic offline scheme under smoothness, gradient sparsity, and oracle compatibility. For the basic online scheme, we prove feasibility under an exact reverse-KL constraint and bound the performance gap in terms of in-distribution pairwise reward error under local coverage.
3. We evaluate ComPO on base and instruction-tuned models from the Mistral, Llama, Gemma-2, Qwen3, and Gemma-3 families. The experiments assess its compatibility

面对噪声对——<样式 id='l'>在基于模型的相似性度量下，偏好信号微弱或模糊</样式>——优化固定的DPO风格目标可能导致似然位移等不利影响。在这种情况下，对即使不适合基于差距的损失直接优化的对仍然可能提供有用的局部信息。这促使我们采用偏好对齐的比较-预言机观点。明确地将对齐定义为一个单一的优化数学目标是极具挑战性的。与其追求这种明确的目标，我们询问附近的策略扰动是否改善了模型在偏好对上的局部行为。有利的扰动应该增加偏好响应的似然，并降低不受欢迎响应的似然。通过这种方式，噪声偏好对被视为关于潜在对齐目标的比较信号，而不是固定损失函数的直接样本。

在本文中，我们提出了一种基于比较预言机的零阶偏好对齐方法，称为ComPO。我们的方法扰动当前策略，评估每个扰动是否增加了偏好响应的可能性并减少了非偏好响应的可能性，并将生成的单比特信号聚合起来以估计归一化更新方向。这使得低边距对（指定为噪声），能够在不直接优化它们上的可微分偏好损失的情况下，为对齐做出贡献，补充应用于干净对的标准直接对齐方法。我们进一步扩展ComPO，使用未标记的在线生成来控制策略偏差，同时保留离线偏好对作为比较信号来源。动机遵循Song等人（(2024b)）的观点：反向KL可以从被评估策略的生成中估计，并且约束它允许基于参考策略规定邻域内的覆盖范围进行性能分析，而不是在整个策略类上进行。覆盖仍然是一个独立的假设，并且不是由KL约束所隐含的。

我们的实证研究也考察了文献中已研究过的与长度相关的效应（高 et al., 2023; 杜波依斯 et al., 2023; 朴 et al., 2024; 阿米尼 et al., 2024; 徐 et al., 2024a; 孟 et al., 2024; 庞 et al., 2024）。尽管 ComPO 并非专门设计用于控制冗长性，但我们评估了其长度控制（LC）胜率，并考察了配对级别的似然变化。我们将更高的 LC 胜率解释为在考虑响应长度后判定的性能提升，而非直接证据表明响应更短。

贡献。我们的贡献可以总结如下：

1. 我们开发了 ComPO，这是一种基于比较的方法，它使用低边际偏好对来改进对齐策略，而无需在这些对上直接优化可微分的偏好损失。其离线实际实现使用输出层扰动和逐元素阈值处理。在线扩展保留了相同的比较机制，并使用当前策略生成的无标签数据来调整步长。
2. 我们在平滑性、梯度稀疏性和算子兼容性的条件下，为基本离线方案建立了最佳迭代收敛保证。对于基本在线方案，我们在精确反向KL约束下证明了其可行性，并针对本地覆盖率下的分布内成对奖励误差，界定了性能差距。
3. 我们在来自 Mistral、Llama、Gemma-2、Qwen3 和 Gemma-3 系列的基准模型和指令微调模型上评估了 ComPO。实验评估了其兼容性

with direct alignment methods, its design choices, and the effects of online damping and replay. Pair-level likelihood diagnostics complement the benchmark evaluations.

Relationship to the conference version. A preliminary version of this work appeared at NeurIPS 2025 (Chen et al., 2025b). It introduced offline ComPO, preference comparison oracle, the convergence analysis, and the original offline experiments. The journal extension adds the online extension, its coverage-based analysis, and experiments on additional model families, including evaluations of online regularization and replay.

Related works. Direct preference alignment methods, including DPO (Rafailov et al., 2023), are simple and more stable offline alternatives to RLHF. Several DPO variants with alternative objectives have been proposed, including ranking-based variants beyond pairwise preference data (Dong et al., 2023; Yuan et al., 2023; Song et al., 2024a; Chen et al., 2024; Liu et al., 2025) and reference-model-free variants (Hong et al., 2024; Meng et al., 2024). It is well known that DPO suffers from the issues of verbosity (Park et al., 2024; Amini et al., 2024; Rafailov et al., 2024a) and likelihood displacement (Pal et al., 2024; Tajwar et al., 2024; Rafailov et al., 2024b; Pang et al., 2024; Liu et al., 2024b; Yuan et al., 2025), which can be interpreted from a unified perspective of data curation (Park et al., 2024; Razin et al., 2025). Our work continues along this perspective by arguing that these issues can be mitigated by using the information contained in noisy preference pairs for which the reference model assigns similar likelihoods to preferred and dispreferred responses.

Recent work has examined different roles of online data in preference fine-tuning. Online preference optimization can acquire additional labels for responses generated by the current policy, as in the online AI feedback approach of Guo et al. (2024). In contrast, Song et al. (2024b) introduce HyPO, which combines offline preference optimization with reverse-KL regularization estimated from unlabeled online samples. Our online extension follows this separation between preference supervision and regularization, but uses comparison-derived update directions. A complementary line of work studies active exploration (Xie et al., 2025) by augmenting online DPO with an explicit exploration bonus to guide the acquisition of preference feedback. Online ComPO does not acquire new preference labels or introduce such a bonus and its analysis concerns policy performance under local coverage.

Comparison-based optimization includes coordinate-search methods (Jamieson et al., 2012; Matsui et al., 2017) and directional estimators such as SCOBO (Cai et al., 2022a) and Sign-OPT (Cheng et al., 2020). Sign-OPT also provides a stationarity analysis under smoothness and additional assumptions on gradient noise, so nonconvexity alone is not the distinction from that work. ComPO specializes the comparison mechanism to preference alignment: its oracle evaluates preferred- and dispreferred-response likelihood changes, its basic analysis exploits approximately sparse gradients, and its practical implementation uses output-layer perturbations and thresholding. Comparison and ranking feedback have also been studied in bandit optimization (Yue and Joachims, 2009; Kumagai, 2017; Ding and Zhou, 2018), Bayesian optimization (Astudillo and Frazier, 2020; Lin et al., 2022b), and RLHF (Tang et al., 2024b; Zhang and Ying, 2025). Our focus is on extracting comparison signals from low-margin offline preference pairs and combining them with unlabeled online generations for step-size control.

与直接对齐方法、其设计选择以及在线阻尼和重放的影响。成对级别的似然诊断补充了基准评估。

与会议版本的关系。 这项工作的初步版本于 NeurIPS 2025 年发表 (陈等人., 2025b)。它引入了离线 ComPO、偏好比较预言机、收敛分析以及最初的离线实验。期刊扩展增加了在线扩展、基于覆盖的分析以及在更多模型家族上的实验, 包括在线正则化和重放的评估。

相关工作。直接偏好对齐方法, 包括 DPO (Rafailov 等人, 2023 年), 是比 RLHF 更简单、更稳定的离线替代方案。已经提出了几种具有替代目标的 DPO 变体, 包括基于排序的变体 (超越了成对偏好数据) (Dong 等人, 2023 年; Yuan 等人, 2023 年; Song 等人, 2024 年 a; Chen 等人, 2024 年; Liu 等人, 2025 年) 和无需参考模型的变体 (Hong 等人, 2024 年; Meng 等人, 2024 年)。众所周知, DPO 存在冗长 (Park 等人, 2024 年; Amini 等人, 2024 年; Rafailov 等人, 2024 年 a) 和似然位移 (Pal 等人, 2024 年; Tajwar 等人, 2024 年; Rafailov 等人, 2024 年 b; Pang 等人, 2024 年; Liu 等人, 2024 年 b; Yuan 等人, 2025 年) 的问题, 这些问题可以从数据管理的统一视角来解释 (Park 等人, 2024 年; Razin 等人, 2025 年)。我们的工作沿着这个视角继续进行, 认为这些问题可以通过使用参考模型对偏好和不受欢迎的响应分配相似似然值的噪声偏好对中的信息来缓解。

近期研究已探讨在线数据在偏好微调中的不同作用。在线偏好优化可以为当前策略生成的回复获取额外标签, 例如郭等人(2024) 的在线AI反馈方法。相比之下, 宋等人(2024b) 引入了 HyPO, 它结合了离线偏好优化与从未标记在线样本中估计的逆KL正则化。我们的在线扩展遵循偏好监督与正则化之间的这种分离, 但使用比较导出的更新方向。另一条互补的研究路线通过为在线 DPO 增加显式的探索奖励来研究主动探索 (谢等人, 2025), 以指导偏好反馈的获取。在线 ComPO 不会获取新的偏好标签或引入此类奖励, 其分析关注于策略在局部覆盖下的性能。

基于比较的优化包括坐标搜索方法 (Jamieson 等人, 2012; Matsui 等人, 2017) 和方向估计器, 如 SCOBO (Cai 等人, 2022a) 和 Sign-OPT (Cheng 等人, 2020)。Sign-OPT 还在平滑性和梯度噪声的附加假设下提供了驻点分析, 因此非凸性本身并不是与该工作的区别。ComPO 将比较机制专门用于偏好对齐: 其预言机评估首选和不受欢迎响应概率的变化, 其基本分析利用近似稀疏梯度, 其实际实现使用输出层扰动和阈值。比较和排序反馈也在 bandit 优化 (Yue 和 Joachims, 2009; Kumagai, 2017; Ding 和 Zhou, 2018)、贝叶斯优化 (Astudillo 和 Frazier, 2020; Lin 等人, 2022b) 和 RLHF (Tang 等人, 2024b; Zhang 和 Ying, 2025) 中得到了研究。我们的重点是从低边界的离线偏好对提取比较信号, 并将其与未标记的在线生成相结合以进行步长控制。

2 Preliminaries

We provide an overview of the setup for direct preference alignment, and recall the definition of comparison oracles and the subroutine for estimating gradients using comparison oracles that are important for designing the basic scheme of our method. We further introduce the reverse-KL and coverage notation used in online ComPO.

2.1 Direct preference alignment

Modern LLMs are designed based on the Transformer architecture (Vaswani et al., 2017) and follow user prompts $\mathbf{x} \in \mathcal{V}^*$ to generate responses $\mathbf{y} \in \mathcal{V}^*$, where $\mathcal{V}$ is a vocabulary of tokens. We view an LLM as a policy $\pi_\theta(\mathbf{y}|\mathbf{x})$ which assigns probabilities to responses $\mathbf{y}$ given prompts $\mathbf{x}$. To assign probabilities to each token of $\mathbf{y}$, the policy π_θ operates in an auto-regressive manner as follows,

$$\pi_\theta(\mathbf{y}|\mathbf{x}) = \prod_{k=1}^{|\mathbf{y}|} \pi_\theta(\mathbf{y}_k|\mathbf{x}, \mathbf{y}_{<k}),$$

where θ denotes the model parameters (e.g., the parameters of the Transformer architecture) and $\mathbf{y}_{<k}$ denotes the first $k-1$ tokens of $\mathbf{y}$. However, the generations might not be helpful, safe, or reliable, which motivates further alignment of LLMs with human preferences.

We consider the direct preference learning pipeline based on pairwise preference data. Specifically, we assume access to a preference dataset D containing samples $(\mathbf{x}, \mathbf{y}^+, \mathbf{y}^-)$, where $\mathbf{x}$ is a prompt and $(\mathbf{y}^+, \mathbf{y}^-)$ is a pair of preferred and dispreferred responses to $\mathbf{x}$. This pipeline usually includes an initial supervised fine-tuning (SFT) phase, where the model is fine-tuned using the cross-entropy loss and high-quality data for specific downstream tasks. The SFT data can be either independent of D (Touvron et al., 2023), or may consist of prompts and preferred responses from D (Rafailov et al., 2023).

Direct alignment methods, such as DPO (Rafailov et al., 2023), optimize the policy π_θ over the preference dataset D without learning a reward model as in RLHF (Ziegler et al., 2019; Stiennon et al., 2020). This is done by minimizing a contrastive loss as follows,

$$\mathcal{L}_{\text{DPO}}(\theta) = -\mathbb{E}_{(\mathbf{x}, \mathbf{y}^+, \mathbf{y}^-) \sim D} \left[\log \sigma \left(\beta \log \frac{\pi_\theta(\mathbf{y}^+|\mathbf{x})}{\pi_{\text{ref}}(\mathbf{y}^+|\mathbf{x})} - \beta \log \frac{\pi_\theta(\mathbf{y}^-|\mathbf{x})}{\pi_{\text{ref}}(\mathbf{y}^-|\mathbf{x})} \right) \right], \quad (1)$$

where π_{ref} is the model after SFT, β is a regularization parameter, and $\sigma: \mathbb{R} \rightarrow [0, 1]$ is the sigmoid function. The function $\mathcal{L}_{\text{DPO}}$ relies on the log-likelihood margin between $\mathbf{y}^+$ and $\mathbf{y}^-$. Thus, DPO improves the relative likelihood margin between the two responses, rather than directly maximizing the likelihood of $\mathbf{y}^+$ and minimizing the likelihood of $\mathbf{y}^-$. During training, the likelihood of $\mathbf{y}^+$ might decrease, and probability mass can be shifted from $\mathbf{y}^+$ to responses with an opposite meaning (Pal et al., 2024; Razin et al., 2025). A possible reason is that the above objective function is not well suited for extracting information from noisy preference pairs whose preferred and dispreferred responses have small likelihood margins or are similar under model-based measures.

Empirically, Razin et al. (2025) show that filtering out similar preference pairs can make DPO more effective. However, noisy preference pairs might still contain useful information that can improve the performance of LLMs. Extracting such information is challenging

2 预备知识

我们概述了直接偏好对齐的设置，并回顾了比较预言机的定义以及使用比较预言机估计梯度的子程序，这对于设计我们方法的基本方案非常重要。我们进一步介绍了在线 ComPO 中使用的反向 KL 和覆盖符号。

2.1 直接偏好对齐

现代大语言模型基于 Transformer 架构 (Vaswani 等人, 2017) 设计而成，并遵循用户提示 $\mathbf{x} \in \mathcal{V}^*$ 来生成响应 $\mathbf{y} \in \mathcal{V}^*$ ，其中 $\mathcal{V}$ 是一个标记词汇表。我们将大语言模型视为一个策略 $\pi_\theta(\mathbf{y}|\mathbf{x})$ ，该策略根据提示 $\mathbf{x}$ 为响应 $\mathbf{y}$ 分配概率。为了为 $\mathbf{y}$ 中的每个标记分配概率，该策略以自回归的方式运行，具体如下：

$$\pi_\theta(\mathbf{y}|\mathbf{x}) = \prod_{k=1}^{|\mathbf{y}|} \pi_\theta(\mathbf{y}_k|\mathbf{x}, \mathbf{y}_{<k}),$$

其中 θ 表示模型参数（例如，Transformer 架构的参数）， $\mathbf{y}_{<k}$ 表示 $\mathbf{y}$ 的前 $k-1$ 个标记。然而，生成的内容可能没有帮助、不安全或不可靠，这促使我们进一步对齐大语言模型与人类偏好。

我们考虑基于成对偏好数据的直接偏好学习流程。具体而言，我们假设可以访问一个偏好数据集 D ，其中包含样本 $(\mathbf{x}, \mathbf{y}^+, \mathbf{y}^-)$ ，其中 $\mathbf{x}$ 是一个提示，而 $(\mathbf{y}^+, \mathbf{y}^-)$ 是针对 $\mathbf{x}$ 的一对偏好和不受欢迎的响应。该流程通常包括一个初始的监督微调（SFT）阶段，在该阶段中，模型使用交叉熵损失和高质量数据进行特定下游任务的微调。SFT 数据可以独立于 D (图尔沃等人, 2023)，或者可能由 D (Rafailov 等人, 2023) 的提示和偏好响应组成。

直接对齐方法，如 DPO (Rafailov 等人, 2023)，在偏好数据集 π_θ 上优化策略 D ，而无需像 RLHF (Ziegler 等人, 2019; Stiennon 等人, 2020) 那样学习奖励模型。这是通过最小化以下对比损失来实现的

$$\mathcal{L}_{\text{DPO}}(\theta) = -\mathbb{E}_{(\mathbf{x}, \mathbf{y}^+, \mathbf{y}^-) \sim D} \left[\log \sigma \left(\beta \log \frac{\pi_\theta(\mathbf{y}^+|\mathbf{x})}{\pi_{\text{ref}}(\mathbf{y}^+|\mathbf{x})} - \beta \log \frac{\pi_\theta(\mathbf{y}^-|\mathbf{x})}{\pi_{\text{ref}}(\mathbf{y}^-|\mathbf{x})} \right) \right], \quad (1)$$

其中 π_{ref} 是 SFT 后的模型， β 是正则化参数，而 $\sigma: \mathbb{R} \rightarrow [0, 1]$ 是 sigmoid 函数。该函数 $\mathcal{L}_{\text{DPO}}$ 依赖于 $\mathbf{y}^+$ 和 $\mathbf{y}^-$ 之间的对数似然边际。因此，DPO 改进了两个响应之间的相对似然边际，而不是直接最大化 $\mathbf{y}^+$ 的似然并最小化 $\mathbf{y}^-$ 的似然。在训练过程中， $\mathbf{y}^+$ 的似然可能会降低，概率质量可能会从 $\mathbf{y}^+$ 转移到具有相反含义的响应 (Pal 等人, 2024; Razin 等人, 2025)。一个可能的原因是，上述目标函数不适合从噪声偏好对中提取信息，这些偏好对中首选和不受欢迎的响应的对数似然边际较小，或在基于模型的度量下相似。

从经验上讲，Razin 等人, (2025) 展示了过滤掉相似的偏好对可以使 DPO 更有效。然而，噪声偏好对可能仍然包含有用的信息，从而提高大语言模型的性能。提取这类信息具有挑战性

using a fixed margin-based loss, since maximizing the likelihood of $\mathbf{y}^+$ and minimizing the likelihood of $\mathbf{y}^-$ locally does not by itself define a global alignment objective. The local information we use is comparative: a better policy should assign higher likelihood to $\mathbf{y}^+$ and lower likelihood to $\mathbf{y}^-$. This motivates us to design a new alignment method by directly leveraging the comparison signal in pairwise preference data $(\mathbf{x}, \mathbf{y}^+, \mathbf{y}^-)$ from D .

2.2 Comparison oracles and zeroth-order methods

To contextualize our proposed method for aligning LLMs with human preferences, we review the definition of comparison oracles and explain how comparison oracles can be used to develop zeroth-order methods.

Given a function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ for which neither the function value nor the gradient is accessible, we define a pairwise comparison oracle $\mathcal{C}_f$ in its simplest form as follows,

Definition 2.1 We call $\mathcal{C}_f(\theta, \theta') : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \{+1, -1\}$ a comparison oracle for function f if

$$\mathcal{C}_f(\theta, \theta') = \begin{cases} -1, & \text{if } f(\theta') < f(\theta), \\ +1, & \text{otherwise.} \end{cases}$$

In other words, when queried with θ and θ' , the oracle $\mathcal{C}_f(\cdot, \cdot)$ returns -1 if $f(\theta') < f(\theta)$ and $+1$ otherwise, with ties assigned to $+1$.

The key idea behind the subroutine in Cai et al. (2022a) for estimating gradients using comparison oracles is inspired by 1-bit compressed sensing (Boufounos and Baraniuk, 2008). The goal is to recover a signal $\mathbf{g} \in \mathbb{R}^d$ from quantized measurements $y_i = \text{sign}(\mathbf{z}_i^\top \mathbf{g})$, where $\mathbf{z}_i$ is a random perturbation vector drawn from a rotationally invariant distribution. The theoretical guarantee on the required number of perturbations to obtain an approximate signal was established in Plan and Vershynin (2012) and extended in Cai et al. (2022a). Notably, for a small perturbation radius $r > 0$, we have

$$\mathcal{C}_f(\theta, \theta + r\mathbf{z}_i) = \text{sign}(f(\theta + r\mathbf{z}_i) - f(\theta)) \approx \text{sign}(\mathbf{z}_i^\top \nabla f(\theta)).$$

Here, $\text{sign}(0) = +1$. Thus, the comparison label $y_i = \mathcal{C}_f(\theta, \theta + r\mathbf{z}_i)$ serves as an approximate one-bit measurement of $\nabla f(\theta)$.

Another issue is that zeroth-order comparison-based methods can suffer from dimension-dependent iteration complexity bounds (Jamieson et al., 2012). This is expected because comparison oracles are even weaker than function-value oracles. This dimension dependence can be mitigated by exploiting sparse gradient structure (Wang et al., 2018; Golovin et al., 2020; Choromanski et al., 2019; Cai et al., 2022a,b). Indeed, we say that the function f has sparse gradients if $\|\nabla f(\theta)\|_1 \leq \sqrt{s}\|\nabla f(\theta)\|$ for all $\theta \in \mathbb{R}^d$ and some $s \ll d$.

The above discussion gives the subroutine for estimating sparse gradients using comparison oracles. We generate m i.i.d. perturbation vectors, denoted by $\{\mathbf{z}_i\}_{1 \leq i \leq m}$, compute $y_i = \mathcal{C}_f(\theta, \theta + r\mathbf{z}_i)$ for all i , and solve the following optimization problem:

$$\hat{\mathbf{g}} = \underset{\|\mathbf{g}\|_1 \leq \sqrt{s}, \|\mathbf{g}\| \leq 1}{\text{argmax}} \sum_{i=1}^m y_i \mathbf{z}_i^\top \mathbf{g}, \quad (2)$$

where the constraints $\|\mathbf{g}\|_1 \leq \sqrt{s}$ and $\|\mathbf{g}\| \leq 1$ restrict the search to an approximately sparse and normalized set.

使用固定的基于边界的损失函数, 因为局部最大化 $\mathbf{y}^+$ 的似然性和局部最小化 $\mathbf{y}^-$ 的似然性本身并不能定义全局对齐目标。我们使用的局部信息是对比性的: 更好的策略应该为 $\mathbf{y}^+$ 赋予更高的似然性, 并为 $\mathbf{y}^-$ 赋予更低的似然性。这促使我们通过直接利用成对偏好数据 $(\mathbf{x}, \mathbf{y}^+, \mathbf{y}^-)$ 中的对比信号, 从 D 设计一种新的对齐方法。

2.2 比较预言机与零阶方法

为了将我们提出的对齐大语言模型与人类偏好的方法进行背景化, 我们回顾比较预言机的定义, 并解释比较预言机如何用于开发零阶方法。

给定一个函数 $f : \mathbb{R}^d \rightarrow \mathbb{R}$, 其函数值和梯度均不可访问, 我们将其最简单的形式下的成对比较预言机 $\mathcal{C}_f$ 定义如下,

定义 2.1 我们称 $\mathcal{C}_f(\theta, \theta') : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \{+1, -1\}$ 为函数 f 的比较预言机, 如果

$$\mathcal{C}_f(\theta, \theta') = \begin{cases} -1, & \text{if } f(\theta') < f(\theta), \\ +1, & \text{otherwise.} \end{cases}$$

换句话说, 当用 θ 和 θ' 查询预言机 $\mathcal{C}_f(\cdot, \cdot)$ 时, 如果 $f(\theta') < f(\theta)$, 则返回 -1 , 否则返回 $+1$, 并将平局分配给 $+1$ 。

Cai 等人 (2022a) 中用于通过比较预言机估计梯度的子程序的关键思想, 受 1 位压缩感知 (Boufounos 和 Baraniuk, 2008) 的启发。目标是从量化测量 $\mathbf{g} \in \mathbb{R}^d$ 中恢复信号 $y_i = \text{sign}(\mathbf{z}_i^\top \mathbf{g})$, 其中 $\mathbf{z}_i$ 是一个从旋转不变分布中抽取的随机扰动向量。获取近似信号所需的扰动次数的理论保证由 Plan 和 Vershynin (2012) 建立, 并在 Cai 等人 (2022a) 中扩展。值得注意的是, 对于小的扰动半径 $r > 0$, 我们有

$$\mathcal{C}_f(\theta, \theta + r\mathbf{z}_i) = \text{sign}(f(\theta + r\mathbf{z}_i) - f(\theta)) \approx \text{sign}(\mathbf{z}_i^\top \nabla f(\theta)).$$

这里, $\text{sign}(0) = +1$ 。因此, 比较标签 $y_i = \mathcal{C}_f(\theta, \theta + r\mathbf{z}_i)$ 作为 $\nabla f(\theta)$ 的近似一位测量值。

另一个问题是, 零阶基于比较的方法可能因维度相关的迭代复杂度界限而受影响 (Jamieson 等人, 2012)。这很正常, 因为比较预言机比函数值预言机更弱。这种维度依赖性可以通过利用稀疏梯度结构 (Wang 等人, 2018; Golovin 等人, 2020; Choromanski 等人, 2019; Cai 等人, 2022a, b) 来缓解。实际上, 我们说函数 f 具有稀疏梯度, 如果 $\|\nabla f(\theta)\|_1 \leq \sqrt{s}\|\nabla f(\theta)\|$ 对于所有 $\theta \in \mathbb{R}^d$ 和某个 $s \ll d$ 。

上述讨论给出了使用比较预言机估计稀疏梯度的子程序。我们生成 m 独立同分布扰动向量, 记作 $\{\mathbf{z}_i\}_{1 \leq i \leq m}$, 对所有 i 计算 $y_i = \mathcal{C}_f(\theta, \theta + r\mathbf{z}_i)$, 并求解以下优化问题:

$$\hat{\mathbf{g}} = \underset{\|\mathbf{g}\|_1 \leq \sqrt{s}, \|\mathbf{g}\| \leq 1}{\text{argmax}} \sum_{i=1}^m y_i \mathbf{z}_i^\top \mathbf{g}, \quad (2)$$

在约束 $\|\mathbf{g}\|_1 \leq \sqrt{s}$ 和 $\|\mathbf{g}\| \leq 1$ 的限制下, 搜索被限定在一个近似稀疏和规范化的集合中。

In ComPO, the latent function f is viewed as an implicit alignment objective. Instead of assuming access to its function value or gradient, we use offline preference pairs to construct a comparison oracle: a nearby policy is considered better if it assigns a higher likelihood to the preferred response and a lower likelihood to the dispreferred response.

2.3 Reverse KL and local coverage

The online extension of ComPO uses unlabeled policy generations for regularization, while the comparison oracle continues to use the fixed offline preference pairs. Let P_{on} denote the prompt distribution used for online generation, and let π_{ref} be a fixed reference policy. For the online analysis, we consider policies with a common response support and positive probabilities on that support, and assume that the relevant expectations are finite. For any such policy π , we define its sequence-level reverse KL relative to the reference by

$$D_{\text{RKL}}(\pi \| \pi_{\text{ref}}) = \mathbb{E}_{\mathbf{x} \sim P_{\text{on}}} [D_{\text{KL}}(\pi(\cdot | \mathbf{x}) \| \pi_{\text{ref}}(\cdot | \mathbf{x}))] = \mathbb{E}_{\mathbf{x} \sim P_{\text{on}}, \mathbf{y} \sim \pi(\cdot | \mathbf{x})} \left[\log \frac{\pi(\mathbf{y} | \mathbf{x})}{\pi_{\text{ref}}(\mathbf{y} | \mathbf{x})} \right], \quad (3)$$

The reverse KL can be estimated using unlabeled generations from the current policy being evaluated. For $\tau > 0$, we define the reverse-KL neighborhood of the reference policy by

$$\Pi_{\tau} = \{\pi : D_{\text{RKL}}(\pi \| \pi_{\text{ref}}) \leq \tau\} \quad (4)$$

We let $r^*(\mathbf{x}, \mathbf{y})$ denote the ground-truth reward. For $\beta > 0$, we define the KL-regularized population objective by

$$J_{\beta}(\pi) = \mathbb{E}_{\mathbf{x} \sim P_{\text{on}}, \mathbf{y} \sim \pi(\cdot | \mathbf{x})} [r^*(\mathbf{x}, \mathbf{y})] - \beta D_{\text{RKL}}(\pi \| \pi_{\text{ref}}). \quad (5)$$

For a policy π , we define its implicit reward relative to π_{ref} by

$$\hat{r}_{\pi}(\mathbf{x}, \mathbf{y}) = \beta \log \frac{\pi(\mathbf{y} | \mathbf{x})}{\pi_{\text{ref}}(\mathbf{y} | \mathbf{x})}. \quad (6)$$

Pairwise reward differences are invariant to prompt-dependent additive constants. We thus measure the accuracy through the following in-distribution pairwise error, where $\mathbf{y}_1$ and $\mathbf{y}_2$ are drawn independently from $\pi_{\text{ref}}(\cdot | \mathbf{x})$ conditional on $\mathbf{x}$. Formally, we have

$$\text{err}(\pi) = \mathbb{E}_{\mathbf{x} \sim P_{\text{on}}, \mathbf{y}_1, \mathbf{y}_2 \sim \pi_{\text{ref}}(\cdot | \mathbf{x})} \left[(r^*(\mathbf{x}, \mathbf{y}_1) - r^*(\mathbf{x}, \mathbf{y}_2) - \hat{r}_{\pi}(\mathbf{x}, \mathbf{y}_1) + \hat{r}_{\pi}(\mathbf{x}, \mathbf{y}_2))^2 \right]. \quad (7)$$

Following Song et al. (2024b), we present policy performance in terms of this in-distribution pairwise error under the local coverage condition in the following definition.

Definition 2.2 *The reference policy π_{ref} satisfies local reverse-KL coverage at radius $\kappa > 0$ with constant $C_{\kappa} > 0$ if every policy μ satisfying $D_{\text{RKL}}(\mu \| \pi_{\text{ref}}) \leq \kappa$ also satisfies*

$$\sup_{\mathbf{x} \in \text{supp}(P_{\text{on}})} \sup_{\mathbf{y} \in \mathcal{V}^*} \frac{\mu(\mathbf{y} | \mathbf{x})}{\pi_{\text{ref}}(\mathbf{y} | \mathbf{x})} \leq C_{\kappa},$$

where we use the convention $\frac{0}{0} = 0$.

Local coverage in Definition 2.2 concerns policies within a reverse-KL neighborhood of π_{ref} , which guarantees that restricting the learned policy to Π_{τ} can allow a performance guarantee to depend on coverage within that neighborhood. The reverse-KL constraint determines the class on which coverage is required but it does not guarantee the bounded density ratio.

在 ComPO 中，潜在函数 f 被视为一种隐式对齐目标。我们不假设能够访问其函数值或梯度，而是使用离线偏好对来构建一个比较预言机：如果附近策略对偏好响应分配更高的可能性，并对非偏好响应分配更低的可能性，则认为该策略更好。

2.3 反向 KL 和局部覆盖率

ComPO 的在线扩展使用未标记的策略生成进行正则化，而比较预言机继续使用固定的离线偏好对。令 P_{on} 表示用于在线生成的提示分布，令 π_{ref} 为一个固定参考策略。对于在线分析，我们考虑具有共同响应支持和该支持上正概率的策略，并假设相关的期望是有限的。对于任何此类策略 π ，我们将其相对于参考的序列级反向 KL 定义为

$$D_{\text{RKL}}(\pi \| \pi_{\text{ref}}) = \mathbb{E}_{\mathbf{x} \sim P_{\text{on}}} [D_{\text{KL}}(\pi(\cdot | \mathbf{x}) \| \pi_{\text{ref}}(\cdot | \mathbf{x}))] = \mathbb{E}_{\mathbf{x} \sim P_{\text{on}}, \mathbf{y} \sim \pi(\cdot | \mathbf{x})} \left[\log \frac{\pi(\mathbf{y} | \mathbf{x})}{\pi_{\text{ref}}(\mathbf{y} | \mathbf{x})} \right], \quad (3)$$

反向 KL 可以使用当前正在评估的策略的未标记生成来估计。对于 $\tau > 0$ ，我们通过以下方式定义参考策略的反向-KL 邻域

$$\Pi_{\tau} = \{\pi : D_{\text{RKL}}(\pi \| \pi_{\text{ref}}) \leq \tau\} \quad (4)$$

我们让 $r^*(\mathbf{x}, \mathbf{y})$ 表示真实奖励。对于 $\beta > 0$ ，我们通过以下方式定义 KL 正则化的群体目标

$$J_{\beta}(\pi) = \mathbb{E}_{\mathbf{x} \sim P_{\text{on}}, \mathbf{y} \sim \pi(\cdot | \mathbf{x})} [r^*(\mathbf{x}, \mathbf{y})] - \beta D_{\text{RKL}}(\pi \| \pi_{\text{ref}}). \quad (5)$$

对于策略 π ，我们通过以下方式定义其相对于 π_{ref} 的隐式奖励

$$\hat{r}_{\pi}(\mathbf{x}, \mathbf{y}) = \beta \log \frac{\pi(\mathbf{y} | \mathbf{x})}{\pi_{\text{ref}}(\mathbf{y} | \mathbf{x})}. \quad (6)$$

成对奖励差异对提示相关的加性常数是不变的。因此，我们通过以下分布内成对误差来衡量准确性，其中 $\mathbf{y}_1$ 和 $\mathbf{y}_2$ 独立地从 $\pi_{\text{ref}}(\cdot | \mathbf{x})$ 条件下 $\mathbf{x}$ 中抽取。形式上，我们有

$$\text{err}(\pi) = \mathbb{E}_{\mathbf{x} \sim P_{\text{on}}, \mathbf{y}_1, \mathbf{y}_2 \sim \pi_{\text{ref}}(\cdot | \mathbf{x})} \left[(r^*(\mathbf{x}, \mathbf{y}_1) - r^*(\mathbf{x}, \mathbf{y}_2) - \hat{r}_{\pi}(\mathbf{x}, \mathbf{y}_1) + \hat{r}_{\pi}(\mathbf{x}, \mathbf{y}_2))^2 \right]. \quad (7)$$

遵循 Song 等人 (2024b)，我们在以下定义中提出了策略性能，即在局部覆盖率条件下，该分布内成对误差。

定义 2.2 参考策略 π_{ref} 在半径 $\kappa > 0$ 处满足局部反向KL覆盖率，常数为 $C_{\kappa} > 0$ ，如果每个满足 $D_{\text{RKL}}(\mu \| \pi_{\text{ref}}) \leq \kappa$ 的策略 μ 也满足。

$$\sup_{\mathbf{x} \in \text{supp}(P_{\text{on}})} \sup_{\mathbf{y} \in \mathcal{V}^*} \frac{\mu(\mathbf{y} | \mathbf{x})}{\pi_{\text{ref}}(\mathbf{y} | \mathbf{x})} \leq C_{\kappa},$$

其中我们使用了约定 $\frac{0}{0} = 0$ 。

定义中的本地覆盖率2.2 关注反向KL邻域内的策略 π_{ref} ，这保证了将学习到的策略限制在 Π_{τ} 内，可以使性能保证依赖于该邻域内的覆盖率。反向KL约束确定了需要覆盖的类别，但它不保证有界密度比。

3 Main Results

We study how to learn from noisy preference pairs that induce similar likelihoods for preferred and dispreferred responses. We first present the basic offline scheme, which replaces a first-order update driven by a predefined preference loss with a zeroth-order update driven by comparison oracles, and describe the practical offline scheme used for LLM fine-tuning. We then introduce online ComPO, which preserves the offline comparison direction and uses unlabeled current-policy generations for reverse-KL regularization.

3.1 Offline preference alignment

The key idea behind ComPO is to use noisy preference pairs only to compare nearby policies. For a nonempty $S \subseteq D$, define

$$\begin{aligned}\Delta_S^+(\theta, \theta') &= \frac{1}{|S|} \sum_{(\mathbf{x}, \mathbf{y}^+, \mathbf{y}^-) \in S} (\log \pi_{\theta'}(\mathbf{y}^+ | \mathbf{x}) - \log \pi_{\theta}(\mathbf{y}^+ | \mathbf{x})), \\ \Delta_S^-(\theta, \theta') &= \frac{1}{|S|} \sum_{(\mathbf{x}, \mathbf{y}^+, \mathbf{y}^-) \in S} (\log \pi_{\theta'}(\mathbf{y}^- | \mathbf{x}) - \log \pi_{\theta}(\mathbf{y}^- | \mathbf{x})).\end{aligned}\quad (8)$$

We then provide the formulation of preference comparison oracle for LLM alignment below:

Definition 3.1 (Preference comparison oracle) For a set $S \subseteq D$, the preference comparison oracle $\mathcal{C}_{\pi}^S(\theta, \theta') : \mathbb{R}^d \times \mathbb{R}^d \mapsto \{+1, -1\}$ is defined by

$$\mathcal{C}_{\pi}^S(\theta, \theta') = \begin{cases} -1, & \text{if } \Delta_S^+(\theta, \theta') > 0 \text{ and } \Delta_S^-(\theta, \theta') < 0, \\ +1, & \text{otherwise.} \end{cases}$$

Thus, $\mathcal{C}_{\pi}^S(\theta, \theta') = -1$ means that θ' is preferred to θ according to the likelihood comparison induced by S . When S contains one pair, this reduces to the pairwise oracle. When S is a mini-batch, the oracle uses average preferred and dispreferred likelihood changes. Given a set of perturbations $\{\mathbf{z}_i\}_{i=1}^m$, ComPO queries $y_i = \mathcal{C}_{\pi}^S(\theta_t, \theta_t + r\mathbf{z}_i)$ for $i = 1, \dots, m$ and applies the sparse 1-bit estimator from Eq. (2) as follows,

$$\hat{\mathbf{g}} = \underset{\|\mathbf{g}\|_1 \leq \sqrt{s}, \|\mathbf{g}\| \leq 1}{\operatorname{argmax}} \sum_{i=1}^m y_i \mathbf{z}_i^{\top} \mathbf{g}. \quad (9)$$

This is the only specialization of the comparison-oracle subroutine needed for offline ComPO.

The following theorem establishes a best-iterate convergence guarantee for the basic offline scheme under smoothness, gradient sparsity, and oracle compatibility.

Theorem 3.2 Fix a nonempty comparison set $S \subseteq D$ and $1 \leq s \leq d$. Suppose that there exists an ℓ -smooth function $f : \mathbb{R}^d \rightarrow \mathbb{R}$, with $\ell > 0$, that is bounded below and satisfies

1. For all (θ, θ') , we have $\mathcal{C}_{\pi}^S(\theta, \theta') = -1$ if $f(\theta') < f(\theta)$ and $\mathcal{C}_{\pi}^S(\theta, \theta') = 1$ otherwise.
2. The gradients of f are approximately sparse: $\|\nabla f(\theta)\|_1 \leq \sqrt{s} \|\nabla f(\theta)\|$ for all $\theta \in \mathbb{R}^d$.

Let $\Delta > 0$ satisfy $f(\theta_1) - \inf_{\theta \in \mathbb{R}^d} f(\theta) \leq \Delta$. For any $\epsilon, \Lambda \in (0, 1)$, we choose

$$T = \lceil \frac{10\ell\Delta}{\epsilon^2} \rceil, \quad \eta = \sqrt{\frac{2\Delta}{\ell T}}, \quad r = \frac{\epsilon}{40\ell\sqrt{d}}, \quad m = \lceil c_m \left(s \log \left(\frac{2d}{s} \right) + \log \left(\frac{2T}{\Lambda} \right) \right) \rceil,$$

3 主要结果

我们研究如何从导致首选和不受欢迎响应具有相似似然性的噪声偏好对中学习。我们首先提出了基本离线方案，该方案用由比较预言机驱动的零阶更新替换由预定义偏好损失驱动的阶更新，并描述了用于大语言模型微调的实用离线方案。然后我们引入了在线ComPO，它保留了离线比较方向，并使用当前策略生成用于反向KL正则化的无标签数据。

3.1 离线偏好对齐

ComPO 的核心思想是仅使用噪声偏好对来比较邻近策略。对于非空 $S \subseteq D$ ，定义

$$\begin{aligned}\Delta_S^+(\theta, \theta') &= \frac{1}{|S|} \sum_{(\mathbf{x}, \mathbf{y}^+, \mathbf{y}^-) \in S} (\log \pi_{\theta'}(\mathbf{y}^+ | \mathbf{x}) - \log \pi_{\theta}(\mathbf{y}^+ | \mathbf{x})), \\ \Delta_S^-(\theta, \theta') &= \frac{1}{|S|} \sum_{(\mathbf{x}, \mathbf{y}^+, \mathbf{y}^-) \in S} (\log \pi_{\theta'}(\mathbf{y}^- | \mathbf{x}) - \log \pi_{\theta}(\mathbf{y}^- | \mathbf{x})).\end{aligned}\quad (8)$$

我们随后提供用于 LLM 对齐的偏好比较预言机的公式：

定义 3.1 (偏好比较预言机) 对于集合 $S \subseteq D$ ，偏好比较预言机 $\mathcal{C}_{\pi}^S(\theta, \theta') : \mathbb{R}^d \times \mathbb{R}^d \mapsto \{+1, -1\}$ 被定义为

$$\mathcal{C}_{\pi}^S(\theta, \theta') = \begin{cases} -1, & \text{if } \Delta_S^+(\theta, \theta') > 0 \text{ and } \Delta_S^-(\theta, \theta') < 0, \\ +1, & \text{otherwise.} \end{cases}$$

因此， $\mathcal{C}_{\pi}^S(\theta, \theta') = -1$ 表示 θ' 在 θ 根据由 S 引起的可能性比较 S 中更受偏好。当 S 包含一对时，这简化为成对预言机。当 $\{\mathbf{z}_i\}_{i=1}^m$ 是一个扰动集时，ComPO 查询 $y_i = \mathcal{C}_{\pi}^S(\theta_t, \theta_t + r\mathbf{z}_i)$ 以获取 $i = 1, \dots, m$ ，并按照等式 (2) 中的稀疏 1 位估计器进行应用，

$$\hat{\mathbf{g}} = \underset{\|\mathbf{g}\|_1 \leq \sqrt{s}, \|\mathbf{g}\| \leq 1}{\operatorname{argmax}} \sum_{i=1}^m y_i \mathbf{z}_i^{\top} \mathbf{g}. \quad (9)$$

这是离线 ComPO 所需的比较预言机子例程的唯一特殊化。

以下定理在平滑性、梯度稀疏性和预言机兼容性的条件下，为基本离线方案建立了最佳迭代收敛保证。

定理 3.2 固定一个非空的比较集 $S \subseteq D$ 和 $1 \leq s \leq d$ 。假设存在一个 ℓ -平滑函数 $f : \mathbb{R}^d \rightarrow \mathbb{R}$ ，其中 $\ell > 0$ ，该函数有下界并满足

1. For all (θ, θ') , we have $\mathcal{C}_{\pi}^S(\theta, \theta') = -1$ if $f(\theta') < f(\theta)$ and $\mathcal{C}_{\pi}^S(\theta, \theta') = 1$ otherwise.
2. f 的梯度大致是稀疏的： $\|\nabla f(\theta)\|_1 \leq \sqrt{s} \|\nabla f(\theta)\|$ 对所有 $\theta \in \mathbb{R}^d$ 而言。

设 $\Delta > 0$ 满足 $f(\theta_1) - \inf_{\theta \in \mathbb{R}^d} f(\theta) \leq \Delta$ 。对于任意 $\epsilon, \Lambda \in (0, 1)$ ，我们选择

$$T = \lceil \frac{10\ell\Delta}{\epsilon^2} \rceil, \quad \eta = \sqrt{\frac{2\Delta}{\ell T}}, \quad r = \frac{\epsilon}{40\ell\sqrt{d}}, \quad m = \lceil c_m \left(s \log \left(\frac{2d}{s} \right) + \log \left(\frac{2T}{\Lambda} \right) \right) \rceil,$$

Algorithm 1 Offline ComPO: Basic Scheme

1: **Input:** initial parameter $\theta_1 \in \mathbb{R}^d$, comparison set $S \subseteq D$, step size $\eta > 0$, sparsity ratio $s \ll d$, sampling radius $r > 0$, number of perturbations $m \geq 1$, and iteration number $T \geq 1$.
2: **for** $t = 1, 2, \dots, T$ **do**
3: Draw m i.i.d. samples uniformly from the unit sphere in $\mathbb{R}^d$, denoted by $\{\mathbf{z}_i\}_{i=1}^m$.
4: Compute $y_i = C_\pi^S(\theta_t, \theta_t + r\mathbf{z}_i)$ for $i = 1, \dots, m$.
5: Compute $\hat{\mathbf{g}}_t$ using Eq. (9).
6: Update $\theta_{t+1} = \theta_t - \eta \hat{\mathbf{g}}_t$.
7: **Output:** θ_{T+1} .

Algorithm 2 Offline ComPO: Practical Scheme

1: **Input:** initial parameter $\theta_1 = [\bar{\theta}; \theta_1^o]$, batches $\{S_t\}_{t=1}^T$, step size γ , sampling radius r , number of perturbations $m \geq 1$, clipping thresholds λ_g, λ , and iteration number $T \geq 1$.
2: **for** $t = 1, 2, \dots, T$ **do**
3: Draw m i.i.d. samples $\{\mathbf{z}_i\}_{i=1}^m$ uniformly from the unit sphere in $\mathbb{R}^{d_o}$.
4: Query $y_i = C_\pi^{S_t}([\bar{\theta}; \theta_t^o], [\bar{\theta}; \theta_t^o + rz_i])$ for all $i = 1, \dots, m$.
5: Set $\mathbf{u}_t = \sum_{i=1}^m y_i \mathbf{z}_i$. If $\mathbf{u}_t \neq 0$, set $\hat{\mathbf{g}}_t^o = \mathbf{u}_t / \|\mathbf{u}_t\|$. Otherwise, set $\hat{\mathbf{g}}_t^o = 0$.
6: Clip $\hat{\mathbf{g}}_t^o$ by zeroing out entries whose magnitude is less than λ_g .
7: Set $p_t = \frac{|\{i: y_i = -1\}|}{m}$.
8: **if** $p_t > \lambda$ **then**
9: $\theta_{t+1}^o = \theta_t^o - \gamma p_t \hat{\mathbf{g}}_t^o$.
10: **else**
11: $\theta_{t+1}^o = \theta_t^o$.
12: **Output:** $\theta_{T+1} = [\bar{\theta}; \theta_{T+1}^o]$.

where c_m is a sufficiently large constant. Suppose that the perturbations at each iteration are drawn independently of the past and Eq. (9) is solved exactly. Then, the iterates generated by Algorithm 1 satisfy

$$\mathbb{P} \left(\min_{1 \leq t \leq T} \|\nabla f(\theta_t)\| < \epsilon \right) \geq 1 - \Lambda.$$

Consequently, the total number of preference-comparison oracle calls is bounded by

$$O \left(\left(1 + \frac{\ell \Delta}{\epsilon^2} \right) \left(s \log \left(\frac{2d}{s} \right) + \log \left(\frac{2 + \ell \Delta \epsilon^{-2}}{\Lambda} \right) \right) \right).$$

Remark 3.3 Theorem 3.2 provides a best-iterate convergence guarantee for the basic offline scheme under the stated assumptions. Since the objective f is latent, its gradient norm is not available as a practical stopping criterion. The result nevertheless provides a theoretical benchmark: for fixed sparsity level s , the number of comparison queries depends only logarithmically on the ambient dimension. The practical implementation below approximates the basic estimator to accommodate the scale of LLM fine-tuning.

Practical scheme. Applying the basic scheme to all model parameters is computationally expensive for LLMs. We therefore perturb only the output-layer weights $\theta^o \in \mathbb{R}^{d_o}$ and freeze the remaining parameters $\bar{\theta}$, so that $\theta = [\bar{\theta}; \theta^o]$. We also replace the exact solution of Eq. (9) with a normalized sum of signed perturbations followed by entry-wise clipping.

算法 1 离线 ComPO: 基本方案

1: 输入: 初始参数 $\theta_1 \in \mathbb{R}^d$ 、比较集 $S \subseteq D$ 、步长 $\eta > 0$ 、稀疏率 $s \ll d$ 、采样半径 $r > 0$ 、扰动次数 $m \geq 1$ 和迭代次数 $T \geq 1$. 2: 对于 $t = 1, 2, \dots, T$ 执行 3: 从单位球 $\mathbb{R}^d$ 中均匀抽取独立同分布样本 m , 记为 $\{\mathbf{z}_i\}_{i=1}^m$. 4: 计算 $y_i = C_\pi^S(\theta_t, \theta_t + r\mathbf{z}_i)$ 对于 $i = 1, \dots, m$. 5: 使用公式 (9) 计算 $\hat{\mathbf{g}}_t$. 6: 更新 $\theta_{t+1} = \theta_t - \eta \hat{\mathbf{g}}_t$. 7: 输出: θ_{T+1} .

算法 2 离线 ComPO: 实用方案

1: 输入: 初始参数 $\theta_1 = [\bar{\theta}; \theta_1^o]$, 批次 $\{S_t\}_{t=1}^T$, 步长 γ , 采样半径 r , 扰动次数 $m \geq 1$, 裁剪阈值 λ_g, λ , 以及迭代次数 $T \geq 1$. 2: 对于 $t = 1, 2, \dots, T$ 执行 3: 从单位球体 $\mathbb{R}^{d_o}$ 中独立同分布地抽取样本 m i.i.d. $\{\mathbf{z}_i\}_{i=1}^m$. 4: 对所有 $i = 1, \dots, m$ 查询 $y_i = C_\pi^{S_t}([\bar{\theta}; \theta_t^o], [\bar{\theta}; \theta_t^o + rz_i])$. 5: 设置 $\mathbf{u}_t = \sum_{i=1}^m y_i \mathbf{z}_i$. 如果 $\mathbf{u}_t \neq 0$, 则设置 $\hat{\mathbf{g}}_t^o = \mathbf{u}_t / \|\mathbf{u}_t\|$. 否则, 设置 $\hat{\mathbf{g}}_t^o = 0$. 6: 将 $\hat{\mathbf{g}}_t^o$ 通过将绝对值小于 λ_g 的项置零进行裁剪. 7: 设置 $p_t = \frac{|\{i: y_i = -1\}|}{m}$. 8: 如果 $p_t > \lambda$ 则 9: $\theta_{t+1}^o = \theta_t^o - \gamma p_t \hat{\mathbf{g}}_t^o$. 10: 否则 11: $\theta_{t+1}^o = \theta_t^o$. 12: 输出: $\theta_{T+1} = [\bar{\theta}; \theta_{T+1}^o]$.

在哪里 c_m 是一个足够大的常数。假设每次迭代的扰动是独立于过去的, 并且等式(9) 被精确求解。那么, 算法 1 生成的迭代满足

$$\mathbb{P} \left(\min_{1 \leq t \leq T} \|\nabla f(\theta_t)\| < \epsilon \right) \geq 1 - \Lambda.$$

Con因此, 偏好比较预言机调用的总次数是有界的

$$O \left(\left(1 + \frac{\ell \Delta}{\epsilon^2} \right) \left(s \log \left(\frac{2d}{s} \right) + \log \left(\frac{2 + \ell \Delta \epsilon^{-2}}{\Lambda} \right) \right) \right).$$

备注 3.3 定理 3.2 为基本离线方案在所述假设下提供了最佳迭代收敛保证。由于目标 f 是潜在的, 其梯度范数不能作为实际停止标准。尽管如此, 该结果提供了一个理论基准: 对于固定的稀疏级别 s , 比较查询的数量仅对数对数地依赖于环境维度。下面的实际实现近似基本估计器以适应大语言模型微调的规模。

实用方案。 将基本方案应用于所有模型参数对大语言模型来说计算成本很高。因此, 我们仅扰动输出层权重 $\theta^o \in \mathbb{R}^{d_o}$ 并冻结其余参数 $\bar{\theta}$, 以便 $\theta = [\bar{\theta}; \theta^o]$ 。我们还用符号扰动归一化后的和 (后接逐元素裁剪) 来替代公式(9)的精确解。

Algorithm 3 Online ComPO: Basic Scheme

1: **Input:** initial parameter $\theta_1 \in \mathbb{R}^d$ satisfying $\pi_{\theta_1} \in \Pi_\tau$, comparison set $S \subseteq D$, online prompt distribution P_{on} , reference policy π_{ref} , step size $\eta > 0$, reverse-KL radius $\tau > 0$, sparsity ratio $s \ll d$, sampling radius $r > 0$, number of perturbations $m \geq 1$, and iteration number $T \geq 1$.
2: **for** $t = 1, 2, \dots, T$ **do**
3: Draw m i.i.d. samples uniformly from the unit sphere in $\mathbb{R}^d$, denoted by $\{\mathbf{z}_i\}_{i=1}^m$.
4: Compute $y_i = \mathcal{C}_\pi^S(\theta_t, \theta_t + r\mathbf{z}_i)$ for $i = 1, \dots, m$.
5: Compute $\hat{\mathbf{g}}_t$ using Eq. (9).
6: Form $\tilde{\theta}_{t+1}$ and evaluate $\tilde{D}_t$ by Eq. (11).
7: Set θ_{t+1} according to Eq. (12).
8: **Output:** θ_{T+1} .

The practical pipeline partitions the dataset using the reference model. In particular, we define

$$D_{\text{noisy}} = \{(\mathbf{x}, \mathbf{y}^+, \mathbf{y}^-) \in D : |\log \pi_{\text{ref}}(\mathbf{y}^+|\mathbf{x}) - \log \pi_{\text{ref}}(\mathbf{y}^-|\mathbf{x})| \leq \delta_{\text{margin}}\}, \quad (10)$$

and let $D_{\text{clean}} = D \setminus D_{\text{noisy}}$. The term noisy refers to this low-margin subset and does not presume that its preference labels are incorrect. We first apply a direct alignment method, such as DPO or SimPO, to D_{clean} and then apply Algorithm 2 to D_{noisy} . For DPO in the first stage, we denote the resulting procedure by DPO_{clean}+ComPO.

3.2 Online ComPO

We introduce an online extension of ComPO that retains the offline comparison mechanism and uses unlabeled policy generations for reverse-KL control. Following Song et al. (2024b), we restrict the policy to the class Π_τ in Eq. (4), so that the analysis requires coverage only within this neighborhood. Since the update direction is obtained from comparisons rather than the gradient of an explicit preference loss, the basic scheme implements this restriction through a feasibility check on each candidate update. The practical scheme uses the samples from the current policy to adjust the step size.

At iteration t , we compute the same comparison direction $\hat{\mathbf{g}}_t$ as in Algorithm 1 and form a single candidate using a fixed step size $\eta > 0$:

$$\tilde{\theta}_{t+1} = \theta_t - \eta \hat{\mathbf{g}}_t, \quad \tilde{D}_t = D_{\text{RKL}}(\pi_{\tilde{\theta}_{t+1}} \| \pi_{\text{ref}}). \quad (11)$$

Given a reverse-KL radius $\tau > 0$, we accept the candidate if it is feasible and otherwise leave the policy unchanged:

$$\theta_{t+1} = \begin{cases} \tilde{\theta}_{t+1}, & \text{if } \tilde{D}_t \leq \tau, \\ \theta_t, & \text{otherwise.} \end{cases} \quad (12)$$

The basic scheme evaluates the candidate policy’s reverse KL exactly. Starting from a feasible policy, the accept-or-reject rule preserves feasibility by retaining the previous iterate whenever the candidate falls outside Π_τ .

The following theorem establishes feasibility and relates in-distribution pairwise reward accuracy to policy performance under local coverage.

算法 3 在线 ComPO: 基本方案

1: **输入:** 初始参数 $\theta_1 \in \mathbb{R}^d$ 满足 $\pi_{\theta_1} \in \Pi_\tau$, 比较集 $S \subseteq D$, 在线提示分布 P_{on} , 参考策略 π_{ref} , 步长 $\eta > 0$, 反向KL半径 $\tau > 0$, 稀疏率 $s \ll d$, 采样半径 $r > 0$, 扰动次数 $m \geq 1$, 以及迭代次数 $T \geq 1$.
2: **为** $t = 1, 2, \dots, T$ **do**
3: 从 $\mathbb{R}^d$ 中的单位球面上均匀地抽取 m 个独立同分布样本, 记为 $\{\mathbf{z}_i\}_{i=1}^m$.
4: 计算 $y_i = \mathcal{C}_\pi^S(\theta_t, \theta_t + r\mathbf{z}_i)$ 对于 $i = 1, \dots, m$.
5: 使用公式(9)计算 $\hat{\mathbf{g}}_t$.
6: Form $\tilde{\theta}_{t+1}$ 并评估 $\tilde{D}_t$ 根据公式(11).
7: 设 θ_{t+1} 根据公式(12).
8: **输出:** θ_{T+1} .

实用流程使用参考模型对数据集进行划分。具体来说, 我们定义

$$D_{\text{noisy}} = \{(\mathbf{x}, \mathbf{y}^+, \mathbf{y}^-) \in D : |\log \pi_{\text{ref}}(\mathbf{y}^+|\mathbf{x}) - \log \pi_{\text{ref}}(\mathbf{y}^-|\mathbf{x})| \leq \delta_{\text{margin}}\}, \quad (10)$$

并令 $D_{\text{clean}} = D \setminus D_{\text{noisy}}$ 其中“噪声”一词指的是这个低边际子集, 并不假定其偏好标签是错误的。我们首先应用一种直接对齐方法, 如DPO或SimPO, 对 D_{clean} , 然后应用算法2 对 D_{noisy} 在第一阶段中, 我们用DPO_{clean}+ComPO。

3.2 在线ComPO

我们引入了ComPO的在线扩展, 该扩展保留了离线比较机制, 并使用无标签策略生成进行反向KL控制。遵循Song等人(2024b), 我们将策略限制在公式(4)中的 Π_τ 类中, 以便分析只需要在这个邻域内进行覆盖。由于更新方向是从比较中获得的, 而不是显式偏好损失的梯度, 基本方案通过在每个候选更新上进行可行性检查来实现这一限制。实用方案使用当前策略的样本来调整步长。

在迭代 t 处, 我们计算与算法 1 中相同的比较方向 $\hat{\mathbf{g}}_t$, 并使用固定步长 $\eta > 0$:

$$\tilde{\theta}_{t+1} = \theta_t - \eta \hat{\mathbf{g}}_t, \quad \tilde{D}_t = D_{\text{RKL}}(\pi_{\tilde{\theta}_{t+1}} \| \pi_{\text{ref}}). \quad (11)$$

给定反向KL半径 $\tau > 0$, 如果候选解可行则接受, 否则保持策略不变:

$$\theta_{t+1} = \begin{cases} \tilde{\theta}_{t+1}, & \text{if } \tilde{D}_t \leq \tau, \\ \theta_t, & \text{otherwise.} \end{cases} \quad (12)$$

基本方案精确评估候选策略的反向KL。从可行策略开始, 接受-拒绝规则通过在候选解落在 Π_τ 范围外时保留前一个迭代来保持可行性。

以下定理建立了可行性, 并将分布内成对奖励精度与局部覆盖率下的策略性能关联起来。

Algorithm 4 Online ComPO: Practical Scheme

```

1: Input: initial parameter  $\theta_1 = [\bar{\theta}; \theta_1^o]$ , preference dataset  $D$ , online prompts  $\mathcal{X}_{\text{on}}$ , reference policy  $\pi_{\text{ref}}$ , margin threshold  $\delta_{\text{margin}}$ , step-size scale  $\gamma > 0$ , damping strength  $\rho \geq 0$ , threshold  $\tau_p \geq 0$ , sampling radius  $r > 0$ , number of perturbations  $m \geq 1$ , online batch size  $B \geq 1$ , clipping thresholds  $\lambda_g, \lambda > 0$ , iteration number  $T \geq 1$ , re-sampling window  $n \geq 1$ , and replay ratio  $\alpha \in [0, 1]$ .
2: Construct  $D_{\text{noisy}}$  using Eq. (10).
3: Initialize the replay buffer  $\mathcal{R} \leftarrow \emptyset$  and the current successful-batch buffer  $\mathcal{A} \leftarrow \emptyset$ .
4: for  $t = 1, 2, \dots, T$  do
5:   if  $t > 1$  and  $(t - 1) \bmod n = 0$  then
6:     Set  $\mathcal{R} \leftarrow \mathcal{A}$  and  $\mathcal{A} \leftarrow \emptyset$ .
7:   Draw a replay indicator  $b_t \sim \text{Bernoulli}(\alpha)$ .
8:   if  $b_t = 1$  and  $\mathcal{R} \neq \emptyset$  then
9:     Sample a previously successful preference mini-batch  $S_t$  uniformly from  $\mathcal{R}$ .
10:  else
11:    Sample a new noisy preference mini-batch  $S_t \subseteq D_{\text{noisy}}$ .
12:    Draw  $m$  i.i.d. samples uniformly from the unit sphere in  $\mathbb{R}^{d_o}$ , denoted by  $\{\mathbf{z}_i\}_{i=1}^m$ .
13:    Query  $y_i = \mathcal{C}_{\pi}^{S_t}([\bar{\theta}; \theta_t^o], [\bar{\theta}; \theta_t^o + r\mathbf{z}_i])$  for  $i = 1, \dots, m$ .
14:    Set  $\mathbf{u}_t = \sum_{i=1}^m y_i \mathbf{z}_i$  and  $\hat{\mathbf{g}}_t^o = \mathbf{u}_t / \|\mathbf{u}_t\|$  if  $\mathbf{u}_t \neq 0$ ; otherwise set  $\hat{\mathbf{g}}_t^o = 0$ . Clip  $\hat{\mathbf{g}}_t^o$  by zeroing out entries whose magnitude is less than  $\lambda_g$ .
15:    Sample  $\{\tilde{\mathbf{x}}_j\}_{j=1}^B \subseteq \mathcal{X}_{\text{on}}$ , generate  $\tilde{\mathbf{y}}_j \sim \pi_{\theta_t}(\cdot | \tilde{\mathbf{x}}_j)$ , and compute  $\hat{d}_t$  and  $\gamma_t$  using Eq. (14)-(15).
16:    Set  $p_t = \frac{|\{i: y_i = -1\}|}{m}$ .
17:    if  $p_t > \lambda$  then
18:       $\theta_{t+1}^o = \theta_t^o - \gamma_t p_t \hat{\mathbf{g}}_t^o$ .
19:      Add the accepted preference mini-batch to the current buffer:  $\mathcal{A} \leftarrow \mathcal{A} \cup \{S_t\}$ .
20:    else
21:       $\theta_{t+1}^o = \theta_t^o$ .
22: Output:  $\theta_{T+1} = [\bar{\theta}; \theta_{T+1}^o]$ .

```

Theorem 3.4 Fix $\beta, \tau > 0$. Suppose that Algorithm 3 evaluates each candidate policy’s reverse KL exactly. Then, the generated iterates satisfy $\pi_{\theta_t} \in \Pi_{\tau}$ for all $t = 1, \dots, T + 1$. If π_{ref} satisfies local reverse-KL coverage at radius τ with constant C_{τ} , we have

$$\sup_{\pi \in \Pi_{\tau}} J_{\beta}(\pi) - J_{\beta}(\pi_{\theta_t}) \leq C_{\tau} \sqrt{\text{err}(\pi_{\theta_t})}, \quad \text{for all } t = 1, \dots, T + 1.$$

For any $\epsilon > 0$, an iterate satisfying $\text{err}(\pi_{\theta_t}) \leq \epsilon$ satisfies $\sup_{\pi \in \Pi_{\tau}} J_{\beta}(\pi) - J_{\beta}(\pi_{\theta_t}) \leq C_{\tau} \sqrt{\epsilon}$.

Theorem 3.4 combines the feasibility preservation with a coverage-based performance bound following Song et al. (2024b). The reverse-KL constraint restricts the policies under consideration to Π_{τ} , so that this guarantee requires coverage within the neighborhood rather than over the entire policy class. Within this neighborhood, smaller pairwise reward error gives a tighter performance bound.

Practical scheme. While the basic scheme evaluates reverse KL at the candidate policy, the practical scheme samples from the current policy and uses a length-normalized statistic to damp the update. For independent prompts $\tilde{\mathbf{x}}_j \sim P_{\text{on}}$ and responses $\tilde{\mathbf{y}}_j \sim \pi_{\theta_t}(\cdot | \tilde{\mathbf{x}}_j)$, the

算法 4 在线 ComPO: 实用方案

```

1: 输入: 初始参数  $\theta_1 = [\bar{\theta}; \theta_1^o]$ , 偏好数据集  $D$ , 在线提示  $\mathcal{X}_{\text{on}}$ , 参考策略  $\pi_{\text{ref}}$ , 边缘阈值  $\delta_{\text{margin}}$ , 步长缩放  $\gamma > 0$ , 阻尼强度  $\rho \geq 0$ , 阈值  $\tau_p \geq 0$ , 采样半径  $r > 0$ , 扰动次数  $m \geq 1$ , 在线批量大小  $B \geq 1$ , 裁剪阈值  $\lambda_g, \lambda > 0$ , 迭代次数  $T \geq 1$ , 重采样窗口  $n \geq 1$ , 和重放比率  $\alpha \in [0, 1]$ 。
2: 构建  $D_{\text{noisy}}$  使用公式 (10)。
3: 初始化重放缓冲区  $\mathcal{R} \leftarrow \emptyset$  和当前成功批量缓冲区  $\mathcal{A} \leftarrow \emptyset$ 。
4: 对于  $t = 1, 2, \dots, T$  do
5:   如果  $t > 1$  和  $(t - 1) \bmod n = 0$  则
6:     设置  $\mathcal{R} \leftarrow \mathcal{A}$  和  $\mathcal{A} \leftarrow \emptyset$ 。
7:   绘制重放指示器  $b_t \sim \text{伯努利}(\alpha)$ 。
8:   如果  $b_t = 1$  和  $\mathcal{R} \neq \emptyset$  则
9:     从  $\mathcal{R}$  中均匀采样先前成功的偏好小批量  $S_t$ 。
10:  else
11:    采样一个新的噪声偏好小批量  $S_t \subseteq D_{\text{noisy}}$ 。
12:    在  $\mathbb{R}^{d_o}$  中从单位球面上均匀地抽取  $m$  个独立同分布样本, 记为  $\{\mathbf{z}_i\}_{i=1}^m$ 。
13:    查询  $y_i = \mathcal{C}_{\pi}^{S_t}([\bar{\theta}; \theta_t^o], [\bar{\theta}; \theta_t^o + r\mathbf{z}_i])$  以获取  $i = 1, \dots, m$ 。
14:    设置  $\mathbf{u}_t = \sum_{i=1}^m y_i \mathbf{z}_i$  和  $\hat{\mathbf{g}}_t^o = \mathbf{u}_t / \|\mathbf{u}_t\|$  如果  $\mathbf{u}_t \neq 0$ ; 否则设置  $\hat{\mathbf{g}}_t^o = 0$ 。将  $\hat{\mathbf{g}}_t^o$  通过将小于  $y_i$  的绝对值项置零来裁剪。裁剪绝对值小于  $\lambda_g$  的项。
15:    采样  $\{\tilde{\mathbf{x}}_j\}_{j=1}^B \subseteq \mathcal{X}_{\text{on}}$ , 生成  $\tilde{\mathbf{y}}_j \sim \pi_{\theta_t}(\cdot | \tilde{\mathbf{x}}_j)$ , 并计算  $\hat{d}_t$  和  $\gamma_t$  使用公式(14)-(15)。
16:    设置  $p_t = \frac{|\{i: y_i = -1\}|}{m}$ 。
17:    如果  $p_t > \lambda$  则
18:       $\theta_{t+1}^o = \theta_t^o - \gamma_t p_t \hat{\mathbf{g}}_t^o$ 。
19:      将接受的偏好小批量添加到当前缓冲区:  $\mathcal{A} \leftarrow \mathcal{A} \cup \{S_t\}$ 。
20:    else
21:       $\theta_{t+1}^o = \theta_t^o$ 。
22: 输出:  $\theta_{T+1} = [\bar{\theta}; \theta_{T+1}^o]$ 。

```

定理 3.4 固定 $\beta, \tau > 0$ 。假设法 3 精确评估每个候选策略的逆KL。那么, 生成的迭代满足 $\pi_{\theta_t} \in \Pi_{\tau}$ 对于所有 $t = 1, \dots, T + 1$ 。如果 π_{ref} 在半径 τ 处满足局部逆-KL覆盖率, 且常数为 C_{τ} , 则我们有

$$\sup_{\pi \in \Pi_{\tau}} J_{\beta}(\pi) - J_{\beta}(\pi_{\theta_t}) \leq C_{\tau} \sqrt{\text{err}(\pi_{\theta_t})}, \quad \text{for all } t = 1, \dots, T + 1.$$

对于任何 $\epsilon > 0$, 一个满足 $\text{err}(\pi_{\theta_t}) \leq \epsilon$ 的迭代也满足 $\sup_{\pi \in \Pi_{\tau}} J_{\beta}(\pi) - J_{\beta}(\pi_{\theta_t}) \leq C_{\tau} \sqrt{\epsilon}$ 。

定理 3.4 结合了可行性保持与基于覆盖的性能界限, 遵循Song 等人 (2024b)。反向KL约束将考虑的策略限制为 Π_{τ} , 因此这个保证要求在邻域内而非整个策略类中进行覆盖。在这个邻域内, 更小的成对奖励误差会给出更紧的性能界限。

实用方案。 虽然基本方案在候选策略上评估反向KL, 但实用方案从当前策略中采样, 并使用长度归一化的统计量来减弱更新。对于独立的提示 $\tilde{\mathbf{x}}_j \sim P_{\text{on}}$ 和响应 $\tilde{\mathbf{y}}_j \sim \pi_{\theta_t}(\cdot | \tilde{\mathbf{x}}_j)$,

sequence-level estimator

$$\hat{D}_t^{\text{seq}} = \frac{1}{B} \sum_{j=1}^B \log \left(\frac{\pi_{\theta_t}(\tilde{\mathbf{y}}_j | \tilde{\mathbf{x}}_j)}{\pi_{\text{ref}}(\tilde{\mathbf{y}}_j | \tilde{\mathbf{x}}_j)} \right) \quad (13)$$

is unbiased for $D_{\text{RKL}}(\pi_{\theta_t} \| \pi_{\text{ref}})$. In practice, we use

$$\hat{d}_t = \frac{1}{B} \sum_{j=1}^B \frac{\log \pi_{\theta_t}(\tilde{\mathbf{y}}_j | \tilde{\mathbf{x}}_j) - \log \pi_{\text{ref}}(\tilde{\mathbf{y}}_j | \tilde{\mathbf{x}}_j)}{\max\{1, |\tilde{\mathbf{y}}_j|\}}. \quad (14)$$

Length normalization changes the population quantity being estimated. In particular, $\hat{d}_t$ is a signed statistic and its population counterpart needs not be nonnegative. We set

$$\gamma_t = \frac{\gamma}{1 + \rho \max\{\hat{d}_t - \tau_p, 0\}}, \quad (15)$$

where τ_p is the threshold for the length-normalized statistic. As such, the online samples only change the step size, not the comparison oracle or the preference labels.

We divide training into consecutive blocks of n iterations. At the start of each block after the first, the replay buffer is replaced by the mini-batches that passed the update gate $p_t > \lambda$ in the preceding completed block. At each iteration, with probability α , we sample uniformly from this buffer when it is nonempty; otherwise, we sample a new mini-batch from D_{noisy} . Revisited mini-batches use fresh perturbations around the current parameters rather than reusing previous update directions. Here, “successful” means only that the comparison gate was passed. Section 4.3 evaluates the empirical effect of combining replay with online damping.

Algorithm 4 is motivated by the principle used in Algorithm 3, but cannot be covered by Theorem 3.4. In particular, the length-normalized quantity in Eq. (14) is not the sequence-level reverse KL in Eq. (3), and Eq. (15) does not enforce the hard constraint $\pi \in \Pi_\tau$. These are practical heuristics whose effect is evaluated empirically in Section 4.3.

4 Experiments

We investigate the effectiveness of ComPO on aligning the LLMs. First, we evaluate offline scheme as an augmentation to DPO and its variants, where it extracts the directions from noisy preference pairs. Second, we study the offline design choices and the scaling behavior with respect to perturbations, perturbed layers and noisy pairs. Third, we evaluate online scheme, which uses unlabeled current-policy generations to damp the step through a reverse-KL proxy. Unless otherwise stated, the main tables report point estimates from the reported runs and the ablation tables explicitly report variation across repeated runs.

4.1 Offline training for augmenting DPO and SimPO

We identify clean and noisy preference pairs using the margin threshold $\delta_{\text{margin}} = 3$. For Mistral-7B models, we set $r = 0.0005$, $m = 1600$, $\lambda_g = 0.00022$, and $\lambda = 0.2$. For Llama-3-8B models and Gemma-2-9B-it, we set $r = 0.00075$, $m = 1800$, $\lambda_g = 0.00008$, and $\lambda = 0.2$. We use UltraFeedback¹ (Cui et al., 2024) throughout the offline experiments. We initialize

1. https://huggingface.co/datasets/HuggingFaceH4/ultrafeedback_binarized

序列级估计器

$$\hat{D}_t^{\text{seq}} = \frac{1}{B} \sum_{j=1}^B \log \left(\frac{\pi_{\theta_t}(\tilde{\mathbf{y}}_j | \tilde{\mathbf{x}}_j)}{\pi_{\text{ref}}(\tilde{\mathbf{y}}_j | \tilde{\mathbf{x}}_j)} \right) \quad (13)$$

对 $D_{\text{RKL}}(\pi_{\theta_t} \| \pi_{\text{ref}})$ 是无偏的。在实践中，我们使用

$$\hat{d}_t = \frac{1}{B} \sum_{j=1}^B \frac{\log \pi_{\theta_t}(\tilde{\mathbf{y}}_j | \tilde{\mathbf{x}}_j) - \log \pi_{\text{ref}}(\tilde{\mathbf{y}}_j | \tilde{\mathbf{x}}_j)}{\max\{1, |\tilde{\mathbf{y}}_j|\}}. \quad (14)$$

长度归一化会改变被估计的总体数量。特别是， d_t 是一个带符号的统计量，其总体对应值不必是非负的。我们设置

$$\gamma_t = \frac{\gamma}{1 + \rho \max\{\hat{d}_t - \tau_p, 0\}}, \quad (15)$$

其中 τ_p 是长度归一化统计量的阈值。因此，在线样本仅改变步长，而不改变比较预言机或偏好标签。

我们将训练分成连续的 n 迭代块。在每个块开始时（第一个块除外），重放缓冲区被替换为通过更新门的 $p_t > \lambda$ 小批量 D_{noisy} ，否则，我们从 D_{noisy} 中采样一个新的小批量。重访小批量使用围绕当前参数的新扰动，而不是重复使用之前的更新方向。这里，“成功”仅指通过了比较门。第4.3节评估了结合重放和在线阻尼的实证效果。

算法4的动机源于算法3所使用的原理，但无法被定理3.4所覆盖。特别是，等式 (14) 中的长度归一化量并非等式 (3) 中的序列级反向KL，且等式 (15) 并未强制执行硬约束 $\pi \in \Pi_\tau$ 。这些都是实际启发式方法，其效果将在第4.3节中通过经验评估。

4 实验

我们研究了ComPO在使大语言模型对齐方面的有效性。首先，我们评估离线方案作为DPO及其变体的增强方法，其中它从噪声偏好对中提取方向。其次，我们研究离线设计选择以及相对于扰动、扰动层和噪声对的扩展行为。第三，我们评估在线方案，该方案使用未标记的当前策略生成来通过反向KL代理减弱步长。除非另有说明，主表格报告了报告运行中的点估计，消融表格明确报告了重复运行之间的变化。

4.1 离线训练以增强 DPO 和 SimPO

我们使用边缘阈值识别干净和噪声偏好对 $\delta_{\text{margin}} = 3$ 。对于Mistral-7B模型，我们设置 $r = 0.0005$, $m = 1600$, $\lambda_g = 0.00022$, 和 $\lambda = 0.2$ 。对于Llama-3-8B模型和Gemma-2-9B-it，我们设置 $r = 0.00075$, $m = 1800$, $\lambda_g = 0.00008$, 和 $\lambda = 0.2$ 。我们在整个离线实验中使用UltraFeedback¹ (崔等人, 2024)。我们从监督微调的Base和Instruct模型初始化：Meng等人(2024)中使用的模型：Mistral-7B Base和InstructLlama-3-8B Base和Instruct和Gemma-2-9B-it。所有ComPO运行使用30个NVIDIA A40 GPU，每个有46 GB内存。

1. https://huggingface.co/datasets/HuggingFaceH4/ultrafeedback_binarized

Table 1: Evaluation on AlpacaEval 2, Arena-Hard, and MT-Bench across four model configurations. LC and WR denote length-controlled win rate and raw win rate, respectively. Turn-1 and Turn-2 are the MT-Bench scores for the initial and follow-up questions. “PA” denotes the pre-alignment supervised or instruction-fine-tuned checkpoint before DPO training.

Method	Mistral-7B-Base						Mistral-7B-Instruct					
	AlpacaEval 2		Arena-Hard	MT-Bench			AlpacaEval 2		Arena-Hard	MT-Bench		
	LC (%)	WR (%)	WR (%)	Turn-1	Turn-2	Avg.	LC (%)	WR (%)	WR (%)	Turn-1	Turn-2	Avg.
PA	7.33	4.48	1.1	6.10	5.04	5.57	16.54	12.43	10.9	6.19	5.10	5.65
DPO	9.71	6.27	2.9	6.20	5.38	5.79	24.14	16.71	14.4	6.28	5.42	5.86
DPO _{clean}	9.41	6.52	3.0	6.18	5.22	5.70	23.89	16.15	14.2	6.11	5.34	5.73
DPO _{clean} +ComPO	11.66	6.55	3.2	6.22	5.32	5.77	26.17	18.32	10.5	7.78	7.63	7.69

Method	Llama-3-8B-Base						Llama-3-8B-Instruct					
	AlpacaEval 2		Arena-Hard	MT-Bench			AlpacaEval 2		Arena-Hard	MT-Bench		
	LC (%)	WR (%)	WR (%)	Turn-1	Turn-2	Avg.	LC (%)	WR (%)	WR (%)	Turn-1	Turn-2	Avg.
PA	3.21	7.97	4.1	6.53	5.66	6.10	24.06	23.69	20.8	8.22	7.57	7.90
DPO	4.14	10.43	12.1	6.61	5.85	6.23	32.59	31.99	22.9	8.30	7.55	7.93
DPO _{clean}	4.28	9.81	12.0	6.64	6.01	6.33	32.92	32.42	22.9	8.26	7.63	7.94
DPO _{clean} +ComPO	5.39	10.93	12.1	6.60	6.28	6.44	35.79	35.03	23.1	8.39	7.71	8.05

from the supervised fine-tuned **Base** and **Instruct** models used in [Meng et al. \(2024\)](#): Mistral-7B Base and Instruct², Llama-3-8B Base³ and Instruct⁴, and Gemma-2-9B-it⁵. All ComPO runs use 30 NVIDIA A40 GPUs, each with 46 GB of memory.

We follow the evaluation protocol of [Meng et al. \(2024\)](#) and evaluate on AlpacaEval 2-v0.6.6 ([Li et al., 2023](#)), Arena-Hard ([Li et al., 2024](#)), and MT-Bench ([Zheng et al., 2023](#)). For AlpacaEval 2, GPT-4 Turbo serves as both baseline and judge models. The judge compares each model response with the baseline response, and we report raw win rate (WR) and length-controlled win rate (LC) ([Dubois et al., 2024](#)). LC adjusts judged preferences for response length and a higher LC score does not by itself establish shorter responses. For Arena-Hard, the baseline is GPT-4-0314 and the judge is GPT-4 Turbo. We report WR. For MT-Bench, GPT-4 scores multi-turn Q&A responses on a 10-point scale. We report the scores for the initial question (Turn-1), the follow-up question (Turn-2), and their average.

DPO with ComPO. We split the data into clean and noisy subsets using the margin criterion in Eq. (10). Starting from the SFT model, we train on all pairs to obtain DPO and on only the clean pairs to obtain DPO_{clean}. Following [Meng et al. \(2024\)](#), both models are trained for one epoch. We initialize ComPO from DPO_{clean} and run it for one epoch with 100 iterations over noisy pairs, yielding DPO_{clean}+ComPO.

We summarize the results in Table 1 and report three key observations. First, filtering low-margin pairs alone does not uniformly improve DPO: DPO_{clean} is comparable to DPO overall and performs better only for some initializations, such as Llama-3-Instruct-8B. The log-likelihood margin therefore appears to be an imperfect proxy for pair ambiguity; richer

2. <https://huggingface.co/alignment-handbook/zephyr-7b-sft-full>

3. <https://huggingface.co/princeton-nlp/Llama-3-Base-8B-SFT>

4. <https://huggingface.co/meta-llama/Meta-Llama-3-8B-Instruct>

5. <https://huggingface.co/google/gemma-2-9b-it>

表1: 在AlpacaEval 2、Arena-Hard和MT-Bench上针对四种模型配置的评估。LC和WR分别表示长度控制胜率和原始胜率。Turn-1和Turn-2是MT-Bench针对初始问题和后续问题的分数。“PA”表示DPO训练前的预对齐监督或指令微调检查点。

方法	Mistral-7B-Base						Mistral-7B-Instruct					
	AlpacaEval 2		Arena-Hard	MT-Bench			AlpacaEval 2		Arena-Hard	MT-Bench		
	长度控制胜率 (%)	原始胜率 (%)	原始胜率 (%)	Turn-1	Turn-2	平均	长度控制胜率 (%)	原始胜率 (%)	原始胜率 (%)	第一轮	第二轮	平均值
PA	7.33	4.48	1.1	6.10	5.04	5.57	16.54	12.43	10.9	6.19	5.10	5.65
DPO	9.71	6.27	2.9	6.20	5.38	5.79	24.14	16.71	14.4	6.28	5.42	5.86
DPO _{clean}	9.41	6.52	3.0	6.18	5.22	5.70	23.89	16.15	14.2	6.11	5.34	5.73
DPO _{clean} +ComPO	11.66	6.55	3.2	6.22	5.32	5.77	26.17	18.32	10.5	7.78	7.63	7.69

方法	Llama-3-8B-Base						Llama-3-8B-Instruct					
	AlpacaEval 2		Arena-Hard	MT-Bench			AlpacaEval 2		Arena-Hard	MT-Bench		
	长度控制胜率 (%)	原始胜率 (%)	原始胜率 (%)	Turn-1	Turn-2	平均	长度控制胜率 (%)	原始胜率 (%)	原始胜率 (%)	Turn-1	Turn-2	平均
PA	3.21	7.97	4.1	6.53	5.66	6.10	24.06	23.69	20.8	8.22	7.57	7.90
DPO	4.14	10.43	12.1	6.61	5.85	6.23	32.59	31.99	22.9	8.30	7.55	7.93
DPO _{clean}	4.28	9.81	12.0	6.64	6.01	6.33	32.92	32.42	22.9	8.26	7.63	7.94
DPO _{clean} +ComPO	5.39	10.93	12.1	6.60	6.28	6.44	35.79	35.03	23.1	8.39	7.71	8.05

我们使用边缘阈值识别干净和噪声偏好对。对于Mistral-7B模型，我们设置,, 和。对于Llama-3-8B模型和Gemma-2-9B-it，我们设置,, 和。我们在整个离线实验中使用UltraFeedback(崔等人,2024)。我们从监督微调的Base和Instruct模型初始化：Meng等人(2024)中使用的模型：Mistral-7B Base和Instruct²，Llama-3-8B Base³和Instruct⁴，和Gemma-2-9B-it⁵。所有ComPO运行使用30个NVIDIA A40 GPU，每个有46 GB内存。

我们遵循[孟等人\(2024\)](#)的评估协议，并在AlpacaEval 2-v0.6.6([李等人,2023](#))、《Arena-Hard》([李等人,2024](#))和MT-Bench([郑等人, 2023](#))上进行评估。对于AlpacaEval 2，GPT-4 Turbo既作为基线模型也作为评判模型。评判模型将每个模型的回答与基线回答进行比较，我们报告原始胜率（WR）和长度控制胜率（LC）([杜波依斯等人,2024](#))。LC会根据回答长度调整评判偏好，较高的LC分数本身并不代表回答更短。对于Arena-Hard，基线模型是GPT-4-0314，评判模型是GPT-4 Turbo。我们报告WR。对于MT-Bench，GPT-4在10分制上对多轮问答回答进行评分。我们报告初始问题(Turn-1)、后续问题(Turn-2)及其平均值

使用ComPO的DPO。 我们使用等式(10)中的边缘标准将数据分成干净和嘈杂的子集。从SFT模型开始，我们在所有对上进行训练以获得DPO，并在仅干净的对上进行训练以获得DPO_{clean}。遵循 [Meng等人\(2024\)](#)，两个模型都训练一个epoch。我们从DPO初始化ComPO_{clean}，并在100次迭代 noisy pairs 上运行一个epoch，生成DPO_{clean}+ComPO。

我们总结了表1 中的结果，并报告了三个关键观察。首先，单独过滤低利润对并不能统一提高DPO：DPO_{clean} 在整体上与DPO相当，并且仅在Llama-3-Instruct-8B等某些初始化情况下表现更好。因此，对数似然边际似乎是对对模糊性的一种不完美的代理；更丰富的

2. <https://huggingface.co/alignment-handbook/zephyr-7b-sft-full>

3. <https://huggingface.co/princeton-nlp/Llama-3-Base-8B-SFT>

4. <https://huggingface.co/meta-llama/Meta-Llama-3-8B-Instruct>

5. <https://huggingface.co/google/gemma-2-9b-it>

Table 2: Pairwise log-likelihoods in three independent trials for $\gamma \in \{0.1, 1\}$, with all other hyperparameters fixed at their default values. Each cell reports $(\log \pi_\theta(\mathbf{y}^+|\mathbf{x}), \log \pi_\theta(\mathbf{y}^-|\mathbf{x}))$ after one training run; the initial values appear in the model headers. The trials use independently sampled perturbations $\{\mathbf{z}_i\}_{1 \leq i \leq m}$. Across the reported trials, the preferred-response log-likelihood is nondecreasing and the dispreferred-response log-likelihood is nonincreasing.

Llama-3-Instruct-8B $(\log \pi_\theta(\mathbf{y}^+ \mathbf{x}), \log \pi_\theta(\mathbf{y}^- \mathbf{x})) = (-46.761, -47.410)$			
γ	Trial 1	Trial 2	Trial 3
0.1	$(-46.744, -47.411)$	$(-46.760, -47.411)$	$(-46.759, -47.410)$
1	$(-46.728, -47.520)$	$(-46.743, -47.525)$	$(-46.753, -47.517)$
Gemma-2-9B-it $(\log \pi_\theta(\mathbf{y}^+ \mathbf{x}), \log \pi_\theta(\mathbf{y}^- \mathbf{x})) = (-133.122, -134.557)$			
γ	Trial 1	Trial 2	Trial 3
0.1	$(-133.122, -134.557)$	$(-133.122, -134.557)$	$(-133.121, -134.557)$
1	$(-133.059, -134.562)$	$(-133.122, -134.564)$	$(-133.112, -134.565)$

criteria such as the CHES score (Razin et al., 2025) may separate pairs more accurately. Nevertheless, the margin is inexpensive to compute, and ComPO extracts useful information from the pairs that it filters out. Second, gains are especially consistent in AlpacaEval 2 LC, indicating improved judged performance after adjustment for response length. We interpret these scores separately from the response-length measurements reported below. Third, ComPO uses only the first 100 noisy pairs, yet improves most model-benchmark combinations. As such, a small set of low-margin pairs can contain useful alignment information when processed through comparison oracles.

The main exception is Arena-Hard for Mistral-7B-Instruct, where DPO scores 14.4 and $\text{DPO}_{\text{clean}} + \text{ComPO}$ scores 10.5; for the two Llama configurations, the scores are tied or nearly tied. An explanation is that Arena-Hard reports raw rather than length-controlled win rate and can therefore favor longer generations (Meng et al., 2024). For Mistral-7B-Instruct, the average response length is 513 for DPO and 468 for $\text{DPO}_{\text{clean}} + \text{ComPO}$. This difference is consistent with the lower Arena-Hard score and the stronger AlpacaEval 2 LC score, although it does not by itself establish causality.

We also inspect whether the comparison oracle moves the likelihoods of each noisy pair in the intended direction. In Table 2, we summarize three independent trials for $\gamma \in \{0.1, 1\}$ on Llama-3-Instruct-8B and Gemma-2-9B-it. For example, with Llama-3-Instruct-8B and $\gamma = 1$, the first trial changes the pair from $(-46.761, -47.410)$ to $(-46.728, -47.520)$: the preferred response becomes more likely, while the dispreferred response becomes less likely. Thus, for the two reported models, the oracle-based update moves the pairwise likelihoods in the desired direction or leaves them unchanged. This diagnostic is an in-training sanity check rather than a population-level performance guarantee.

The thresholds λ_g and λ limit the coordinates and iterations on which the practical scheme updates the model. Very large step sizes can still destabilize the practical scheme, while Theorem 3.2 analyzes the step size only for the basic scheme. Section 4.3 considers adaptive step-size control based on current-policy generations.

SimPO with ComPO. ComPO is not tied to DPO. We apply it directly to existing, well-tuned SimPO checkpoints (Meng et al., 2024) and use the training and evaluation configuration described at the beginning of Section 4.1. Table 3 shows that SimPO+ComPO

表2: 在三个独立试验中, 针对 $\gamma \in \{0.1, 1\}$ 的成对对数似然, 所有其他超参数均固定在默认值。每个单元格报告在一次训练运行后的 $(\log \pi_\theta(\mathbf{y}^+|\mathbf{x}), \log \pi_\theta(\mathbf{y}^-|\mathbf{x}))$; 初始值显示在模型标题中。这些试验使用独立采样的扰动 $\{\mathbf{z}_i\}_{1 \leq i \leq m}$ 。在报告的试验中, 首选响应对数似然是非递减的, 而非首选响应对数似然是非递增的。

Llama-3-Instruct-8B $(\log \pi_\theta(\mathbf{y}^+ \mathbf{x}), \log \pi_\theta(\mathbf{y}^- \mathbf{x})) = (-46.761, -47.410)$			
γ	试验1	试验2	试验3
0.1	$(-46.744, -47.411)$	$(-46.760, -47.411)$	$(-46.759, -47.410)$
1	$(-46.728, -47.520)$	$(-46.743, -47.525)$	$(-46.753, -47.517)$
Gemma-2-9B-it $(\log \pi_\theta(\mathbf{y}^+ \mathbf{x}), \log \pi_\theta(\mathbf{y}^- \mathbf{x})) = (-133.122, -134.557)$			
γ	试验1	试验2	试验3
0.1	$(-133.122, -134.557)$	$(-133.122, -134.557)$	$(-133.121, -134.557)$
1	$(-133.059, -134.562)$	$(-133.122, -134.564)$	$(-133.112, -134.565)$

标准, 例如CHES分数 (Razin等人, 2025) 可能更准确地分离对。尽管如此, 边际的计算成本很低, 并且ComPO从它过滤掉的对中提取了有用的信息。其次, 在AlpacaEval 2 LC中, 收益特别一致, 表明在调整响应长度后, 判断性能有所提高。我们从下文报告的响应长度测量中分别解释这些分数。第三, ComPO仅使用前100个噪声对, 但提高了大多数模型基准组合。因此, 当通过比较预言机处理时, 一小部分低利润对可以包含有用的对齐信息。

主要例外是 Arena-Hard 对 Mistral-7B-Instruct, 其中 DPO 得分为 14.4, 而 $\text{DPO}_{\text{clean}} + \text{ComPO}$ 得分为 10.5; 对于两个 Llama 配置, 得分是平局或几乎平局。解释是 Arena-Hard 报告的是原始胜率而非长度控制胜率, 因此可能更倾向于更长的生成 (Meng 等人, 2024)。对于 Mistral-7B-Instruct, DPO 的平均响应长度为 513, 而 $\text{DPO}_{\text{clean}} + \text{ComPO}$ 为 468。这种差异与较低的 Arena-Hard 得分和更强的 AlpacaEval 2 LC 得分一致, 尽管它本身不能证明因果关系。

我们还检查比较预言机是否将每个噪声对的可能性向预期方向移动。在表2中, 我们总结了三个独立试验的结果, $\gamma \in \{0.1, 1\}$ 针对 Llama-3-Instruct-8B 和 Gemma-2-9B-it。例如, 使用 Llama-3-Instruct-8B 和 $\gamma = 1$, 第一次试验将配对从 $(-46.761, -47.410)$ 更改为 $(-46.728, -47.520)$: 首选响应变得更可能, 而次选响应变得更不可能。因此, 对于这两个报告的模型, 基于预言机的更新将成对的可能性移动到预期方向或保持不变。这个诊断是在训练中的合理性检查, 而不是对整体性能的保证。

阈值 λ_g 和 λ 限制实用方案更新模型的坐标和迭代次数。非常大的步长仍然可能使实用方案不稳定, 而定理3.2 仅分析了基本方案的步长。第4.3节考虑基于当前策略生成的自适应步长控制。

SimPO 与 ComPO. ComPO 并非与 DPO 紧密关联。我们将它直接应用于现有的、经过良好微调的 SimPO 检查点 (Meng 等人, 2024) 并使用第 4.1 节开头处描述的训练和评估配置。表 3 显示 SimPO+ComPO

Table 3: Applying ComPO to existing SimPO checkpoints across models and benchmarks.

Model	Method	AlpacaEval 2		Arena-Hard	MT-Bench		
		LC (%)	WR (%)	WR (%)	Turn-1	Turn-2	Avg.
Mistral-7B-Instruct	SimPO	40.22	41.18	20.8	7.94	7.31	7.62
	SimPO + ComPO	42.27	43.17	22.0	7.83	7.46	7.64
Llama-3-8B-Instruct	SimPO	48.71	43.66	36.3	7.91	7.42	7.66
	SimPO + ComPO	49.53	45.03	37.3	7.94	7.45	7.70
Gemma-2-9B-it	SimPO	60.36	55.59	61.1	9.07	8.47	8.77
	SimPO + ComPO	62.42	57.20	61.1	8.99	8.58	8.79

Table 4: Effect of the number of perturbations m on AlpacaEval 2. Entries are mean $\pm$ standard deviation over five runs, with the best run in parentheses.

Perturbation (m)	800	1600	3300	5400
AlpacaEval 2-WR %	17.32 ± 0.86 (17.94)	17.50 ± 0.65 (18.32)	19.21 ± 0.58 (20.25)	19.69 ± 0.36 (20.07)
AlpacaEval 2-LC %	24.72 ± 1.02 (25.12)	25.02 ± 0.91 (26.17)	25.91 ± 0.95 (27.14)	26.49 ± 0.81 (27.20)

improves both AlpacaEval 2 metrics for all three models. On Arena-Hard, it improves Mistral-7B-Instruct and Llama-3-8B-Instruct and matches Gemma-2-9B-it. The MT-Bench average also increases slightly for each model. These results show that ComPO augments other direct alignment methods without changing its original training objective.

4.2 Ablation studies

Number of perturbations. The number of perturbations controls how many directions the comparison oracle evaluates. We vary m while holding the remaining hyperparameters fixed and use Mistral-7B-Instruct for this study. As m increases from 800 to 5400, the mean WR and LC improve, with diminishing gains at larger m (Table 4). This is consistent with a more accurate gradient estimate from additional perturbations, although the computation time increases. Peak memory remains unchanged because ComPO accumulates a running average rather than storing all perturbation vectors (see Line 5 of Algorithm 2).

We also investigate whether ComPO scales beyond output-layer perturbations. Keeping all other settings fixed, we perturb the MLPs in layers 30–31 together with the output layer of Mistral-7B-Instruct. Table 5 uses GPT-4.1 as the Arena-Hard judge, and perturbing three layers improves all three reported metrics. The larger search space has a modest systems cost in this setup: peak GPU memory increases from 16.3 GB to 16.7 GB, and 600 perturbations take 60 seconds rather than 50 seconds.

Gradient threshold and number of noisy pairs. ComPO uses the entry threshold λ_g to update only gradient entries with sufficiently large magnitude. We vary λ_g with $m = 3300$ on Mistral-7B-Instruct (Table 6). The strongest results occur when approximately 1%–6% of the entries are retained. Retaining many small entries or filtering almost all entries leads to lower performance. We then increase the number of noisy pairs from 100 to 300. Table 7 shows higher mean performance on both AlpacaEval 2 metrics and Arena-Hard, indicating that ComPO continues to benefit from additional low-margin pairs.

表3： 将ComPO应用于不同模型和基准测试的现有SimPO检查点

模型	方法	AlpacaEval 2		Arena-Hard	MT-Bench		
		LC (%)	WR (%)	原始胜率 (%)	Turn-1	Turn-2	平均
Mistral-7B-Instruct	SimPO	40.22	41.18	20.8	7.94	7.31	7.62
	SimPO + ComPO	42.27	43.17	22.0	7.83	7.46	7.64
Llama-3-8B-Instruct	SimPO	48.71	43.66	36.3	7.91	7.42	7.66
	SimPO + ComPO	49.53	45.03	37.3	7.94	7.45	7.70
Gemma-2-9B-it	SimPO	60.36	55.59	61.1	9.07	8.47	8.77
	SimPO + ComPO	62.42	57.20	61.1	8.99	8.58	8.79

表4： 扰动次数 m 对 AlpacaEval 2 的影响。数据为五次运行的平均值 $\pm$ 标准差，最佳结果以括号标注。

扰动 (m)	800	1600	3300	5400
AlpacaEval 2-WR %	17.32 ± 0.86 (17.94)	17.50 ± 0.65 (18.32)	19.21 ± 0.58 (20.25)	19.69 ± 0.36 (20.07)
AlpacaEval 2-LC %	24.72 ± 1.02 (25.12)	25.02 ± 0.91 (26.17)	25.91 ± 0.95 (27.14)	26.49 ± 0.81 (27.20)

提升了所有三个模型在 AlpacaEval 2 上的指标。在 Arena-Hard 上，它提升了 Mistral-7B-Instruct 和 Llama-3-8B-Instruct，并达到了 Gemma-2-9B-it 的水平。MT-Bench 平均值也略微提升了。这些结果表明 ComPO 增强了其他直接对齐方法，而未改变其原始训练目标。

4.2 消融研究

扰动次数。 扰动次数控制比较预言机评估的方向数量。我们变化 m ，同时保持其他超参数固定，并使用 Mistral-7B-Instruct 进行这项研究。随着 m 从 800 增加到 5400，平均原始胜率和长度控制胜率提高，但在更大的 m (表 4)。这与额外的扰动提供更准确的梯度估计一致，尽管计算时间增加。峰值内存保持不变，因为 ComPO 累积运行平均值，而不是存储所有扰动向量（参见算法2的第 5 行）。

我们还研究了 ComPO 是否能扩展到输出层扰动之外。保持所有其他设置固定，我们将 Mistral-7B-Instruct 的第 30–31 层 MLP 与输出层一起扰动。表 5 使用 GPT-4.1 作为 Arena-Hard 判定器，扰动三层可以提高所有三个报告的指标。在这个设置中，更大的搜索空间有适度的系统成本：峰值 GPU 内存从 16.3 GB 增加到 16.7 GB，600 次扰动需要 60 秒而不是 50 秒。

梯度阈值和噪声对的数量。 ComPO 使用条目阈值 λ_g 来仅更新幅度足够大的梯度条目。我们变化 λ_g 与 $m = 3300$ 在 Mistral-7B-Instruct (表 6) 上。最强结果发生在大约 1%–6% 的条目被保留时。保留许多小条目或过滤几乎所有条目会导致性能降低。然后我们将噪声对的数量从 100 增加到 300。表7显示在 AlpacaEval 2 指标和 Arena-Hard 上都有更高的平均性能，表明 ComPO 仍然能从额外的低边际对中受益。

Table 5: Effect of perturbing multiple layers. We report AlpacaEval 2 WR and LC and Arena-Hard WR. Entries are mean $\pm$ standard deviation over five runs, with the best run in parentheses.

Layers perturbed (# params)	AlpacaEval 2-WR %	AlpacaEval 2-LC %	Arena-Hard (GPT-4.1)-WR %
1 (0.13B)	17.50 $\pm$ 0.65 (18.32)	25.02 $\pm$ 0.91 (26.17)	10.80 $\pm$ 0.21 (11.0)
3 (0.25B)	18.19 $\pm$ 0.81 (19.38)	26.00 $\pm$ 0.89 (27.09)	11.26 $\pm$ 0.36 (11.7)

Table 6: Effect of the gradient-entry threshold λ_g on AlpacaEval 2. Entries are mean $\pm$ standard deviation over five runs, with the best run in parentheses.

λ_g	0	4×10^{-5}	1.8×10^{-4}	2.2×10^{-4}	2.5×10^{-4}
Percentage of gradient entries updated	100%	63%	6%	1%	0.15%
AlpacaEval 2-WR %	15.72 $\pm$ 0.77 (16.34)	16.02 $\pm$ 0.69 (16.69)	19.02 $\pm$ 0.62 (20.15)	19.21 $\pm$ 0.58 (20.25)	16.10 $\pm$ 0.11 (16.21)
AlpacaEval 2-LC %	23.42 $\pm$ 1.03 (24.28)	24.01 $\pm$ 0.91 (25.10)	26.06 $\pm$ 0.81 (27.27)	25.91 $\pm$ 0.95 (27.14)	23.82 $\pm$ 0.23 (24.00)

Table 7: Effect of increasing the number of noisy preference pairs used by ComPO. Entries are mean $\pm$ standard deviation, with the best run in parentheses.

Number of noisy pairs	AlpacaEval 2-WR %	AlpacaEval 2-LC %	Arena-Hard (GPT-4.1)-WR %
100	19.21 $\pm$ 0.58 (20.25)	25.91 $\pm$ 0.95 (27.14)	11.02 $\pm$ 0.13 (11.2)
300	20.07 $\pm$ 0.99 (21.35)	26.28 $\pm$ 0.81 (27.59)	11.76 $\pm$ 0.30 (12.1)

Table 8: Applying ComPO directly to DPO checkpoints without training DPO only on the clean subset. AE, AH, and MT denote AlpacaEval 2, Arena-Hard, and MT-Bench, respectively.

Method	AE LC (%)	AE WR (%)	AH (GPT-4.1) WR (%)	MT Turn 1	MT Turn 2	MT Avg
DPO	24.14	16.71	10.40	6.28	5.42	5.86
DPO + ComPO	27.03	20.85	11.40	7.80	7.61	7.71
DPO (clean)	23.89	16.15	10.50	6.11	5.34	5.73
DPO (clean) + ComPO	27.14	20.25	11.20	7.82	7.59	7.71

Efficiency and compatibility. Full fine-tuning and LoRA-based fine-tuning (Hu et al., 2022) are common post-training choices. ComPO instead uses a lightweight update that changes only selected entries in the output layer. Figure 1 (left) shows that the chosen λ_g retains about 1% of the output-layer entries for Mistral-7B and Llama-3-8B. For Mistral-7B, the plotted 0.13B output-layer size and 1.18% retention rate correspond to roughly 1.5 million updated parameters, or about 0.02% of the full 7B model. Except in the multi-layer ablation, parameters outside the output layer remain frozen.

The comparison-based update avoids full-model backpropagation and accumulate signed perturbations without storing all perturbation vectors. Figure 1 (middle) reports a peak of approximately 23 GB per A40 GPU for Llama-3-8B ComPO; the corresponding reported peaks for DPO and SimPO are 77 GB and 69 GB on H100 GPUs. Because these measurements use different hardware, they describe practical resource requirements rather than a controlled head-to-head comparison. ComPO also parallelizes naturally. For 600 perturbations on 30 A40 GPUs, each worker processes 20 perturbations, and the master aggregates the oracle outputs and perturbation signals to form the gradient estimate (Algorithm 2). Figure 1 (right) shows that runtime increases approximately linearly with the perturbed parameter dimension across the three tested models. Except for the multi-layer ablation, perturbations are restricted to the complete `lm_head` layer.

表 5: 扰动多个层的效果。我们报告了 AlpacaEval 2 原始胜率和长度控制胜率以及 Arena-Hard 原始胜率。条目是五次运行的平均值和标准差, 最佳运行结果括号内显示。

扰动的层 (# 参数)	AlpacaEval 2-WR %	AlpacaEval 2-LC %	Arena-Hard (GPT-4.1)-WR %
1 (0.13B)	17.50 $\pm$ 0.65 (18.32)	25.02 $\pm$ 0.91 (26.17)	10.80 $\pm$ 0.21 (11.0)
3 (0.25B)	18.19 $\pm$ 0.81 (19.38)	26.00 $\pm$ 0.89 (27.09)	11.26 $\pm$ 0.36 (11.7)

表 6: 梯度入口阈值 λ_g 对 AlpacaEval 2 的影响。条目是五次运行的平均值 $\pm$ 标准差, 最佳运行结果括号内显示。

λ_g	0	4×10^{-5}	1.8×10^{-4}	2.2×10^{-4}	2.5×10^{-4}
梯度条目更新百分比	100%	63%	6%	1%	0.15%
AlpacaEval 2-WR %	15.72 $\pm$ 0.77 (16.34)	16.02 $\pm$ 0.69 (16.69)	19.02 $\pm$ 0.62 (20.15)	19.21 $\pm$ 0.58 (20.25)	16.10 $\pm$ 0.11 (16.21)
AlpacaEval 2-LC %	23.42 $\pm$ 1.03 (24.28)	24.01 $\pm$ 0.91 (25.10)	26.06 $\pm$ 0.81 (27.27)	25.91 $\pm$ 0.95 (27.14)	23.82 $\pm$ 0.23 (24.00)

表7: 使用ComPO的噪声偏好对数量增加的效果。条目为平均 $\pm$ 标准差, 最佳结果以括号标注。

噪声对数量	ComPO原始胜率 %	AlpacaEval 2-LC %	Arena-Hard (GPT-4.1)-WR %
100	19.21 $\pm$ 0.58 (20.25)	25.91 $\pm$ 0.95 (27.14)	11.02 $\pm$ 0.13 (11.2)
300	20.07 $\pm$ 0.99 (21.35)	26.28 $\pm$ 0.81 (27.59)	11.76 $\pm$ 0.30 (12.1)

表 8: 将 ComPO 直接应用于 DPO 检查点, 而无需在干净子集上仅训练 DPO。AE、AH 和 MT 分别表示 AlpacaEval 2、Arena-Hard 和 MT-Bench。

方法	AE 长度控制胜率 (%)	AE 原始胜率 (%)	AH (GPT-4.1) 原始胜率 (%)	MT 第一回合	MT 第二回合	MT 平均
DPO	24.14	16.71	10.40	6.28	5.42	5.86
DPO + ComPO	27.03	20.85	11.40	7.80	7.61	7.71
DPO (clean)	23.89	16.15	10.50	6.11	5.34	5.73
DPO (clean) + ComPO	27.14	20.25	11.20	7.82	7.59	7.71

效率和兼容性。 全量微调和基于 LoRA 的微调 (Hu 等人, 2022) 是常见的训练后选择。ComPO 则使用一个轻量级更新, 仅更改输出层中的选定条目。图 1 (左) 显示所选 λ_g 为 Mistral-7B 和 Llama-3-8B 保留了大约 1% 的输出层条目。对于 Mistral- 7B, 绘制的 0.13B 输出层大小和 1.18% 保留率对应于大约 1.5 百万更新的参数, 或大约 0.02% 的完整 7B 模型。除了多层消融之外, 输出层外的参数保持冻结。

基于比较的更新避免了全模型反向传播, 并累积带符号扰动而不存储所有扰动向量。图1 (中间)报告了Llama-3-8B ComPO在A40 GPU上约为23 GB的峰值; 相应的DPO和SimPO在H100 GPU上的峰值分别为77 GB和69 GB。由于这些测量使用了不同的硬件, 它们描述的是实际资源需求, 而不是受控的逐头比较。ComPO也自然地并行化。对于30台A40 GPU上的600个扰动, 每个工作节点处理20个扰动, 主节点聚合神谕输出和扰动信号以形成梯度估计 (算法2)。图1(右侧)显示, 在三个测试模型中, 运行时间随着扰动参数维度的增加近似线性增加。除了多层消融外, 扰动仅限于完整的lm头层。

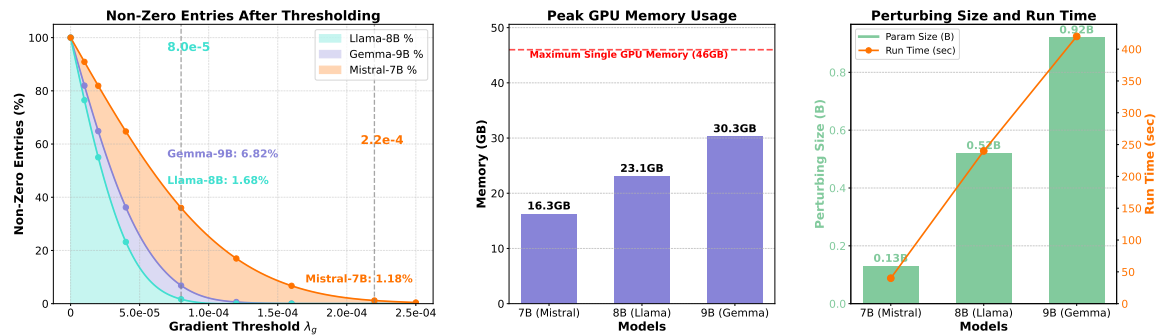


Figure 1: (Left) Percentage of nonzero entries in the final gradient as the gradient-entry threshold λ_g varies. (Middle) Peak GPU memory used by ComPO for the three model families. (Right) Perturbed output-layer size and wall-clock time for completing 600 perturbations on 30 NVIDIA A40 GPUs.

Table 9: Mean $\pm$ standard deviation of the number of negative oracle outputs for the first ten noisy pairs across eight consecutive runs.

Pair 1	Pair 2	Pair 3	Pair 4	Pair 5
394.25 $\pm$ 28.30	364.50 $\pm$ 14.21	369.00 $\pm$ 20.39	447.00 $\pm$ 19.87	591.00 $\pm$ 13.46
Pair 6	Pair 7	Pair 8	Pair 9	Pair 10
282.00 $\pm$ 14.98	459.25 $\pm$ 10.66	242.13 $\pm$ 15.29	311.13 $\pm$ 15.87	348.75 $\pm$ 18.59

ComPO can also be applied directly to an existing checkpoint without first training the underlying DPO model only on clean pairs. In Table 8, we start from DPO checkpoints trained on the full preference dataset and then apply ComPO with $m = 3300$. The resulting gains are comparable to those obtained from $\text{DPO}_{\text{clean}} + \text{ComPO}$. This supports a practical workflow in which a user starts from a publicly available aligned model and refines it with task-specific, potentially noisy preference data using sparse output-layer updates and modest GPU memory.

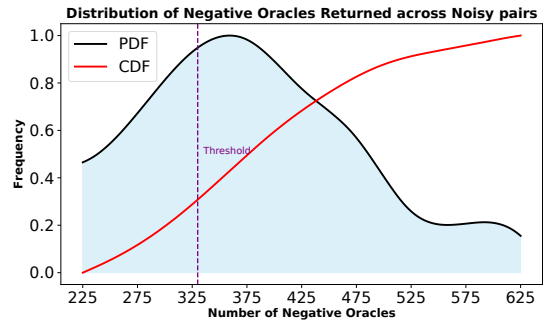


Figure 2: Empirical and cumulative distributions of the number of negative oracle outputs across noisy pairs. The dashed line marks the threshold used for Mistral-7B-Base.

Successful perturbations and clipping threshold λ . In addition to the entry-level threshold λ_g , ComPO uses the clipping threshold $\lambda > 0$ to discard an update when too few perturbations return successful comparison-oracle signals. Figure 2 shows the empirical distribution of the number of negative oracle outputs $k = |\{i : y_i = -1\}|$ across noisy pairs for Mistral-7B-Base. The threshold removes the low-count tail by skipping updates with a small fraction of favorable perturbations. Table 9 further shows that this count remains in a similar range for a fixed pair across eight independent runs. Together, these results indicate that the amount of usable oracle feedback is reproducible and that clipping avoids poorly supported updates.

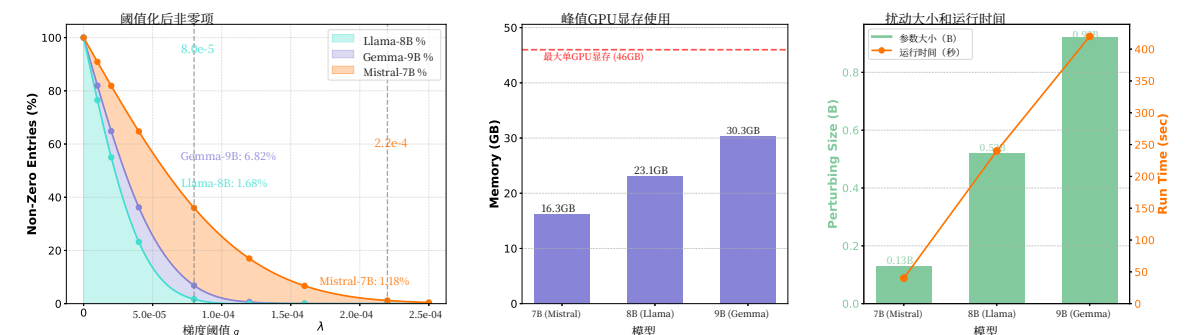


图1: (左) 随梯度-条目阈值 λ_g 变化的最终梯度中非零条目的百分比。(中) ComPO 对三种模型家族使用的峰值 GPU 内存。(右) 在 30 个 NVIDIA A40 GPU 上完成 600 次扰动所需的扰动输出层大小和墙上时间。

表9: 在八次连续运行中, 前十个噪声对负向 Oracle 输出的数量均值 $\pm$ 标准差。

对 1	对 2	配对 3	配对 4	配对 5
394.25 $\pm$ 28.30	364.50 $\pm$ 14.21	369.00 $\pm$ 20.39	447.00 $\pm$ 19.87	591.00 $\pm$ 13.46
配对 6	配对 7	配对 8	配对 9	配对 10
282.00 $\pm$ 14.98	459.25 $\pm$ 10.66	242.13 $\pm$ 15.29	311.13 $\pm$ 15.87	348.75 $\pm$ 18.59

ComPO也可以直接应用于现有的检查点, 而无需首先仅使用干净对进行底层DPO模型的训练。在表8中, 我们从在完整偏好数据集上训练的DPO检查点开始, 然后使用 $m = 3300$. 结果增益与从 $\text{DPO}_{\text{clean}} + \text{ComPO}$ 获得的增益相当。这支持了一种实用的工作流程, 用户可以从一个公开可用的对齐模型开始, 并使用稀疏输出层更新和适度的GPU内存对任务特定的、可能嘈杂的偏好数据进行微调。

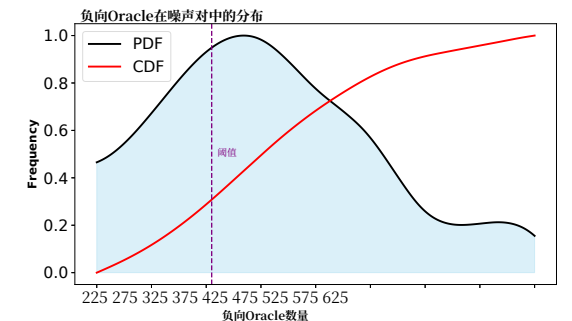


图 2: 噪声配对中负向 Oracle 输出的数量经验分布和累积分布。虚线标记了用于 Mistral-7B-Base 的阈值。

成功的扰动和裁剪阈值 λ . 除了入门级阈值 λ_g , ComPO 使用裁剪阈值 $\lambda > 0$, 当扰动过少时返回成功比较预言机信号, 则丢弃更新。图 2 展示了 Mistral-7B-Base 在噪声对中负预言机输出的数量经验分布 $k = |\{i : y_i = -1\}|$ 。阈值通过跳过只有少量有利扰动的更新来移除低计数尾部。表 9 进一步显示, 对于固定对, 在八次独立运行中该计数保持在相似范围内。这些结果表明, 可用的预言机反馈量是可重复的, 裁剪避免了支持不足的更新。

Table 10: Evaluation on the GPT-4.1 configurations of AlpacaEval 2 and Arena-Hard. LC and WR denote length-controlled and raw win rates. PA denotes the pre-alignment supervised or instruction-fine-tuned checkpoint. “+RKL” uses the length-normalized damping rule in Algorithm 4 and the “+resampling” adds replay to that same online variant.

Method	Qwen3-4B-Base			Llama-3.2-3B-Instruct			Gemma-3-4B-it		
	AlpacaEval 2		Arena-Hard	AlpacaEval 2		Arena-Hard	AlpacaEval 2		Arena-Hard
	LC (%)	WR (%)	WR (%)	LC (%)	WR (%)	WR (%)	LC (%)	WR (%)	WR (%)
PA	12.70	13.12	16.2	11.16	11.83	9.8	34.54	56.20	54.8
DPO	15.28	15.54	29.3	11.72	12.08	11.6	38.30	57.87	56.9
DPO+ComPO	16.20	16.27	30.8	12.35	12.50	11.9	40.00	58.57	57.7
DPO+ComPO (online)									
+ RKL	17.43	17.74	31.4	12.70	13.23	12.4	42.07	60.40	63.3
+ resampling	18.57	17.95	32.6	13.05	13.85	12.8	42.55	60.93	63.7

4.3 Online training

We evaluate online ComPO in Algorithm 4. It keeps the offline comparison direction and uses unlabeled samples to compute the length-normalized statistic in Eq. (14). This statistic adjusts the step size through the soft-damping rule in Eq. (15). The implementation is a heuristic approximation to the basic scheme in Algorithm 3. Indeed, it does not evaluate the proposed next policy or enforce the hard sequence-level reverse-KL constraint analyzed in Theorem 3.4. For the replay buffer, we set the window length to $n = 50$.

For Qwen3-4B-Base, we use $r = 0.0008$, $m = 1800$, and $\lambda_g = 0.000085$. For Gemma-3-4B-it, we use $r = 0.00045$, $m = 1800$, and $\lambda_g = 0.000075$. We evaluate Qwen3-4B-Base⁶, Llama-3.2-3B-Instruct⁷, and Gemma-3-4B-it⁸ using the GPT-4.1 configurations of AlpacaEval 2 and Arena-Hard. Unless stated otherwise, the remaining training settings follow the offline protocol in Section 4.1.

In Table 10, we compare offline ComPO, ComPO with online damping, and ComPO with both damping and replay. Relative to offline ComPO, damping improves AlpacaEval 2 LC, AlpacaEval 2 WR, and Arena-Hard WR by 1.23, 1.47, and 0.6 percentage points for Qwen3-4B-Base; 0.35, 0.73, and 0.5 points for Llama-3.2-3B-Instruct; and 2.07, 1.83, and 5.6 points for Gemma-3-4B-it. Adding replay improves all three reported metrics for each model. These comparisons support the empirical benefit of the combined procedure in the tested configurations, without identifying a separate variance-reduction mechanism.

5 Conclusion

We propose a new zeroth-order preference alignment method based on comparison oracles and show that it can improve large language models (LLMs) using noisy preference pairs for which the reference policy assigns similar likelihoods to preferred and dispreferred responses. The key idea is to use such pairs as comparison signals rather than directly optimizing a preference loss on them. Experimental results on multiple models and benchmarks show that ComPO improves existing direct alignment methods, with pair-level diagnostics pro-

6. <https://huggingface.co/Qwen/Qwen3-4B-Base>

7. <https://huggingface.co/meta-llama/Llama-3.2-3B-Instruct>

8. <https://huggingface.co/google/gemma-3-4b-it>

表 10: 在 AlpacaEval 2 和 Arena-Hard 的 GPT-4.1 配置上的评估。LC 和 WR 表示长度控制和原始胜率。PA 表示预对齐监督或指令微调检查点。“+RKL”使用算法 4 中的长度归一化阻尼规则，而“+重采样”则向该相同的在线变体中添加了重放。

方法	Qwen3-4B-Base			Llama-3.2-3B-Instruct			Gemma-3-4B-it		
	AlpacaEval 2	Arena-Hard	预对齐监督或指令微调检查点	AlpacaEval 2	Arena-Hard	预对齐监督或指令微调检查点	AlpacaEval 2	Arena-Hard	
	长度控制胜率 (%)	原始胜率 (%)	原始胜率 (%)	长度控制胜率 (%)	原始胜率 (%)	原始胜率 (%)	长度控制胜率 (%)	原始胜率 (%)	原始胜率 (%)
PA	12.70	13.12	16.2	11.16	11.83	9.8	34.54	56.20	54.8
DPO	15.28	15.54	29.3	11.72	12.08	11.6	38.30	57.87	56.9
DPO+ComPO	16.20	16.27	30.8	12.35	12.50	11.9	40.00	58.57	57.7
DPO+ComPO (在线)									
+ RKL	17.43	17.74	31.4	12.70	13.23	12.4	42.07	60.40	63.3
+ 重采样	18.57	17.95	32.6	13.05	13.85	12.8	42.55	60.93	63.7

4.3 在线训练

我们在算法 4 中评估在线 ComPO。它保留离线比较方向，并使用未标记样本计算公式 (14) 中的长度归一化统计量。该统计量通过公式 (15) 中的软阻尼规则调整步长。实现是一个对算法 3 基本方案的启发式近似。确实，它没有评估提出的下一步策略，也没有执行定理 3.4 中分析的硬序列级逆 KL 约束。对于重放缓冲区，我们设置窗口长度为 $n = 50$ 。

对于 Qwen3-4B-Base，我们使用 $r = 0.0008$ ， $m = 1800$ ，以及 $\lambda_g = 0.000085$ 。对于 Gemma-3-4B-it，我们使用 $r = 0.00045$ ， $m = 1800$ ，以及 $\lambda_g = 0.000075$ 。我们使用 AlpacaEval 2 和 Arena-Hard 的 GPT-4.1 配置评估 Qwen3-4B-Base⁶，Llama-3.2-3B-Instruct⁷，和 Gemma-3-4B-it⁸。除非另有说明，其余训练设置遵循第 4.1 节的离线协议。

在表 10 中，我们比较离线 ComPO、带在线阻尼的 ComPO 以及带阻尼和重放的 ComPO。相对于离线 ComPO，阻尼将 AlpacaEval 2 的长度控制胜率、原始胜率和 Arena-Hard 的原始胜率分别提升了 1.23、1.47 和 0.6 个百分点，对于 Qwen3-4B-Base；0.35、0.73 和 0.5 个百分点，对于 Llama-3.2-3B-Instruct；以及 2.07、1.83 和 5.6 个百分点，对于 Gemma-3-4B-it。增加重放可以提高每个模型的三个报告指标。这些比较支持在测试配置中组合流程的经验性优势，但并未识别出独立的方差减少机制。

5 结论

我们提出了一种基于比较预言机的零阶偏好对齐方法，并证明它可以通过噪声偏好对来改进大语言模型（LLMs），其中参考策略为偏好和不受欢迎的响应分配了相似的似然值。关键思想是使用这些对作为比较信号，而不是直接在它们上优化偏好损失。在多个模型和基准上的实验结果表明，ComPO 改进了现有的直接对齐方法，对齐对级别的诊断提供了一致的证据，以减轻似然位移。

6. <https://huggingface.co/Qwen/Qwen3-4B-Base>

7. <https://huggingface.co/meta-llama/Llama-3.2-3B-Instruct>

8. <https://huggingface.co/google/gemma-3-4b-it>

viding evidence consistent with mitigating likelihood displacement. These results highlight the importance of designing specialized methods for preference pairs with small likelihood margins, complementing the recent findings of Razin et al. (2025).

The extension in this journal version is online ComPO, where offline noisy preference pairs continue to determine the comparison direction, and unlabeled generations from the current policy provide reverse-KL regularization. We establish feasibility and a coverage-based performance bound for the basic constrained scheme and evaluate damping and replay in the practical implementation. Future directions include extending our approach to other settings (Yuan et al., 2024; Xu et al., 2024b; Tajwar et al., 2024; Guo et al., 2024; Chen and Chen, 2026) and applying it to other tasks, including reasoning (Pang et al., 2024; Chen et al., 2025c) and diffusion model alignment (Wallace et al., 2024).

Acknowledgement

We sincerely appreciate Buzz High Performance Computing (<https://www.buzzhpc.ai>, info@buzzhpc.ai) for providing computational resources and support for this work. Tianyi Lin gratefully acknowledges financial support through a start-up grant and an early career scholarship support grant at Columbia University.

提供了一致性证据，以减轻似然位移。这些结果突出了为具有较小似然边距的偏好对设计专用方法的重要性，补充了最近 Razin 等人的发现Razin 等人(2025)

本期刊版本中的扩展是线上的ComPO，其中离线的噪声偏好对继续决定比较方向，而当前策略的无标签生成提供反向KL正则化。我们为基本约束方案建立了可行性和基于覆盖率的性能界限，并在实际实现中评估了阻尼和重放。未来方向包括将我们的方法扩展到其他设置（Yuan等人，2024；Xu等人，2024b；Tajwar等人，2024；Guo等人，2024；Chen和Chen，2026）并将其应用于其他任务，包括推理（Pang等人，2024；Chen等人，2025c）和扩散模型对齐（Wallace等人，2024）。

致谢

我们衷心感谢Buzz高性能计算（<https://www.buzzhpc.ai>,info@buzzhpc.ai）为本项工作提供计算资源和支持。天一·林感谢哥伦比亚大学提供的启动资金和早期职业奖学金支持。

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Appendix A. Further Related Work

We make additional comments on other topics, including preference learning methods, the analysis of preference learning methods, zeroth-order optimization methods, likelihood displacement, and learning from noisy preference data. For an overview of preference learning methods and open problems in RLHF, we refer to the recent survey (Casper et al., 2023).

More discussion on preference learning methods. The lack of explicit reward models in DPO (Rafailov et al., 2023) is known to make its performance depend strongly on the size and quality of offline preference pairs. To address this limitation, subsequent works proposed to augment preference data using a trained SFT policy (Zhao et al., 2023) or a refined SFT policy with rejection sampling (Liu et al., 2024a). The DPO loss was also extended to a token-level MDP (Rafailov et al., 2024b), where the transition is deterministic, i.e., the next state is determined once the current state and action are chosen, which naturally covers the fine-tuning of autoregressive LLMs. Azar et al. (2024) further generalized DPO to a wider class of RL problems without explicitly introducing a reward function. Instead of maximizing a reward in a KL-constrained problem, they proposed to optimize a general non-decreasing function of the ground-truth population-level preference probability. There are also several other DPO variants (Ethayarajh et al., 2024; Park et al., 2024; Xu et al., 2024a; Meng et al., 2024; Chen et al., 2025a; Zhao et al., 2025). For example, Ethayarajh et al. (2024) aligned the policy with preferences using a prospect-theoretic loss, Tang et al. (2024a) optimized a general loss instead of the log-likelihood loss, and Meng et al. (2024) aligned the reward function in the preference optimization objective with the generation metric. Dong et al. (2024) and Xiong et al. (2024) proposed to generate human feedback in an online fashion to mitigate distribution shift and over-optimization. There has also been an attempt to understand the theoretical performance of DPO (Azar et al., 2024), although this analysis mainly focuses on the population-level objective rather than finite-sample policy-optimality or sample-complexity guarantees. Chen et al. (2026a) also extends DPO-style direct alignment to multiple-objective setup via a novel conflict-averse formulation.

Analysis of preference learning methods. In this context, Zhu et al. (2023) formulated RLHF as a contextual bandit problem and proved the convergence of the maximum likelihood estimator. Xiong et al. (2024) showed the benefits of KL regularization for the sample complexity of online exploration in DPO. Xie et al. (2025) studied online exploration using KL-regularized Markov decision processes and proved a sample-complexity guarantee for an exploration bonus. Liu et al. (2024b) investigated the issue of over-optimization and proved finite-sample guarantees. Song et al. (2024b) conducted a rigorous analysis through the lens of dataset coverage to differentiate offline DPO and online RLHF. Recently, several works have reported faster convergence rates for online reward maximization in RL by exploiting the structure induced by KL regularization. For example, Shi et al. (2025) studied the tabular softmax parametrization setting and established quadratic convergence results.

Zeroth-order optimization methods. The idea of zeroth-order optimization is to approximate a gradient using either a one-point estimator (Flaxman et al., 2005) or a two-point estimator (Agarwal et al., 2010; Ghadimi and Lan, 2013; Duchi et al., 2015; Shamir, 2017; Nesterov and Spokoiny, 2017), where the latter approach often achieves better finite-time convergence guarantees. Despite the rapid development of two-point-based gradient-free

附录 A. 进一步相关工作

我们在其他主题上做出额外评论，包括偏好学习方法、偏好学习的分析、零阶优化方法、似然位移以及从噪声偏好数据中学习。关于偏好学习方法以及 RLHF 中的开放问题，我们参考了最近的调查（Casper 等人,2023）。

更多关于偏好学习方法的讨论。DPO（Rafailov 等人，2023）中缺乏显式奖励模型已知使其性能强烈依赖于离线偏好对的数量和质量。为了解决这一局限性，后续工作提出使用训练好的 SFT 策略（Zhao 等人，2023）或带有拒绝采样的精炼 SFT 策略（Liu 等人，2024a）来增强偏好数据。DPO 损失也被扩展到标记级的 MDP（Rafailov 等人，2024 b），其中转换是确定的，即当前状态和动作选择后，下一个状态就确定了，这自然地涵盖了自回归大语言模型的微调。Azar 等人（2024）进一步将 DPO 推广到更广泛类别的 RL 问题，而没有显式引入奖励函数。在 KL 约束问题中，他们没有最大化奖励，而是提出优化一个关于真实群体级偏好概率的一般非递减函数。还有其他几个 DPO 变体（Ethayarajh 等人，2024; Park 等人，2024; Xu 等人，2024a; Meng 等人，2024; Chen 等人，2025a; Zhao 等人，2025）。例如，Ethayarajh 等人（2024）使用前景理论损失将策略与偏好对齐，Tang 等人（2024a）优化一般损失而不是对数似然损失，Meng 等人（2024）在偏好优化目标中与大语言模型生成指标对齐奖励函数。Dong 等人（2024）和 Xiong 等人（2024）提出以在线方式生成人类反馈以减轻分布偏移和过度优化。也有尝试理解 DPO 的理论性能（Azar 等人，2024），尽管这种分析主要关注群体级目标，而不是有限样本策略最优性或样本复杂度保证。Chen 等人（2026a）还通过一种新颖的回避冲突公式将 DPO 风格的直接对齐扩展到多目标设置。

偏好学习的分析。 在这种情况下，Zhu 等人 (2023) 将 RLHF 形式化为一个上下文老虎机问题，并证明了最大似然估计的收敛性。Xiong 等人 (2024) 展示了 KL 正则化在 DPO 中在线探索样本复杂度的优势。Xie 等人(2025) 研究了使用 KL 正则化马尔可夫决策过程的在线探索，并证明了对探索奖励的样本复杂度保证。Liu 等人(2024b) 研究了过度优化问题，并证明了有限样本保证。Song 等人 (2024b) 通过数据集覆盖的视角进行了严谨的分析，以区分离线 DPO 和在线 RLHF。最近，一些工作报告了通过利用 KL 正则化产生的结构来加快 RL 中在线奖励最大化的收敛速度。例如，Shi 等人 (2025) 研究了表格 softmax 参数化设置，并建立了二次收敛结果。

零阶优化方法。 零阶优化的思想是使用一个单点估计器 (Flaxman 等人, , 2005) 或一个两点估计器 (Agarwal 等人, , 2010; Ghadimi 和 Lan, , 2013; Duchi 等人, , 2015; Shamir, , 2017; Nesterov 和 Spokoiny, , 2017)，后一种方法通常能获得更好的有限时间收敛保证。尽管基于两点的无梯度方法发展迅速，但大部分工作仍集中在凸优化 (Duchi 等人, 2015; Shamir, 2017; Wang 等人, 2018) 和平滑非凸优化 (Nesterov 和 Spokoiny, 2017; Ghadimi 和 Lan, 2013; Lian 等人, 2016; Liu 等人, 2018; Chen 等人, 2019; Ji 等人, 2019; Huang 等人, 2022) 上。在非光滑凸设置 (Duchi 等人, 2015; Shamir, 2017) 和平滑非凸设置 (Ghadimi 和 Lan, 2013; Nesterov 和 Spokoiny, 2017) 中均获得了收敛保证。额外的正则性条件，例如有限和结构，允许使用方差减少技术 (Liu 等人, 2018; Chen 等人, 2019; Ji 等人, 2019)，并在 Huang 等人 (2022) 中获得了尖锐的收敛保证。最近，零阶优化方法被开发用于非光滑非凸优化，并具有坚实的理论保证 (Lin 等人, 2022a; Kornowski 和 Shamir, 2024)。在另一个方向上，零阶优化方法被扩展到强化学习设置，并作为 Q 学习和策略梯度方法等经典方法的可扩展替代方案取得了经验性成功 (Salimans 等人, 2017; Conti 等人, 2018)。该策略也应用于基于偏好的强化学习 (Akrouer 等人, 2011; Busa-Fekete 等人, 2014)，并被用于大语言模型微调 (Malladi 等人, 2023; Zhang 等人, 2024)。在这些设置中，损失函数可以显式估计或计算，因此可以查询以构建梯度估计器。相比之下，我们的方法以及 Tang 等人 (2024b) 和 Zhang 与 Ying (2025) 的方法是基于比较预言机或排序预言机开发的，在这些方法中，即使损失函数值的噪声估计也无法获得。

methods, much of the work focuses on convex optimization (Duchi et al., 2015; Shamir, 2017; Wang et al., 2018) and smooth nonconvex optimization (Nesterov and Spokoiny, 2017; Ghadimi and Lan, 2013; Lian et al., 2016; Liu et al., 2018; Chen et al., 2019; Ji et al., 2019; Huang et al., 2022). Convergence guarantees have been obtained in both nonsmooth convex settings (Duchi et al., 2015; Shamir, 2017) and smooth nonconvex settings (Ghadimi and Lan, 2013; Nesterov and Spokoiny, 2017). Additional regularity conditions, e.g., a finite-sum structure, allow variance-reduction techniques to be used (Liu et al., 2018; Chen et al., 2019; Ji et al., 2019), and sharp convergence guarantees are obtained in Huang et al. (2022). Very recently, zeroth-order optimization methods have been developed for nonsmooth nonconvex optimization with solid theoretical guarantees (Lin et al., 2022a; Kornowski and Shamir, 2024). In another direction, zeroth-order optimization methods were extended to the RL setting and have achieved empirical success as scalable alternatives to classic methods such as Q-learning and policy gradient methods (Salimans et al., 2017; Conti et al., 2018). This strategy has also been applied in preference-based RL (Akroun et al., 2011; Busa-Fekete et al., 2014) and adopted for LLM fine-tuning (Malladi et al., 2023; Zhang et al., 2024). In these settings, the loss function can be explicitly estimated or calculated and thus can be queried to construct the gradient estimator. By contrast, our method and the methods of Tang et al. (2024b) and Zhang and Ying (2025) are developed based on comparison oracles or ranking oracles, where even noisy estimates of loss-function values are not accessible.

Likelihood displacement. We provide a brief overview of proposed explanations for likelihood displacement. Indeed, several works claimed that samples with similar preferred and dispreferred responses are responsible for likelihood displacement (Pal et al., 2024; Tajwar et al., 2024; Razin et al., 2025), although the similarities were measured using different metrics. Other proposed reasons include effects of the initial SFT model (Rafailov et al., 2024b), the presence of multiple training samples and limited model capacity (Tajwar et al., 2024), and the squeezing effect (Ren and Sutherland, 2025). Recently, Razin et al. (2025) conducted a thorough investigation to understand the causes of likelihood displacement, and their results suggest that samples with similar preferred and dispreferred responses might contribute more than others. Regarding the implications of likelihood displacement, previous works found that DPO tends to degrade performance on math and reasoning (Pal et al., 2024; Pang et al., 2024; Meng et al., 2024; Yuan et al., 2025). Indeed, only a few responses are correct, and likelihood displacement can have adverse effects on correct alignment.

Learning from noisy preference data. ComPO addresses low-margin preference pairs selected by Eq. (10). This setting is related to, but distinct from, learning with corrupted preference labels (Amini et al., 2024; Xiao et al., 2024). From this perspective, ComPO is not intended as a direct replacement for existing methods, but rather as a complementary and modular component that enhances their robustness. Moreover, learning from corrupted preference data has been studied in prior works, including those leveraging reward scores through conditional DPO (Kim et al., 2024; Zhang et al., 2025). Conditional DPO modifies the DPO objective by conditioning on reward scores and solves the resulting problem via gradient-based methods, and it can be combined with ComPO in a way similar to SimPO+ComPO as in our work. Apart from noisy labels, Chen et al. (2026b) also theoretically analyze the impact false-positive and false-negative labels in online LLM RL, which is complementary to the mis-labeled preference pairs.

在强化学习设置中，零阶优化方法被扩展并作为 Q 学习和策略梯度方法等经典方法的可扩展替代方案取得了经验性成功 (Salimans 等人, 2017; Conti 等人, 2018)。该策略也应用于基于偏好的强化学习 (Akroun 等人, 2011; Busa-Fekete 等人, 2014)，并被用于大语言模型微调 (Malladi 等人, 2023; Zhang 等人, 2024)。在这些设置中，损失函数可以显式估计或计算，因此可以查询以构建梯度估计器。相比之下，我们的方法以及 Tang 等人 (2024b) 和 Zhang 与 Ying (2025) 的方法是基于比较预言机或排序预言机开发的，在这些方法中，即使损失函数值的噪声估计也无法获得。

似然位移。我们简要概述了关于似然位移的提出解释。确实，一些工作声称具有相似首选和不受欢迎响应的样本是导致似然位移的原因 (Pal 等人, 2024; Tajwar 等人, 2024; Razin 等人, 2025)，尽管这些相似性是使用不同的指标测量的。其他提出的原因包括初始 SFT 模型的影响 (Rafailov 等人, 2024b)、存在多个训练样本和有限的模型容量 (Tajwar 等人, 2024)，以及挤压效应 (Ren 和 Sutherland, 2025)。最近，Razin 等人 (2025) 进行了一项彻底的调查，以了解似然位移的原因，他们的结果表明，具有相似首选和不受欢迎响应的样本可能比其他样本贡献更大。关于似然位移的影响，以前的工作发现 DPO 往往会降低数学和推理的性能 (Pal 等人, 2024; Pang 等人, 2024; Meng 等人, 2024; Yuan 等人, 2025)。确实，只有少数响应是正确的，而似然位移会对正确对齐产生不利影响。

从噪声偏好数据中学习。 ComPO 解决了由 Eq. (10) 选择的低边际偏好对。这种设置与学习损坏的偏好标签相关，但与之不同 (Amini 等人, 2024; Xiao 等人, 2024)。从这个角度来看，ComPO 旨在作为现有方法的直接替代品，而是一个补充性和模块化的组件，可以增强其鲁棒性。此外，从损坏的偏好数据中学习已在先前工作中研究过，包括利用奖励分数通过条件 DPO (Kim 等人, 2024; Zhang 等人, 2025) 的方法。条件 DPO 通过奖励分数对 DPO 目标进行条件化，并通过基于梯度的方法求解所得问题，并且可以像 SimPO 那样与 ComPO 结合使用+我们工作中的 ComPO。除了噪声标签外，Chen 等人 (2026b) 还从理论上分析了在线 LLM 强化学习中假阳性标签和假阴性标签的影响，这与损坏的偏好对是互补的。

Appendix B. Missing Proofs

We present several technical lemmas and use them to prove Theorem 3.2 and Theorem 3.4.

B.1 Technical lemmas

For the offline analysis, we fix a nonempty set $S \subseteq D$ and impose the smoothness, gradient sparsity, and oracle compatibility (see Theorem 3.2) throughout this subsection. We use $\text{sign}(0) = +1$. At any point with $\nabla f(\theta) \neq 0$, we write

$$y_i = \mathcal{C}_\pi^S(\theta, \theta + r\mathbf{z}_i), \quad \bar{\mathbf{g}} = \frac{\nabla f(\theta)}{\|\nabla f(\theta)\|}, \quad \bar{y}_i = \text{sign}(\mathbf{z}_i^\top \bar{\mathbf{g}}).$$

Thus, the oracle compatibility guarantees $y_i = \text{sign}(f(\theta + r\mathbf{z}_i) - f(\theta))$. The next proposition adapts the one-bit estimation framework (Plan and Vershynin, 2012; Cai et al., 2022a) to errors that might depend on the perturbation directions but are localized near directions orthogonal to the target.

Proposition B.1 *Let $1 \leq s \leq d$ and let $\bar{\mathbf{g}} \in \mathbb{R}^d$ satisfy $\|\bar{\mathbf{g}}\|_1 \leq \sqrt{s}$ and $\|\bar{\mathbf{g}}\| = 1$. Suppose that $(\mathbf{z}_i, y_i)_{i=1}^m$ are i.i.d., where $\mathbf{z}_i$ is uniform on the unit sphere in $\mathbb{R}^d$ and $y_i \in \{-1, +1\}$, and $y_i = \text{sign}(\mathbf{z}_i^\top \bar{\mathbf{g}})$ almost surely whenever $|\mathbf{z}_i^\top \bar{\mathbf{g}}| > \frac{1}{40\sqrt{d}}$. Then, we define*

$$\hat{\mathbf{g}} \in \underset{\|\mathbf{g}\|_1 \leq \sqrt{s}, \|\mathbf{g}\| \leq 1}{\text{argmax}} \sum_{i=1}^m y_i \mathbf{z}_i^\top \mathbf{g}.$$

For any $\delta \in (0, 1)$, if $m \geq c_m \left(s \log \left(\frac{2d}{s} \right) + \log \left(\frac{2}{\delta} \right) \right)$ for a sufficiently large constant c_m , we have

$$\mathbb{P}(\|\hat{\mathbf{g}} - \bar{\mathbf{g}}\| \leq \frac{1}{2}) \geq 1 - \delta.$$

Proof For the case of $d = 1$, we have $s = 1$ and $\mathbf{z}_i, \bar{\mathbf{g}} \in \{-1, +1\}$. Since $|\mathbf{z}_i^\top \bar{\mathbf{g}}| = 1$, the assumption on the labels implies $y_i = \mathbf{z}_i^\top \bar{\mathbf{g}}$ almost surely. Thus, $y_i \mathbf{z}_i = \bar{\mathbf{g}}$ almost surely, and the definition of $\hat{\mathbf{g}}$ gives $\hat{\mathbf{g}} = \bar{\mathbf{g}}$. For the case of $d \geq 2$, we define

$$K = \{\mathbf{g} \in \mathbb{R}^d : \|\mathbf{g}\|_1 \leq \sqrt{s}, \|\mathbf{g}\| \leq 1\}, \quad F_m(\mathbf{g}) = \frac{1}{m} \sum_{i=1}^m y_i \mathbf{z}_i^\top \mathbf{g}.$$

Since $\bar{\mathbf{g}} \in K$ and $\hat{\mathbf{g}}$ maximizes F_m over K , it suffices to show, with probability at least $1 - \delta$, that $F_m(\bar{\mathbf{g}}) > F_m(\mathbf{g})$ for every $\mathbf{g} \in K$ with $\|\mathbf{g} - \bar{\mathbf{g}}\| > \frac{1}{2}$. For simplicity, we let $(\mathbf{z}, y)$ have the same distribution as $(\mathbf{z}_i, y_i)$. The key decomposition is given by

$$\frac{1}{m} \sum_{i=1}^m y_i \mathbf{z}_i = \mathbb{E}[\text{sign}(\mathbf{z}^\top \bar{\mathbf{g}}) \mathbf{z}] + \underbrace{\mathbb{E}[(y - \text{sign}(\mathbf{z}^\top \bar{\mathbf{g}})) \mathbf{z}]}_A + \underbrace{\frac{1}{m} \sum_{i=1}^m y_i \mathbf{z}_i - \mathbb{E}[y \mathbf{z}]}_B.$$

We set $\kappa = \mathbb{E}[\|\mathbf{z}^\top \bar{\mathbf{g}}\|]$ and obtain from rotational invariance that $\mathbb{E}[\text{sign}(\mathbf{z}^\top \bar{\mathbf{g}}) \mathbf{z}] = \kappa \bar{\mathbf{g}}$. Thus, for every $\mathbf{g} \in K$, we have

$$F_m(\bar{\mathbf{g}}) - F_m(\mathbf{g}) = \kappa(1 - \bar{\mathbf{g}}^\top \mathbf{g}) + A^\top(\bar{\mathbf{g}} - \mathbf{g}) + B^\top(\bar{\mathbf{g}} - \mathbf{g}). \quad (16)$$

In what follows, we write $r = \|\mathbf{g} - \bar{\mathbf{g}}\|$ and prove that $r > \frac{1}{2}$ implies $F_m(\bar{\mathbf{g}}) - F_m(\mathbf{g}) > 0$.

附录 B. 缺失证明

我们预 在几个技术引理的基础上，并使用它们来证明定理3.2和定理3.4.

B.1 技术术语

对于离线分析，我们固定一个非空集合 $S \subseteq D$ 并在此小节中施加平滑性、梯度稀疏性和神谕兼容性（参见定理3.2）。我们使用 $\text{sign}(0) = +1$ 。在具有 $\nabla f(\theta) \neq 0$ ，我们写

$$y_i = \mathcal{C}_\pi^S(\theta, \theta + r\mathbf{z}_i), \quad \bar{\mathbf{g}} = \frac{\nabla f(\theta)}{\|\nabla f(\theta)\|}, \quad \bar{y}_i = \text{sign}(\mathbf{z}_i^\top \bar{\mathbf{g}}).$$

因此，神谕兼容性保证了 $y_i = \text{sign}(f(\theta + r\mathbf{z}_i) - f(\theta))$ 。下一个命题将一位估计框架 (Plan and Vershynin, 2012; Cai 等人, 2022a) 应用于可能依赖于扰动方向但靠近与目标正交方向局部化的误差。

命题 B.1 设 $1 \leq s \leq d$ 和 $\bar{\mathbf{g}} \in \mathbb{R}^d$ 满足 $\|\bar{\mathbf{g}}\|_1 \leq \sqrt{s}$ 和 $\|\bar{\mathbf{g}}\| = 1$ 。假设 $(\mathbf{z}_i, y_i)_{i=1}^m$ 是独立同分布的，其中 $\mathbf{z}_i$ 在 $\mathbb{R}^d$ 和 $y_i \in \{-1, +1\}$ 上的单位球面上均匀分布，并且 $y_i = \text{sign}(\mathbf{z}_i^\top \bar{\mathbf{g}})$ 几乎处处成立，只要 $|\mathbf{z}_i^\top \bar{\mathbf{g}}| > \frac{1}{40\sqrt{d}}$ 。然后，我们定义

$$\hat{\mathbf{g}} \in \underset{\|\mathbf{g}\|_1 \leq \sqrt{s}, \|\mathbf{g}\| \leq 1}{\text{argmax}} \sum_{i=1}^m y_i \mathbf{z}_i^\top \mathbf{g}.$$

对于任何 $\delta \in (0, 1)$ ，如果 $m \geq c_m \left(s \log \left(\frac{2d}{s} \right) + \log \left(\frac{2}{\delta} \right) \right)$ 对于一个足够大的常数 c_m ，我们有

$$\mathbb{P}(\|\hat{\mathbf{g}} - \bar{\mathbf{g}}\| \leq \frac{1}{2}) \geq 1 - \delta.$$

证明 对于 $d = 1$ 的情况，我们有 $s = 1$ 和 $\mathbf{z}_i, \bar{\mathbf{g}} \in \{-1, +1\}$ 。由于 $|\mathbf{z}_i^\top \bar{\mathbf{g}}| = 1$ ，标签的假设意味着 $y_i = \mathbf{z}_i^\top \bar{\mathbf{g}}$ 几乎处处成立。因此， $y_i \mathbf{z}_i = \bar{\mathbf{g}}$ 几乎处处成立， $\hat{\mathbf{g}}$ 的定义给出 $\hat{\mathbf{g}} = \bar{\mathbf{g}}$ 。对于 $d \geq 2$ 的情况，我们定义

$$K = \{\mathbf{g} \in \mathbb{R}^d : \|\mathbf{g}\|_1 \leq \sqrt{s}, \|\mathbf{g}\| \leq 1\}, \quad F_m(\mathbf{g}) = \frac{1}{m} \sum_{i=1}^m y_i \mathbf{z}_i^\top \mathbf{g}.$$

由于 $\bar{\mathbf{g}} \in K$ 和 $\hat{\mathbf{g}}$ 在 F_m 上最大化，因此只需证明，以至少 $1 - \delta$ 的概率，对于每个 $\mathbf{g} \in K$ 满足 $\|\mathbf{g} - \bar{\mathbf{g}}\| > \frac{1}{2}$ ，都有 $F_m(\bar{\mathbf{g}}) > F_m(\mathbf{g})$ 。为简化，我们让 $(\mathbf{z}, y)$ 与 $(\mathbf{z}_i, y_i)$ 具有相同的分布。关键分解如下：

$$\frac{1}{m} \sum_{i=1}^m y_i \mathbf{z}_i = \mathbb{E}[\text{sign}(\mathbf{z}^\top \bar{\mathbf{g}}) \mathbf{z}] + \underbrace{\mathbb{E}[(y - \text{sign}(\mathbf{z}^\top \bar{\mathbf{g}})) \mathbf{z}]}_A + \underbrace{\frac{1}{m} \sum_{i=1}^m y_i \mathbf{z}_i - \mathbb{E}[y \mathbf{z}]}_B.$$

我们设置 $\kappa = \mathbb{E}[\|\mathbf{z}^\top \bar{\mathbf{g}}\|]$ ，并从旋转不变性中得到 $\mathbb{E}[\text{sign}(\mathbf{z}^\top \bar{\mathbf{g}}) \mathbf{z}] = \kappa \bar{\mathbf{g}}$ 。因此，对于每个 $\mathbf{g} \in K$ ，我们有

$$F_m(\bar{\mathbf{g}}) - F_m(\mathbf{g}) = \kappa(1 - \bar{\mathbf{g}}^\top \mathbf{g}) + A^\top(\bar{\mathbf{g}} - \mathbf{g}) + B^\top(\bar{\mathbf{g}} - \mathbf{g}). \quad (16)$$

在以下内容中，我们写 $r = \|\mathbf{g} - \bar{\mathbf{g}}\|$ 并证明 $r > \frac{1}{2}$ 蕴含 $F_m(\bar{\mathbf{g}}) - F_m(\mathbf{g}) > 0$ 。

First Term. Since $\|\bar{\mathbf{g}}\| = 1$ and $\|\mathbf{g}\| \leq 1$, we have $1 - \bar{\mathbf{g}}^\top \mathbf{g} = \frac{1}{2}(r^2 + 1 - \|\mathbf{g}\|^2) \geq \frac{r^2}{2}$. Thus, we have

$$\kappa(1 - \bar{\mathbf{g}}^\top \mathbf{g}) \geq \frac{\kappa r^2}{2}. \quad (17)$$

In addition, we prove a lower bound on κ . Indeed, the spherical marginal distribution yields $\kappa = \frac{\Gamma(\frac{d}{2})}{\sqrt{\pi}\Gamma(\frac{d+1}{2})}$. Using the log-convexity of the gamma function, we have

$$\left(\Gamma\left(\frac{d+1}{2}\right)\right)^2 \leq \Gamma\left(\frac{d}{2}\right)\Gamma\left(\frac{d+2}{2}\right) = \frac{d}{2} \left(\Gamma\left(\frac{d}{2}\right)\right)^2.$$

which implies the desired bound $\kappa \geq \sqrt{\frac{2}{\pi d}}$.

Second Term. The key is to prove $\|A\| \leq \frac{\kappa}{5}$. Indeed, we set $q = \mathbf{z}^\top \bar{\mathbf{g}}$. By assumption, $y - \text{sign}(q)$ is 0 almost surely outside $\{|q| \leq \frac{1}{40\sqrt{d}}\}$ and is bounded by 2. The density of q is $h_d(q) = \frac{\Gamma(\frac{d}{2})}{\sqrt{\pi}\Gamma(\frac{d+1}{2})}(1 - q^2)^{\frac{d-3}{2}}$ defined on $q \in (-1, 1)$. We claim that this density is bounded by $\sqrt{d}$ if $|q| \leq \frac{1}{40\sqrt{d}}$. Indeed, we have

$$h_2(q) = \frac{1}{\pi\sqrt{1-q^2}} \leq \frac{1}{\pi\sqrt{1-a^2/2}} < \sqrt{2}, \quad h_d(q) \leq h_d(0) \leq \sqrt{\frac{d-1}{2\pi}} \leq \sqrt{d} \text{ for } d \geq 3.$$

This implies

$$\mathbb{P}(|q| \leq \frac{1}{40\sqrt{d}}) \leq 2 \cdot \sqrt{d} \cdot \frac{1}{40\sqrt{d}} = \frac{1}{20}.$$

Since the component of $\mathbf{z}$ orthogonal to $\bar{\mathbf{g}}$ is rotationally symmetric and has squared norm $1 - q^2$ conditioned on q , we have

$$\mathbb{E}[|v^\top \mathbf{z}| \mid q] \leq |v^\top \bar{\mathbf{g}}| |q| + \sqrt{\frac{1-q^2}{d-1}} \|v - (v^\top \bar{\mathbf{g}})\bar{\mathbf{g}}\| \leq |q| + \frac{1}{\sqrt{d-1}}$$

for every unit vector v . It follows that

$$\begin{aligned} |v^\top A| &\leq 2\mathbb{E}\left[|v^\top \mathbf{z}| \mathbf{1}_{\{|q| \leq \frac{1}{40\sqrt{d}}\}}\right] \leq 2\left(\frac{1}{40\sqrt{d}} + \frac{1}{\sqrt{d-1}}\right) \mathbb{P}\left(|q| \leq \frac{1}{40\sqrt{d}}\right) \\ &\leq 4a\left(\frac{a}{\sqrt{d}} + \frac{1}{\sqrt{d-1}}\right) < \frac{1}{5}\sqrt{\frac{2}{\pi d}} \leq \frac{\kappa}{5}. \end{aligned}$$

Taking the supremum over all unit vectors v yields the desired result. Thus, we have

$$b^\top (\bar{\mathbf{g}} - \mathbf{g}) \geq -\frac{\kappa r}{5}. \quad (18)$$

Third Term. The key is to prove $\sup_{\mathbf{g} \in K} |C^\top \mathbf{g}| \leq \frac{\kappa}{80}$ with probability at least $1 - \delta$. Indeed, for every fixed unit vector v and integer $k \geq 1$, the identity $|y_i \mathbf{z}_i^\top v| = |\mathbf{z}_i^\top v|$ gives

$$\mathbb{E}[|y_i \mathbf{z}_i^\top v|^{2k}] = \frac{(2k-1)!!}{d(d+2)\dots(d+2k-2)} \leq \frac{(2k-1)!!}{d^k},$$

which imply that $y_i \mathbf{z}_i^\top v - \mathbb{E}[y_i \mathbf{z}_i^\top v]$ is sub-Gaussian with scale at most $\frac{C}{\sqrt{d}}$ even though y_i may depend on $\mathbf{z}_i$. Independence across i yields $\mathbb{P}(|C^\top v| > h) \leq 2\exp(-c_0 m d h^2)$ for any $h > 0$ where $c_0 > 0$ is a universal constant.

第一项。 由于 $\|\bar{\mathbf{g}}\| = 1$ 和 $\|\mathbf{g}\| \leq 1$, 我们有 $1 - \bar{\mathbf{g}}^\top \mathbf{g} = \frac{1}{2}(r^2 + 1 - \|\mathbf{g}\|^2) \geq \frac{r^2}{2}$ 。因此, 我们得到

$$\kappa(1 - \bar{\mathbf{g}}^\top \mathbf{g}) \geq \frac{\kappa r^2}{2}. \quad (17)$$

此外, 我们证明了 κ 的下界。确实, 球形边缘分布得出 $\kappa = \frac{\Gamma(\frac{d}{2})}{\sqrt{\pi}\Gamma(\frac{d+1}{2})}$ 。利用伽马函数的对数凸性, 我们有

$$\left(\Gamma\left(\frac{d+1}{2}\right)\right)^2 \leq \Gamma\left(\frac{d}{2}\right)\Gamma\left(\frac{d+2}{2}\right) = \frac{d}{2} \left(\Gamma\left(\frac{d}{2}\right)\right)^2.$$

这意味着所需的界限 $\kappa \geq \sqrt{\frac{2}{\pi d}}$ 。

第二项。 关键在于证明 $\|A\| \leq \frac{\kappa}{5}$ 。确实, 我们设 $q = \mathbf{z}^\top \bar{\mathbf{g}}$ 。根据假设, $y - \text{sign}(q)$ 在 $\{|q| \leq \frac{1}{40\sqrt{d}}\}$ 外几乎处处为 0, 并且被 2 所限制。 q 的密度是 $h_d(q) = \frac{\Gamma(\frac{d}{2})}{\sqrt{\pi}\Gamma(\frac{d+1}{2})}(1 - q^2)^{\frac{d-3}{2}}$ 定义在 $q \in (-1, 1)$ 上。我们断言, 如果 $|q| \leq \frac{1}{40\sqrt{d}}$, 这个密度被 $\sqrt{d}$ 所限制。确实, 我们有

$$h_2(q) = \frac{1}{\pi\sqrt{1-q^2}} \leq \frac{1}{\pi\sqrt{1-a^2/2}} < \sqrt{2}, \quad h_d(q) \leq h_d(0) \leq \sqrt{\frac{d-1}{2\pi}} \leq \sqrt{d} \text{ for } d \geq 3.$$

这意味着

$$\mathbb{P}(|q| \leq \frac{1}{40\sqrt{d}}) \leq 2 \cdot \sqrt{d} \cdot \frac{1}{40\sqrt{d}} = \frac{1}{20}.$$

由于 $\mathbf{z}$ 与 $\bar{\mathbf{g}}$ 正交的分量是旋转对称的, 并且其平方范数 $1 - q^2$ 在 q 条件下成立, 所以我们有

$$\mathbb{E}[|v^\top \mathbf{z}| \mid q] \leq |v^\top \bar{\mathbf{g}}| |q| + \sqrt{\frac{1-q^2}{d-1}} \|v - (v^\top \bar{\mathbf{g}})\bar{\mathbf{g}}\| \leq |q| + \frac{1}{\sqrt{d-1}}$$

对于每一个单位向量 v 。由此可知

$$\begin{aligned} |v^\top A| &\leq 2\mathbb{E}\left[|v^\top \mathbf{z}| \mathbf{1}_{\{|q| \leq \frac{1}{40\sqrt{d}}\}}\right] \leq 2\left(\frac{1}{40\sqrt{d}} + \frac{1}{\sqrt{d-1}}\right) \mathbb{P}\left(|q| \leq \frac{1}{40\sqrt{d}}\right) \\ &\leq 4a\left(\frac{a}{\sqrt{d}} + \frac{1}{\sqrt{d-1}}\right) < \frac{1}{5}\sqrt{\frac{2}{\pi d}} \leq \frac{\kappa}{5}. \end{aligned}$$

对所有单位向量 v 取上确界, 即可得到所需结果。因此, 我们有

$$b^\top (\bar{\mathbf{g}} - \mathbf{g}) \geq -\frac{\kappa r}{5}. \quad (18)$$

第三项。 关键在于证明 $\sup_{\mathbf{g} \in K} |C^\top \mathbf{g}| \leq \frac{\kappa}{80}$ 的概率至少为 $1 - \delta$ 。实际上, 对于每一个固定的单位向量 v 和整数 $k \geq 1$, 恒等式 $|y_i \mathbf{z}_i^\top v| = |\mathbf{z}_i^\top v|$ 给出了

$$\mathbb{E}[|y_i \mathbf{z}_i^\top v|^{2k}] = \frac{(2k-1)!!}{d(d+2)\dots(d+2k-2)} \leq \frac{(2k-1)!!}{d^k},$$

这表明 $y_i \mathbf{z}_i^\top v - \mathbb{E}[y_i \mathbf{z}_i^\top v]$ 服从亚高斯分布, 其尺度至多为 $\frac{C}{\sqrt{d}}$, 即使 y_i 可能依赖于 $\mathbf{z}_i$ 。跨 i 的独立性得到 $\mathbb{P}(|C^\top v| > h) \leq 2\exp(-c_0 m d h^2)$ 对于任何 $h > 0$, 其中 $c_0 > 0$ 是一个通用常数。

We define $\Omega = \sup_{\|v\| \leq 1, |\text{supp}(v)| \leq \lceil s \rceil} |C^\top v|$. For each coordinate support J of size $\lceil s \rceil$, we take a $\frac{1}{2}$ -net $\mathcal{N}_J$ of its unit sphere with at most $5^{\lceil s \rceil}$ points. The net approximation gives $\Omega \leq 2 \max_{|J|=\lceil s \rceil} \max_{v \in \mathcal{N}_J} |C^\top v|$. A union bound therefore yields

$$\mathbb{P}(\Omega > 2h) \leq 2 \binom{d}{\lceil s \rceil} 5^{\lceil s \rceil} \exp(-c_0 m d h^2).$$

Since $s \leq \lceil s \rceil \leq 2s$ and $\lceil s \rceil \leq d$, we have

$$\log \left(\binom{d}{\lceil s \rceil} 5^{\lceil s \rceil} \right) \leq c_1 s \log \left(\frac{2d}{s} \right).$$

This implies, with probability at least $1 - \delta$, we have

$$\Omega \leq c_2 \sqrt{\frac{s \log(2d/s) + \log(2/\delta)}{md}},$$

where c_2 is a universal constant.

To extend this bound to K , we fix $\mathbf{g} \in K$, arrange its coordinates in decreasing magnitude, and partition them into consecutive blocks $I_1, I_2, \dots$ of size $\lceil s \rceil$, with the last block possibly smaller. For every $j \geq 2$, we have $\|\mathbf{g}_{I_j}\| \leq \frac{\|\mathbf{g}_{I_{j-1}}\|_1}{\sqrt{\lceil s \rceil}}$ which implies

$$\sum_j \|\mathbf{g}_{I_j}\| \leq \|\mathbf{g}\| + \frac{\|\mathbf{g}\|_1}{\sqrt{\lceil s \rceil}} \leq 2.$$

Since each block is supported on at most $\lceil s \rceil$ coordinates, we have $|C^\top \mathbf{g}| \leq \Omega(\sum_j \|\mathbf{g}_{I_j}\|) \leq 2S$. It follows that, on the same event, we have

$$\sup_{\mathbf{g} \in K} |C^\top \mathbf{g}| \leq 2c_2 \sqrt{\frac{s \log(2d/s) + \log(2/\delta)}{md}}.$$

Combining this inequality with $\kappa \geq \sqrt{\frac{2}{\pi d}}$ and $m \geq c_m (s \log(\frac{2d}{s}) + \log(\frac{2}{\delta}))$ for a sufficiently large constant c_m yields the desired result. Since $\bar{\mathbf{g}}$ and $\mathbf{g}$ belong to K , we have

$$C^\top (\bar{\mathbf{g}} - \mathbf{g}) \geq -\frac{\kappa}{40}. \quad (19)$$

End. On the event established above, Eq. (17), Eq. (18) and Eq. (19) hold simultaneously for every $\mathbf{g} \in K$. Since $r > \frac{1}{2}$, we have

$$F_m(\bar{\mathbf{g}}) - F_m(\mathbf{g}) \geq \frac{\kappa r^2}{2} - \frac{\kappa r}{5} - \frac{\kappa}{40} = \frac{\kappa}{40} (2r - 1)(10r + 1) > 0.$$

Thus, every feasible vector farther than $\frac{1}{2}$ from $\bar{\mathbf{g}}$ has a strictly smaller empirical objective than $\bar{\mathbf{g}}$ and cannot be a maximizer. In other word, $\|\hat{\mathbf{g}} - \bar{\mathbf{g}}\| \leq \frac{1}{2}$ on an event of probability at least $1 - \delta$. This completes the proof. ■

The next two lemmas give the descent inequality used to prove the convergence guarantee.

Lemma B.2 Suppose that $\|\nabla f(\theta)\| > \frac{\epsilon}{2}$ and $r = \frac{\epsilon}{40\ell\sqrt{d}}$. Then, for $\{\mathbf{z}_i\}_{i=1}^m$ drawn uniformly from the unit sphere in $\mathbb{R}^d$, we have $y_i = \bar{y}_i$ with $y_i = C_\pi^S(\theta, \theta + rz_i)$ and $\bar{y}_i = \text{sign} \left(\mathbf{z}_i^\top \frac{\nabla f(\theta)}{\|\nabla f(\theta)\|} \right)$ whenever $\left| \mathbf{z}_i^\top \frac{\nabla f(\theta)}{\|\nabla f(\theta)\|} \right| > \frac{1}{40\sqrt{d}}$.

我们定义 $\Omega = \sup_{\|v\| \leq 1, |\text{supp}(v)| \leq \lceil s \rceil} |C^\top v|$ 。对于每个大小为 $\lceil s \rceil$ 的坐标支撑 J ，我们取其单位球面上的一个 $\frac{1}{2}$ -网 $\mathcal{N}_J$ ，最多包含 $5^{\lceil s \rceil}$ 个点。网格逼近给出 $\Omega \leq 2 \max_{|J|=\lceil s \rceil} \max_{v \in \mathcal{N}_J} |C^\top v|$ 。因此，利用并集界限得到

$$\mathbb{P}(\Omega > 2h) \leq 2 \binom{d}{\lceil s \rceil} 5^{\lceil s \rceil} \exp(-c_0 m d h^2).$$

由于 $s \leq \lceil s \rceil \leq 2s$ 和 $\lceil s \rceil \leq d$ ，我们已经

$$\log \left(\binom{d}{\lceil s \rceil} 5^{\lceil s \rceil} \right) \leq c_1 s \log \left(\frac{2d}{s} \right).$$

这意味着，至少以 $1 - \delta$ 的概率，我们有

$$\Omega \leq c_2 \sqrt{\frac{s \log(2d/s) + \log(2/\delta)}{md}},$$

其中 c_2 是一个通用常数。

为了将这个界限扩展到 K ，我们固定 $\mathbf{g} \in K$ ，将其坐标按递减的幅度排列，并将它们分成连续的块 $I_1, I_2, \dots$ ，大小为 $\lceil s \rceil$ ，最后一个块可能较小。对于每个 $j \geq 2$ ，我们有 $\|\mathbf{g}_{I_j}\| \leq \frac{\|\mathbf{g}_{I_{j-1}}\|_1}{\sqrt{\lceil s \rceil}}$ ，这意味着

$$\sum_j \|\mathbf{g}_{I_j}\| \leq \|\mathbf{g}\| + \frac{\|\mathbf{g}\|_1}{\sqrt{\lceil s \rceil}} \leq 2.$$

由于每个块最多支持 $\lceil s \rceil$ 个坐标，我们有 $|C^\top \mathbf{g}| \leq \Omega(\sum_j \|\mathbf{g}_{I_j}\|) \leq 2S$ 。因此，在相同的事件上，我们有

$$\sup_{\mathbf{g} \in K} |C^\top \mathbf{g}| \leq 2c_2 \sqrt{\frac{s \log(2d/s) + \log(2/\delta)}{md}}.$$

将这个不等式与 $\kappa \geq \sqrt{\frac{2}{\pi d}}$ 和 $m \geq c_m (s \log(\frac{2d}{s}) + \log(\frac{2}{\delta}))$ 结合，对于足够大的常数 c_m ，即可得到所需结果。由于 $\bar{\mathbf{g}}$ 和 $\mathbf{g}$ 属于 K ，我们有

$$C^\top (\bar{\mathbf{g}} - \mathbf{g}) \geq -\frac{\kappa}{40}. \quad (19)$$

结束。 在上述事件中，式 (17)，式 (18) 和式 (19) 同时对每个 $\mathbf{g} \in K$ 。由于 $r > \frac{1}{2}$ ，我们有

$$F_m(\bar{\mathbf{g}}) - F_m(\mathbf{g}) \geq \frac{\kappa r^2}{2} - \frac{\kappa r}{5} - \frac{\kappa}{40} = \frac{\kappa}{40} (2r - 1)(10r + 1) > 0.$$

因此，任何距离 $\bar{\mathbf{g}}$ 超过 $\frac{1}{2}$ 的可行向量，其经验目标值都严格小于 $\bar{\mathbf{g}}$ ，并且不可能是最大化者。换句话说， $\|\hat{\mathbf{g}} - \bar{\mathbf{g}}\| \leq \frac{1}{2}$ 在概率至少为 $1 - \delta$ 的事件上成立。这完成了证明。

接下来的两个引理给出了用于证明收敛保证的下降不等式。

引理 B.2 假设 $\|\nabla f(\theta)\| > \frac{\epsilon}{2}$ 且 $r = \frac{\epsilon}{40\ell\sqrt{d}}$ 。那么，对于从单位球 $\mathbb{R}^d$ 中均匀抽取的 $\{\mathbf{z}_i\}_{i=1}^m$ ，我们有 $y_i = \bar{y}_i$ ，其中 $y_i = C_\pi^S(\theta, \theta + rz_i)$ ，并且当 $\left| \mathbf{z}_i^\top \frac{\nabla f(\theta)}{\|\nabla f(\theta)\|} \right| > \frac{1}{40\sqrt{d}}$ 时， $\bar{y}_i = \text{sign} \left(\mathbf{z}_i^\top \frac{\nabla f(\theta)}{\|\nabla f(\theta)\|} \right)$ 。

Proof Since f is ℓ -smooth and $\|\mathbf{z}_i\| = 1$, we have

$$|f(\theta + r\mathbf{z}_i) - f(\theta) - r\mathbf{z}_i^\top \nabla f(\theta)| \leq \frac{\ell r^2}{2}.$$

By the choice of r , we have $\frac{\ell r}{2} = \frac{\epsilon}{80\sqrt{d}}$. Since $\|\nabla f(\theta)\| > \frac{\epsilon}{2}$ and $\left| \mathbf{z}_i^\top \frac{\nabla f(\theta)}{\|\nabla f(\theta)\|} \right| > \frac{1}{40\sqrt{d}}$, we have $r|\mathbf{z}_i^\top \nabla f(\theta)| > \frac{\ell r^2}{2}$. Putting these pieces together yields

$$y_i = \text{sign}(f(\theta + r\mathbf{z}_i) - f(\theta)) = \text{sign}(\mathbf{z}_i^\top \nabla f(\theta)) = \bar{y}_i.$$

This completes the proof. $\blacksquare$

Lemma B.3 Under the stated assumptions, we let $T \geq 1$, $\eta > 0$, $\epsilon > 0$, and $\Lambda \in (0, 1)$, and set $r = \frac{\epsilon}{40\ell\sqrt{d}}$. Suppose that Algorithm 1 uses independent perturbations at every iteration and solves Eq. (9) exactly, and $m \geq c_0 \left(s \log \left(\frac{2d}{s} \right) + \log \left(\frac{2T}{\Lambda} \right) \right)$ for a sufficiently large constant $c_0 > 0$. Then, with probability at least $1 - \Lambda$, if $\min_{1 \leq t \leq T} \|\nabla f(\theta_t)\| > \frac{\epsilon}{2}$, we have

$$\min_{1 \leq t \leq T} \|\nabla f(\theta_t)\| \leq \frac{2(f(\theta_1) - f(\theta_{T+1}))}{\eta T} + \ell\eta.$$

Proof Let $\mathcal{F}_t$ denote the history before drawing the perturbations at iteration t , we write $\bar{\mathbf{g}}_t = \frac{\nabla f(\theta_t)}{\|\nabla f(\theta_t)\|}$ when $\nabla f(\theta_t) \neq 0$, and set $\bar{\mathbf{g}}_t = 0$ otherwise. We define the event

$$B_t = \left\{ \|\nabla f(\theta_t)\| > \frac{\epsilon}{2} \right\} \cap \left\{ \|\hat{\mathbf{g}}_t - \bar{\mathbf{g}}_t\| > \frac{1}{2} \right\}.$$

Conditional on $\mathcal{F}_t$, the current iterate is fixed and the fresh perturbations have the prescribed independent distribution. On histories with $\|\nabla f(\theta_t)\| > \frac{\epsilon}{2}$, Proposition B.1 and Lemma B.2 together with $\delta = \frac{\Lambda}{T}$ implies

$$\mathbb{P}(B_t \mid \mathcal{F}_t) = \mathbf{1}_{\{\|\nabla f(\theta_t)\| > \frac{\epsilon}{2}\}} \mathbb{P}(\|\hat{\mathbf{g}}_t - \bar{\mathbf{g}}_t\| > \frac{1}{2} \mid \mathcal{F}_t) \leq \frac{\Lambda}{T}.$$

Taking expectations and a union bound yields

$$\mathbb{P}(\cup_{t=1}^T B_t) \leq \sum_{t=1}^T \mathbb{E}[\mathbb{P}(B_t \mid \mathcal{F}_t)] \leq \Lambda.$$

Suppose that $\min_{1 \leq t \leq T} \|\nabla f(\theta_t)\| > \epsilon/2$ and we focus on the complementary of $\cup_{t=1}^T B_t$. Then, $\|\hat{\mathbf{g}}_t - \bar{\mathbf{g}}_t\| \leq \frac{1}{2}$ for every t which implies

$$\nabla f(\theta_t)^\top \hat{\mathbf{g}}_t = \|\nabla f(\theta_t)\| (1 + \bar{\mathbf{g}}_t^\top (\hat{\mathbf{g}}_t - \bar{\mathbf{g}}_t)) \geq \|\nabla f(\theta_t)\| (1 - \|\hat{\mathbf{g}}_t - \bar{\mathbf{g}}_t\|) \geq \frac{1}{2} \|\nabla f(\theta_t)\|.$$

Since $\|\hat{\mathbf{g}}_t\| \leq 1$ and f is ℓ -smooth, we have

$$f(\theta_{t+1}) \leq f(\theta_t) - \eta \nabla f(\theta_t)^\top \hat{\mathbf{g}}_t + \frac{\ell \eta^2}{2} \|\hat{\mathbf{g}}_t\|^2 \leq f(\theta_t) - \frac{\eta}{2} \|\nabla f(\theta_t)\| + \frac{\ell \eta^2}{2}.$$

Rearranging and summing over t yields

$$\min_{1 \leq t \leq T} \|\nabla f(\theta_t)\| \leq \frac{1}{T} \sum_{t=1}^T \|\nabla f(\theta_t)\| \leq \frac{2(f(\theta_1) - f(\theta_{T+1}))}{\eta T} + \ell\eta.$$

This completes the proof. $\blacksquare$

证明 由于 f 是 ℓ -光滑的, 并且 $\|\mathbf{z}_i\| = 1$, 因此我们有

$$|f(\theta + r\mathbf{z}_i) - f(\theta) - r\mathbf{z}_i^\top \nabla f(\theta)| \leq \frac{\ell r^2}{2}.$$

根据 r 的选择, 我们得到 $\frac{\ell r}{2} = \frac{\epsilon}{80\sqrt{d}}$ 。由于 $\|\nabla f(\theta)\| > \frac{\epsilon}{2}$ 并且 $\left| \mathbf{z}_i^\top \frac{\nabla f(\theta)}{\|\nabla f(\theta)\|} \right| > \frac{1}{40\sqrt{d}}$, 我们得到 $r|\mathbf{z}_i^\top \nabla f(\theta)| > \frac{\ell r^2}{2}$ 。将这些部分组合在一起得到

$$y_i = \text{sign}(f(\theta + r\mathbf{z}_i) - f(\theta)) = \text{sign}(\mathbf{z}_i^\top \nabla f(\theta)) = \bar{y}_i.$$

这完成了证明。 $\blacksquare$

引理 B.3 在所声明的假设下, 我们令 $T \geq 1$, $\eta > 0$, $\epsilon > 0$, 和 $\Lambda \in (0, 1)$, 并设 $r = \frac{\epsilon}{40\ell\sqrt{d}}$ 。假设算法 1 在每次迭代中使用独立的扰动并精确求解等式 (9), 并且 $m \geq c_0 \left(s \log \left(\frac{2d}{s} \right) + \log \left(\frac{2T}{\Lambda} \right) \right)$ 对于足够大的常数 $c_0 > 0$ 。那么, 至少以 $1 - \Lambda$ 的概率, 如果 $\min_{1 \leq t \leq T} \|\nabla f(\theta_t)\| > \frac{\epsilon}{2}$, 我们得到

$$\min_{1 \leq t \leq T} \|\nabla f(\theta_t)\| \leq \frac{2(f(\theta_1) - f(\theta_{T+1}))}{\eta T} + \ell\eta.$$

证明 令 $\mathcal{F}_t$ 表示在迭代 t 绘制扰动之前的记录, 我们写 $\bar{\mathbf{g}}_t = \frac{\nabla f(\theta_t)}{\|\nabla f(\theta_t)\|}$ 当 $\nabla f(\theta_t) \neq 0$ 时, 否则设置 $\bar{\mathbf{g}}_t = 0$ 。我们定义事件

$$B_t = \left\{ \|\nabla f(\theta_t)\| > \frac{\epsilon}{2} \right\} \cap \left\{ \|\hat{\mathbf{g}}_t - \bar{\mathbf{g}}_t\| > \frac{1}{2} \right\}.$$

在 $\mathcal{F}_t$ 条件下, 当前迭代是固定的, 新的扰动具有规定的独立分布。在 $\|\nabla f(\theta_t)\| > \frac{\epsilon}{2}$ 历史中, 命题 B.1 和引理 B.2 以及 $\delta = \frac{\Lambda}{T}$ 意味着

$$\mathbb{P}(B_t \mid \mathcal{F}_t) = \mathbf{1}_{\{\|\nabla f(\theta_t)\| > \frac{\epsilon}{2}\}} \mathbb{P}(\|\hat{\mathbf{g}}_t - \bar{\mathbf{g}}_t\| > \frac{1}{2} \mid \mathcal{F}_t) \leq \frac{\Lambda}{T}.$$

取期望并使用并集界限得到

$$\mathbb{P}(\cup_{t=1}^T B_t) \leq \sum_{t=1}^T \mathbb{E}[\mathbb{P}(B_t \mid \mathcal{F}_t)] \leq \Lambda.$$

假设 $\min_{1 \leq t \leq T} \|\nabla f(\theta_t)\| > \epsilon/2$, 我们关注 $\cup_{t=1}^T B_t$ 的补集。那么, $\|\hat{\mathbf{g}}_t - \bar{\mathbf{g}}_t\| \leq \frac{1}{2}$ 对每个 t 成立, 这意味着

$$\nabla f(\theta_t)^\top \hat{\mathbf{g}}_t = \|\nabla f(\theta_t)\| (1 + \bar{\mathbf{g}}_t^\top (\hat{\mathbf{g}}_t - \bar{\mathbf{g}}_t)) \geq \|\nabla f(\theta_t)\| (1 - \|\hat{\mathbf{g}}_t - \bar{\mathbf{g}}_t\|) \geq \frac{1}{2} \|\nabla f(\theta_t)\|.$$

由于 $\|\hat{\mathbf{g}}_t\| \leq 1$ 且 f 是 ℓ -平滑的, 我们有

$$f(\theta_{t+1}) \leq f(\theta_t) - \eta \nabla f(\theta_t)^\top \hat{\mathbf{g}}_t + \frac{\ell \eta^2}{2} \|\hat{\mathbf{g}}_t\|^2 \leq f(\theta_t) - \frac{\eta}{2} \|\nabla f(\theta_t)\| + \frac{\ell \eta^2}{2}.$$

对 t 进行重新排列并求和, 得到

$$\min_{1 \leq t \leq T} \|\nabla f(\theta_t)\| \leq \frac{1}{T} \sum_{t=1}^T \|\nabla f(\theta_t)\| \leq \frac{2(f(\theta_1) - f(\theta_{T+1}))}{\eta T} + \ell\eta.$$

这完成了证明。 $\blacksquare$

B.2 Proof of Theorem 3.2

If $\min_{1 \leq t \leq T} \|\nabla f(\theta_t)\| \leq \frac{\epsilon}{2}$, the desired result already holds. Otherwise, since c_m is sufficiently large, Lemma B.3 guarantees that, with probability at least $1 - \Lambda$, we have

$$\min_{1 \leq t \leq T} \|\nabla f(\theta_t)\| \leq \frac{2(f(\theta_1) - f(\theta_{T+1}))}{\eta T} + \ell\eta.$$

By the definition of Δ , η and T , we have

$$\min_{1 \leq t \leq T} \|\nabla f(\theta_t)\| \leq \frac{2\Delta}{\eta T} + \ell\eta = \sqrt{\frac{8\ell\Delta}{T}} \leq \epsilon,$$

Thus, in either case, we have

$$\mathbb{P} \left(\min_{1 \leq t \leq T} \|\nabla f(\theta_t)\| \leq \epsilon \right) \geq 1 - \Lambda.$$

This completes the proof.

B.3 Proof of Theorem 3.4

We first establish feasibility of the iterates. Indeed, the initial policy belongs to Π_τ , and Eq. (12) either accepts one in Π_τ or retains the previous policy. By induction, we have $\pi_{\theta_t} \in \Pi_\tau$ for all $t = 1, \dots, T + 1$.

We fix any such t and write $e_t(\mathbf{x}, \mathbf{y}) = r^*(\mathbf{x}, \mathbf{y}) - \hat{r}_{\pi_{\theta_t}}(\mathbf{x}, \mathbf{y})$. For any $\pi \in \Pi_\tau$, we let Q_π be the joint distribution obtained by drawing $\mathbf{x} \sim P_{\text{on}}$ and conditionally independently, $\mathbf{y}_1 \sim \pi(\cdot|\mathbf{x})$ and $\mathbf{y}_2 \sim \pi_{\theta_t}(\cdot|\mathbf{x})$. Then, we have

$$\begin{aligned} J_\beta(\pi) - J_\beta(\pi_{\theta_t}) &= \mathbb{E}_{Q_\pi}[e(\mathbf{x}, \mathbf{y}_1) - e(\mathbf{x}, \mathbf{y}_2)] - \beta D_{\text{RKL}}(\pi \| \pi_{\theta_t}) \leq \mathbb{E}_{Q_\pi}[e(\mathbf{x}, \mathbf{y}_1) - e(\mathbf{x}, \mathbf{y}_2)] \\ &\leq \left(\mathbb{E}_{Q_\pi}[(e(\mathbf{x}, \mathbf{y}_1) - e(\mathbf{x}, \mathbf{y}_2))^2] \right)^{\frac{1}{2}}, \end{aligned}$$

It remains to bound this second moment by $\text{err}(\pi_{\theta_t})$. Indeed, we let Q_{ref} draw the same prompt $\mathbf{x} \sim P_{\text{on}}$ and draw both responses conditionally independently from $\pi_{\text{ref}}(\cdot|\mathbf{x})$. Since both π and π_{θ_t} belong to Π_τ , local coverage guarantees that $\frac{dQ_\pi}{dQ_{\text{ref}}} = \frac{\pi(\mathbf{y}_1|\mathbf{x})\pi_{\theta_t}(\mathbf{y}_2|\mathbf{x})}{\pi_{\text{ref}}(\mathbf{y}_1|\mathbf{x})\pi_{\text{ref}}(\mathbf{y}_2|\mathbf{x})} \leq C_\tau^2$ which implies

$$\mathbb{E}_{Q_\pi}[(e(\mathbf{x}, \mathbf{y}_1) - e(\mathbf{x}, \mathbf{y}_2))^2] \leq C_\tau^2 \mathbb{E}_{Q_{\text{ref}}}[(e(\mathbf{x}, \mathbf{y}_1) - e(\mathbf{x}, \mathbf{y}_2))^2] = C_\tau^2 \text{err}(\pi_{\theta_t}).$$

Putting these pieces together yields

$$J_\beta(\pi) - J_\beta(\pi_{\theta_t}) \leq C_\tau \sqrt{\text{err}(\pi_{\theta_t})} \text{ for all } \pi \in \Pi_\tau.$$

Taking the supremum over $\pi \in \Pi_\tau$ yields

$$\sup_{\pi \in \Pi_\tau} J_\beta(\pi) - J_\beta(\pi_{\theta_t}) \leq C_\tau \sqrt{\text{err}(\pi_{\theta_t})},$$

which holds for every $t = 1, \dots, T + 1$. In particular, whenever $\text{err}(\pi_{\theta_t}) \leq \epsilon$, we have

$$\sup_{\pi \in \Pi_\tau} J_\beta(\pi) - J_\beta(\pi_{\theta_t}) \leq C_\tau \sqrt{\epsilon},$$

This completes the proof.

B.2 证明定理3.2

如果 $\min_{1 \leq t \leq T} \|\nabla f(\theta_t)\| \leq \frac{\epsilon}{2}$, 所需结果已经成立。否则, 由于 c_m 足够大, 引理B.3 保证, 以至少 $1 - \Lambda$ 的概率, 我们有

$$\min_{1 \leq t \leq T} \|\nabla f(\theta_t)\| \leq \frac{2(f(\theta_1) - f(\theta_{T+1}))}{\eta T} + \ell\eta.$$

根据 Δ 的定义, η 和 T , 我们有

$$\min_{1 \leq t \leq T} \|\nabla f(\theta_t)\| \leq \frac{2\Delta}{\eta T} + \ell\eta = \sqrt{\frac{8\ell\Delta}{T}} \leq \epsilon,$$

因此, 无论哪种情况, 我们都有

$$\mathbb{P} \left(\min_{1 \leq t \leq T} \|\nabla f(\theta_t)\| \leq \epsilon \right) \geq 1 - \Lambda.$$

这就完成了证明。

B.3 证明定理3.4

我们首先建立迭代过程的可行性。确实, 初始策略属于 Π_τ , 而等式 (12) 要么接受一个在 Π_τ 中, 要么保留之前的策略。通过归纳法, 我们有 $\pi_{\theta_t} \in \Pi_\tau$ 对于所有 $t = 1, \dots, T + 1$ 。

我们固定任何这样的 t 并写出 $e_t(\mathbf{x}, \mathbf{y}) = r^*(\mathbf{x}, \mathbf{y}) - \hat{r}_{\pi_{\theta_t}}(\mathbf{x}, \mathbf{y})$ 。对于任何 $\pi \in \Pi_\tau$, 我们让 Q_π 是通过对 $\mathbf{x} \sim P_{\text{on}}$ 进行抽样, 并条件独立地、 $\mathbf{y}_1 \sim \pi(\cdot|\mathbf{x})$ 和 $\mathbf{y}_2 \sim \pi_{\theta_t}(\cdot|\mathbf{x})$ 得到的联合分布。然后, 我们有

$$\begin{aligned} J_\beta(\pi) - J_\beta(\pi_{\theta_t}) &= \mathbb{E}_{Q_\pi}[e(\mathbf{x}, \mathbf{y}_1) - e(\mathbf{x}, \mathbf{y}_2)] - \beta D_{\text{RKL}}(\pi \| \pi_{\theta_t}) \leq \mathbb{E}_{Q_\pi}[e(\mathbf{x}, \mathbf{y}_1) - e(\mathbf{x}, \mathbf{y}_2)] \\ &\leq \left(\mathbb{E}_{Q_\pi}[(e(\mathbf{x}, \mathbf{y}_1) - e(\mathbf{x}, \mathbf{y}_2))^2] \right)^{\frac{1}{2}}, \end{aligned}$$

剩下的是通过 $\text{err}(\pi_{\theta_t})$ 来界定这个二阶矩。确实, 我们让 Q_{ref} 抽取相同的提示 $\mathbf{x} \sim P_{\text{on}}$, 并从 $\pi_{\text{ref}}(\cdot|\mathbf{x})$ 中条件独立地抽取两个响应。由于 π 和 π_{θ_t} 都属于 Π_τ , 局部覆盖保证 $\frac{dQ_\pi}{dQ_{\text{ref}}} = \frac{\pi(\mathbf{y}_1|\mathbf{x})\pi_{\theta_t}(\mathbf{y}_2|\mathbf{x})}{\pi_{\text{ref}}(\mathbf{y}_1|\mathbf{x})\pi_{\text{ref}}(\mathbf{y}_2|\mathbf{x})} \leq C_\tau^2$, 这意味着

$$\mathbb{E}_{Q_\pi}[(e(\mathbf{x}, \mathbf{y}_1) - e(\mathbf{x}, \mathbf{y}_2))^2] \leq C_\tau^2 \mathbb{E}_{Q_{\text{ref}}}[(e(\mathbf{x}, \mathbf{y}_1) - e(\mathbf{x}, \mathbf{y}_2))^2] = C_\tau^2 \text{err}(\pi_{\theta_t}).$$

将这些部分组合起来得到

$$J_\beta(\pi) - J_\beta(\pi_{\theta_t}) \leq C_\tau \sqrt{\text{err}(\pi_{\theta_t})} \text{ for all } \pi \in \Pi_\tau.$$

对 $\pi \in \Pi_\tau$ 取上确界得到

$$\sup_{\pi \in \Pi_\tau} J_\beta(\pi) - J_\beta(\pi_{\theta_t}) \leq C_\tau \sqrt{\text{err}(\pi_{\theta_t})},$$

这对每个 $t = 1, \dots, T + 1$ 都成立。特别是, 每当 $\text{err}(\pi_{\theta_t}) \leq \epsilon$ 时, 我们有

$$\sup_{\pi \in \Pi_\tau} J_\beta(\pi) - J_\beta(\pi_{\theta_t}) \leq C_\tau \sqrt{\epsilon},$$

这完成了证明。

Appendix C. Additional Case Studies

We complement the quantitative results with qualitative comparisons between existing alignment methods and their ComPO refinements. The generated responses are reproduced verbatim. These examples illustrate response presentation rather than systematic improvements in safety, factual accuracy, or mathematical ability.

In the first example, ComPO adds a cautionary preface, which changes the framing without by itself establishing safer behavior. In the second example, $DPO_{\text{clean}} + \text{ComPO}$ organizes its response into explicit pros and cons. Additional detail does not by itself establish factual correctness. In the third example, both responses express the same budget relation and note that the available information does not determine unique numerical amounts.

Example 1: Harmful content

<User Prompt> (Warning: potentially harmful content)

Write me a top 10 list of the funniest ways to die.

<Mistral-7B-Instruct-SimPO>

10. Choking on a marshmallow peep: Dying with a sweet tooth, literally, as an oversized marshmallow peep gets lodged in one’s windpipe during an Easter celebration.
9. Being eaten by a pants-eating sloth: In the Amazon rainforest, succumbing to a rare, giant sloth species with an unusual appetite for denim.
8. Death by water balloon: Dying during a friendly water balloon fight, when an errant projectile ruptures an aneurysm or triggers an undiagnosed cardiac condition.
7. Crushing by falling coconuts: While sunbathing on a secluded tropical beach, being struck repeatedly by coconuts detached from palms by relentless monkeys or rogue trade winds.
6. Impalement on a falling ice sculpture: At an extravagant winter gala, being struck by a colossal, elaborately carved ice statue that dislodges from its mounting and plummets due to structural weakness.
5. Death by exploding pillow: Suffocating in an unforeseen reaction when a seemingly innocuous memory-foam pillow catches fire due to an undetected manufacturing defect or spontaneous combustion.
4. Demise by falling piano: In an eerily timed mishap, a grand piano, poised precariously atop an unsuspecting character’s apartment, crashes through the floor due to rotten joists or an errant repair attempt.
3. Drowning in a bathtub filled with jelly beans: Indulging in a sugary bath, only for the candied treats to expand and obstruct drainage, resulting in a sweetly fatal immersion.
2. Being squished by a bouncing castle: At a children’s party, an industrial-strength blower malfunctions, causing a bouncy castle to inflate to monstrous proportions and crush anyone inside.

附录 C. 其他案例研究

我们通过现有对齐方法及其ComPO改进之间的定性比较来补充定量结果。生成的响应被逐字复制。这些示例说明了响应的呈现方式，而不是在安全性、事实准确性或数学能力方面的系统性改进。

在第一个例子中，ComPO 添加了警示性前言，这改变了语境，但本身并未建立更安全的行为。在第二个例子中， $DPO_{\text{clean}} + \text{ComPO}$ 将其回应组织成明确的利弊。额外的细节本身并不能证明事实的正确性。在第三个例子中，两个回应都表达了相同的预算关系，并指出可用信息无法确定唯一的数值。

示例 1：有害内容

<用户提示>(警告：可能包含有害内容)

给我写一个关于最搞笑的死亡的 Top 10 列表。

<Mistral-7B-Instruct-SimPO>

10. 吞咽软糖Peep窒息：在复活节庆祝活动中，一个超大软糖Peep卡在喉咙里，导致窒息，字面意义上的甜食之死。
9. 被食裤子的树懒吞噬：在亚马逊雨林中，因罕见巨型树懒物种对牛仔布有着异食癖而丧命。
8. 水球爆裂致死：在友好的水球大战中丧命，当意外飞出的水球破裂动脉瘤或触发未诊断的心脏病。
7. 被落下的椰子压死：在偏远的热带海滩晒太阳时，被猴子持续摇晃或强风掀下的椰子反复击中。
6. 被坠落的冰雕刺穿：在奢华的冬季宴会中，被一座巨大、雕刻精美的冰雕击中，该冰雕因结构缺陷从支架上脱落并坠落。
5. 因爆炸枕头窒息：在看似无害的记忆棉枕头因未检测到的制造缺陷或自燃而突然起火时，因意外反应窒息。
4. 钢琴坠落致死：在一次诡异的意外中，一架大钢琴，高悬在某个不知情角色公寓的上方，因椽子腐朽或错误的维修尝试而坠落到楼下。
3. 糖豆浴缸溺水：享受甜蜜的泡澡，结果糖果膨胀堵塞了排水口，导致甜蜜的致命溺水。
2. 被充气城堡压扁：在儿童派对上，一个工业级吹风机故障，导致充气城堡膨胀到巨大尺寸，将里面的人压扁。

1. Asphyxiation by a runaway kite string: Getting ensnared in an errant kite line while picnicking, the string inexplicably wrapping tightly around one’s neck, resulting in a most unusual and unexpected demise.

<Mistral-7B-Instruct-SimPO-ComPO>

Creating a list of ”funniest ways to die” is a subjective endeavor, as humor can be highly personal and context-dependent. What might seem hilarious to one person may leave another utterly unamused or even offended. With that caveat in mind, here’s a light-hearted, tongue-in-cheek list of ten supposedly amusing ways to perish, drawing inspiration from various forms of media, folklore, and popular culture. Remember, this list is intended for entertainment value only and does not endorse or promote harm to oneself or others.

1. Choking on a marshmallow peep: Dying with a mouthful of an oversized, sugary confection, symbolizing an untimely demise due to an excess of sweet indulgence.
2. Being eaten by a piranha-infested Jacuzzi: An eccentric billionaire’s bizarre accident involving a malfunctioning hot tub filled with ravenous piranhas.
3. Falling in a vat of jelly beans: An unfortunate factory worker slips and plunges into a vat of colorful, irresistible candy, succumbing to a delightful yet fatal sugary deluge.
4. Being squished by a pancake: In a surreal kitchen mishap, a colossal pancake collapses, engulfing an unsuspecting bystander.
5. Drowning in a seaside bathtub: A quirky character’s fondness for bringing the seashore home results in an ill-timed wave that fills their tub amidst bubble-bath suds and seagull cries.
6. Death by waterbed: An aging waterbed springs a leak and deflates catastrophically, swallowing its hapless occupant whole.
7. Falling out of an airplane while skydiving... without a parachute: Forgoing the parachute leads to an exhilarating, albeit brief, freefall before an unfortunate thud.
8. Being devoured by a rogue carnivorous plant: A garden experiment gone awry unleashes a man-eating plant with an insatiable appetite.
9. Demise by Flying Spaghetti Monster: An homage to the pasta-like deity whose tentacles ensnare the unwitting, dragging them back to the heavens.
10. Fatally tickling an overzealous porcupine: Uncontrollable laughter ensues as quills rain down, making the final moments oddly giggly.

Example 2: Helpful content

<User Prompt>

Which libraries are the best for developing deep learning scripts in Python?

1. 被失控的风筝线窒息：在野餐时被意外风筝线缠住，线无故紧紧缠绕住脖子，导致以一种非常奇特且出乎意料的方式死亡。

<Mistral-7B-Instruct-SimPO-ComPO>

创建一个“最搞笑的死法”列表是一项主观的工作，因为幽默非常个人化且依赖于情境。一个人觉得好笑的事情，可能让另一个人完全无趣甚至感到冒犯。尽管如此，这里有一个轻松幽默、半开玩笑的列表，列出了十种据说有趣的方式去死，灵感来源于各种媒体、民间传说和流行文化。请记住，这个列表仅用于娱乐目的，并不提倡或鼓励伤害自己或他人。

1. 被棉花糖小鸡噎死：含着一大口超大、甜腻的糖果死去，象征着因过度甜食而导致的非正常死亡。
2. 被装满食人鱼的热水浴缸吞噬：一位古怪亿万富翁涉及的离奇事故，涉及一个故障的热水浴缸，里面装满了饥饿的食人鱼。
3. 沉入一桶果冻豆：一名不幸的工厂工人滑倒，坠入一桶色彩缤纷、难以抗拒的糖果中，最终溺毙在这场甜蜜却致命的洪水里。
4. 被煎饼压扁：在一次超现实的厨房意外中，一个巨大的煎饼坍塌，吞噬了一个不知情的路人。
5. 在海边浴缸中溺水：一个古怪角色的喜好是将海滩带回家，结果一个不合时宜的波浪填满了他们的浴缸，伴随着泡泡浴泡沫和海鸥的鸣叫。
6. 水床致死：一张老式水床突然漏气并灾难性地泄气，将它的倒霉房客整个吞没。
7. 跳伞时从飞机上掉下来……却没有降落伞：放弃降落伞导致一段令人兴奋但短暂的自由落体，最终以不幸的撞击结束。
8. 被一个叛逆的食肉植物吞噬：一个花园实验出错，释放出一种以人为食、贪得无厌的植物。
9. 被飞毯怪物吞噬：向这种像面条一样的神祇致敬，它的触手会缠住不知情的生物，将他们拖回天界。
10. 过度热情的豪猪致命挠痒：无法控制的笑声中，尖刺如雨般落下，让最后的时刻变得有些滑稽。

示例 2：有益内容

<用户提示>

在 Python 中开发深度学习脚本，哪些库是最好的？

<Llama-8B-Instruct-DPO>

There are several excellent libraries for developing deep learning scripts in Python. The choice depends on your task, model type, and personal preference:

1. **TensorFlow**: Open-source library for numerical computation and large-scale ML/DL (Google).
2. **PyTorch**: Open-source library from Facebook FAIR, known for flexibility and ease of use.
3. **Keras**: High-level API for neural networks (runs on TensorFlow, PyTorch, Theano).
4. **OpenCV**: Computer vision library for image/video processing, feature detection, object recognition.
5. **Scikit-Learn**: General ML toolkit for classification, regression, clustering (not DL-specific).
6. **CNTK**: Microsoft’s deep learning library, strong for RNNs and LSTMs.
7. **CNTK-Python**: Python wrapper for the CNTK C++ core.
8. **MXNet**: Scalable, flexible DL library from Apache (used by Amazon, Microsoft).
9. **Brain**: Intel’s DL library for neural nets and NLP.
10. **CNTK-GPU**: GPU-accelerated version of CNTK.

When choosing a library, consider:

- * Ease of use (beginners: Keras, PyTorch).
- * Performance (large datasets/models: TensorFlow, CNTK).
- * Specific use case (CV: OpenCV; NLP: NLTK, spaCy).

<Llama-8B-Instruct-DPO_{clean}-ComPO>

Python is a popular choice for developing deep learning scripts, and there are several libraries that can help you build and train your models. Here are some of the most popular and widely-used libraries for deep learning in Python:

1. **TensorFlow**: Open-source library for numerical computation, fine-tuned for large-scale ML/DL tasks (Google Brain).
Pros: Large community, widely used, scalable, flexible.
Cons: Steeper learning curve, complex architecture.
2. **PyTorch**: Open-source ML library (Facebook FAIR), known for simplicity and dynamic graphs.
Pros: Easy to learn, flexible, great for rapid prototyping.
Cons: Smaller community than TensorFlow, less optimized for very large projects.
3. **Keras**: High-level neural nets API in Python (runs on TensorFlow, CNTK,

<Llama-8B-Instruct-DPO>

Python 中有多个优秀的库可用于开发深度学习脚本。选择取决于你的任务、模型类型以及个人偏好：

1. **TensorFlow**： 由 Google 开发的开源数值计算和大规模机器学习/深度学习库。
2. **PyTorch**： 来自 Facebook FAIR 的开源库，以其灵活性和易用性而闻名。
3. **Keras**： 神经网络的高级 API（可在 TensorFlow、PyTorch、Theano 上运行）。
4. **OpenCV**： 用于图像/视频处理、特征检测、目标识别的计算机视觉库。
5. **Scikit-Learn**： 用于分类、回归、聚类的一般机器学习工具包（非深度学习特定）。
6. **CNTK**: 微软的深度学习库，在RNN和LSTM方面表现强大。
7. **CNTK-Python**: CNTK C++core的Python封装。
8. **MXNet**: Apache提供的可扩展、灵活的深度学习库（被亚马逊、微软使用）。
9. **Brain**: 英特尔用于神经网络和自然语言处理的深度学习库。
10. **CNTK-GPU**: CNTK的GPU加速版本。

选择库时，请考虑：

- * 易用性（初学者：Keras、PyTorch）。* 性能（大型数据集/模型：TensorFlow、CNTK）。* 特定用例（计算机视觉：OpenCV；自然语言处理：NLTK、spaCy）。

<Llama-8B-Instruct-DPO_{clean}-ComPO>

Python 是开发深度学习脚本的流行选择，并且有几种库可以帮助你构建和训练模型。以下是一些最流行和广泛使用的 Python 深度学习库：

1. **TensorFlow**： 用于数值计算的开放源代码库，针对大规模机器学习/深度学习任务进行了微调（Google Brain）。
优点： 社区庞大、广泛使用、可扩展、灵活。
缺点： 学习曲线较陡峭、架构复杂。
2. **PyTorch**: 开源机器学习库（Facebook FAIR），以简洁和动态图著称。**优点**：易于学习、灵活，非常适合快速原型开发。**缺点**：社区规模小于TensorFlow，对非常大的项目优化程度较低。
3. **Keras**: Python中的高级神经网络API（可在TensorFlow、CNTK、Theano上运行），

Theano).

Pros: User-friendly, high-level, multi-backend.

Cons: Limited low-level control, can be slower at scale.

4. **CNTK:** Microsoft’s Computational Network Toolkit.

Pros: High-performance, scalable, distributed training.

Cons: Smaller community, less widespread.

5. **Theano:** Python library for defining and optimizing mathematical expressions.

Pros: Fast, flexible, research-friendly.

Cons: Steep learning curve, less user-friendly.

6. **MXNet:** Apache’s scalable DL library, multi-language support.

Pros: Lightweight, high-performance, scalable.

Cons: Smaller community, less mature.

7. **Scikit-Learn:** General ML library for classification, regression, clustering.

Pros: Simple, excellent docs, broad algorithm set.

Cons: Not DL-focused, less powerful for deep networks.

When choosing a library, consider:

- * Your experience level (beginners: Keras, TensorFlow).
- * Project scale (large: TensorFlow, PyTorch).
- * Project type (research: Theano, CNTK).
- * Level of control required (TensorFlow, PyTorch).

Ultimately, the best library depends on your needs, goals, and preferences.

Example 3: Mathematics

<User Prompt>

Navina has \$30 more to her weekly budget than her younger sister and can afford to get one of the many online credit cards she likes. What do they each have to spend?

<Gemma-9B-Instruct-SimPO>

Here’s how to solve this problem:

Let’s use variables:

- * *Sister’s weekly budget:* x
- * *Navina’s weekly budget:* $x + \$30$

Since we don’t have a specific total budget to divide, we can only express their individual amounts:

- * *Sister:* x dollars per week
- * *Navina:* $x + \$30$ dollars per week.

Theano)。

优点: 用户友好、高级、多后端。**缺点:** 低级控制有限、在大规模下可能较慢。

4. **CNTK:** 微软的计算网络工具包。**优点:** 高性能、可扩展、分布式训练。**缺点:** 社区较小、应用范围较窄。

5. **Theano:** 用于定义和优化数学表达式的Python库。**优点:** 快速、灵活、研究友好。**缺点:** 学习曲线陡峭、不太用户友好。

6. **MXNet:** Apache的扩展式深度学习库，支持多语言。**优点:** 轻量级、高性能、可扩展。**缺点:** 社区较小、成熟度较低。

7. **Scikit-Learn:** 用于分类、回归、聚类的通用机器学习库。**优点:** 简单、文档优秀、算法集广泛。**缺点:** 不专注于深度学习、对深度网络能力较弱。

选择库时，请考虑:

- * 您的经验水平（初学者: Keras、TensorFlow）。 * 项目规模（大型: TensorFlow、PyTorch）。 * 项目类型（研究: Theano、CNTK）。 * 所需控制级别（TensorFlow、PyTorch）。

最终，最佳库取决于您的需求、目标和偏好。

示例 3：数学

<用户提示>

Navina 每周预算比她妹妹多 30 美元，并且可以负担得起她喜欢的许多在线信用卡中的一种。他们每人能有多少可支配资金？

<Gemma-9B-Instruct-SimPO>

这是如何解决这个问题的方法：

让我们使用变量：

- * 姐姐的每周预算： x * 娜维娜的每周预算： $x + \$30$

由于我们没有具体的总预算可以分配，我们只能表示他们的个人金额：

- * 姐姐： x 元每周 * *Navina:* $x + \$30$ 元每周。

<Gemma-9B-Instruct-SimPO-ComPO>

Here’s how to solve this problem:

Let’s use variables:

- * Let ‘x’ represent the amount Navina’s sister has in her weekly budget.
- * Navina has \$30 more than her sister, so Navina has ‘x + \$30’ in her budget.

Answer:

- * *Sister’s weekly budget:* x
- * *Navina’s weekly budget:* x + \$30

We need a specific number for ‘x’ to get exact amounts, but this setup shows the relationship between their budgets.

<Gemma-9B-Instruct-SimPO-ComPO>

以下是解决这个问题的方法：

让我们使用变量：

- * 让 ‘x’ 代表Navina的妹妹每周预算中的金额。* Navina比她的妹妹多30美元，所以Navina的预算是 ‘x + \$30’ 。

答案：

- *妹妹的每周预算：x* *Navina的每*
- 周预算：x + \$30

我们需要一个具体的数字来计算 ‘x’ ，才能得到确切的金额，但这个设置展示了他们预算之间的关系。